

Master of Science in Chemistry

Department of Chemistry

The overall credits structure

Category	PC	PE	OC	Total
Credits	60	9	6	75

Program Core

CML511	Quantum Chemistry	3	0	0	3
CML512	Stereochemistry & Organic Reaction Mechanisms	3	0	0	3
CML513	Photochemistry & Pericyclic Reactions	3	0	0	3
CML514	Main Group Chemistry	3	0	0	3
CML515	Instrumental Methods of Analysis	3	0	0	3
CML521	Molecular Thermodynamics	3	0	0	3
CML522	Chemical Dynamics & Surface Chemistry	3	0	0	3
CML523	Organic Synthesis	3	0	0	3
CML524	Transition and Inner Transition Metal Chemistry	3	0	0	3
CML525	Basic Organometallic Chemistry	3	0	0	3
CML526	Structure & Function of Cellular Biomolecules	3	0	0	3
CML631	Molecular Biochemistry	3	0	0	3
CMP511	Lab-I	0	0	4	2
CMP512	Lab-II	0	0	4	2
CMP521	Lab-III	0	0	4	2
CMP522	Lab-IV	0	0	4	2
CMD631	Project Part-I	0	0	12	6
CMD641	Project Part-II	0	0	20	10
Total Credits		60			

Program Electives

CML661	Solid State Chemistry	3	0	0	3
CML662	Statistical Mechanics & Molecular Simulation Methods	3	0	0	3
CML663	Selected Topics in Spectroscopy	3	0	0	3
CML664	Group Theory & Spectroscopy	3	0	0	3
CML665	Biophysical Chemistry	3	0	0	3
CML671	Supramolecular Chemistry	3	0	0	3
CML672	Recent Trends in Organic Chemistry	3	0	0	3
CML673	Bio-organic and Medicinal Chemistry	3	0	0	3
CML674	Physical Methods of Structure Determination of Organic Compounds	3	0	0	3
CML675/740	Chemistry of Heterocyclic Compounds	3	0	0	3
CML681	Physical Methods in Inorganic Chemistry	3	0	0	3
CML682	Inorganic Polymers	3	0	0	3
CML683	Applied Organometallic Chemistry	3	0	0	3
CML684	Bio-inorganic Chemistry	3	0	0	3
CML691	Microbial Biochemistry	3	0	0	3
CML692	Food Chemistry and Biochemistry	3	0	0	3
CML693/739	Applied Biocatalysis	3	0	0	3
CML696	Physical Organic Chemistry	3	0	0	3
CML670	Organic Chemistry of Biological Systems	3	0	0	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)								Lecture courses	Contact h/week				Credits
	L	T	P	Total										
I	CML511 Quantum Chemistry (3-0-0) 3	CML512 Stereo- chemistry & Organic Reaction Mechanisms (3-0-0) 3	CML513 Photochemistry & Pericyclic Reactions (3-0-0) 3	CML514 Main Group Chemistry (3-0-0) 3	CML515 Instrumen- tal Methods of Analysis (3-0-0) 3		CMP511 Lab-I (0-0-4) 2	CMP512 Lab-II (0-0-4) 2	5	15	0	8	23	19
II	CML521 Molecular Thermo- dynamics (3-0-0) 3	CML522 Chemical Dynamics & Surface Chemistry (3-0-0) 3	CML523 Organic Synthesis (3-0-0) 3	CML524 Transition and Inner Transition Metal Chemistry (3-0-0) 3	CML525 Basic Organo- metallic Chemistry (3-0-0) 3	CML526 Structure & Function of Cellular Bio- molecules (3-0-0) 3	CMP521 Lab-III (0-0-4) 2	CMP522 Lab-IV (0-0-4) 2	6	18	0	8	26	22
Summer														
III	CML631 Molecular Bio- chemistry (3-0-0) 3	PE-1 (3-0-0) 3	PE-2 (3-0-0) 3	PE-3 (3-0-0) 3	CMD631 Project Part-I (0-0-12) 6				4	12	0	12	24	18
IV	OE-1 (3-0-0) 3	OE-2 (3-0-0) 3	CMD641 Project Part-II (0-0-20) 10						2	6	0	20	26	16

Total = 75

Master of Science in Cognitive Science

Department of Humanities and Social Sciences

The overall credits structure

Category	PC	PE	OC	Total
Credits	60	9	6	75

Program Core

HSL521	Introduction to Cognitive Science	3	0	2	4	HUL763	Cognitive Psychology	3	0	0	3
HSS521	Independent Study in Cognitive Science (Bridge course)	0	2	0	2	HUL742	Transformational Theories of Language	3	0	0	3
HSL522	Basics of Programming for Cognitive Science	1	0	2	2	HUL743	Language Acquisition, Teaching and Assessment	3	0	0	3
HSL541	Language in the Mind	3	1	0	4	HUL745	Psycholinguistics	3	0	0	3
HSL561	Cognitive Neuroscience	3	0	2	4	HSL727	Advances in Social Cognition	3	0	2	4
HSL621	Mathematical Foundations for Cognitive Science	2	1	0	3	HSL767	Emotion and Decision making	3	0	0	3
HSL622	Computation and Cognition	3	0	2	4	HSL768	Judgment and Decision Making	3	0	0	3
HSL651	Philosophy of Mind and Cognition	3	1	0	4	HSL769	Number Cognition	3	0	2	4
HSL661	Cognitive Processes: From Labs to Fields	3	0	2	4	HSL780	Social and Cultural Construction of Emotions	3	0	0	3
HSL721	Research Methods in Cognitive Science	1	0	4	3	HSL722	Data Analysis for Behavioral Research using R	3	0	2	4
HSL747	Language Computations and Mental Architecture	3	0	0	3	HSL723	Advanced Computational Methods	0.5	0	2	1.5
HSP700	Workshop on Scientific Writing	0	2	8	2	HSL724	Advanced Experimental Methods	0.5	0	2	1.5
HSD621	Cognitive Science Project-I	0	0	12	6	HSL725	Advanced Qualitative Methods	0.5	0	2	1.5
HSD622	Cognitive Science Project-II	0	0	18	9	HSL748	Natural Language Understanding	3	0	0	3
HSD623	Mini Project in Cognitive Science	0	0	4	2	HSL749	Optimality Theory and Harmonic Grammar	3	0	0	3
HSP522	Cognitive Science Lecture Series-I	0	0	4	2	HSL821	Eye Movement and Cognitive Processes	2	0	2	3
HSP523	Cognitive Science Lecture Series-II	0	0	4	2	HSL822	Advanced Data Analysis for Behavioral Research using R	3	0	2	4
Total Credits		60				HSL844	Computational Models of Meaning	3	0	0	3
						HUL843	Reading and Sentence Processing	2	1	0	3

Program Electives

HSL726	Culture and Cognition	3	0	0	3
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Sem.	Courses (Number, Abbreviated Title, L-T-P, credits)						Lecture courses	Contact h/week				Credits
								L	T	P	Total	
I	Programme Core											
	HSL521 Introduction to Cognitive Science (3-0-2) 4	HSS521 Independent Study in Cognitive Science (Bridge course) (0-2-0) 2	HSL522 Basics of Programming for Cognitive Science (1-0-2) 2	HSL541 Language in the Mind (3-1-0) 4	HSL561 Cognitive Neuroscience (3-0-2) 4	HSL721 Research Methods in Cognitive Science (1-0-4) 3	5	11	3	10	24	19
II	Programme Core				Programme Elective							
	HSL621 Mathematical Foundations for Cognitive Science (2-1-0) 3	HSL622 Computation and Cognition (3-0-2) 4	HSL651 Philosophy of Mind and Cognition (3-1-0) 4	HSL661 Cognitive Processes: From Labs to Fields (3-0-2) 4		PE-1 (3-0-0) 3	5	11	2	4	17	18
Summer		HSD623 Mini Project in Cognitive Science (0-0-4) 2										
Programme Core					Open Elective	Programme Elective						
III	HSL747 Language Computations and Mental Architecture (3-0-0) 3	HSP522 Cognitive Science Lecture Series-I (0-0-4) 2	HSP700 Workshop on Scientific Writing (0-2-0) 2	HSD621 Cognitive Science Project-I (0-0-12) 6	OE-I (3-0-0) 3	PE-2 (3-0-0) 3	3	9	2	16	27	19
Programme Core					Open Elective	Programme Elective						
IV	HSD622 Cognitive Science Project II (0-0-18) 9		HSP523 Cognitive Science Lecture-II (0-0-4) 2		OE-II (3-0-0) 3	PE-3 (3-0-0) 3	2	6	0	22	28	17

Total = 75

Master of Science in Economics

Department of Humanities and Social Sciences

The overall credits structure

Category	PC	PE	OC	Total
Credits	60	9	6	75

Program Core

HSL511	Microeconomics-I	3	1	0	4
HSL512	Macroeconomics-I	3	1	0	4
HSL513	Probability and Statistics for Economics	3	0	2	4
HSL516	Mathematical Economics	3	1	0	4
HSL514	Issues in Development	3	0	0	3
HSL611	Microeconomics-II	3	1	0	4
HSL612	Macroeconomics-II	3	1	0	4
HSL613	Econometrics	3	0	2	4
HSL614	Development Economics	3	0	0	3
HSL515	Indian Economy	3	0	0	3
HSP612	Research Seminar in Economics	1	0	4	3
HSP511	Economics Lab	0	0	8	4
HSP611	Advanced Economics Lab	0	0	8	4
HSP520	Research Project in Economics-I	0	0	6	3
HSP620	Research Project in Economics-II	0	0	18	9
Total Credits		60			

Program Electives

Microeconomics Track

HSL816	Game Theory	3	0	0	3
HSL815	Theory of Market Design	3	0	0	3
HSL716	Industrial Economics	3	0	0	3
HSL813	Foundations of Decision Theory	3	0	0	3

Macroeconomics Track

HSL714	International Economics	3	0	0	3
HSL711	Macro Development Economics	3	0	0	3
HSL812	Advanced International Trade	3	0	0	3
HSL811	Advanced Economic Growth Theory	3	0	0	3

Quantitative Economics Track

HSL719	Advanced Econometrics	3	0	0	3
HSL715	Time Series Econometrics and Forecasting	3	0	0	3
MTL625	Principles of Optimization Theory	3	0	0	3
MTL725	Stochastic Processes and its Applications	3	0	0	3

MTL732	Financial Mathematics	3	0	0	3
MTL733	Stochastic of Finance	3	0	0	3
MTL766	Multivariate Statistical Methods	3	0	0	3
MTL843	Mathematical Modeling of Credit Risk	3	0	0	3
MTL505	Computer Programming	3	1	0	4
MTL508	Mathematical Programming	3	1	0	4
COL774	Machine Learning	3	0	2	4
ELL885	Machine Learning for Computational Finance	3	0	0	3
COL671	Principles of Artificial Intelligence	3	0	2	4
Development Economics Track					
HSL818	Labour Economics	3	0	0	3
HSL817	Health Economics	3	0	0	3
HSL814	Research Methods in Economics	1	0	2	2
HSL717	Perspectives on Indian Economy	3	0	0	3
HSL718	Political Economy of Development	3	0	0	3
HSL775	Agrarian Societies and Rural Development	3	0	0	3
HSL874	Civil Society and Democracy in India	3	0	0	3
HSL878	Globalization	3	0	0	3
HSL783	Science, Technology and Society	3	0	0	3
HSL701	Introduction to Science and Technology Policy Studies	1.50	0	1.5	
HSL702	Approaches to Science and Technology Policy Studies	1.50	0	1.5	
HSL703	Perspectives on climate change: Implications for policy	3	0	0	3
HSL704	Inclusive Innovation: Theory and Practice	2	0	4	4
HSL762	Social Issues: Analysis and Policy	3	0	0	3
HSL772	Sociology of India	3	0	0	3
HSL776	Capitalism: Theory and Practice	3	0	0	3
HSL779	Gender and Society	3	0	0	3
HSL781	Potential and Perils of the Digital Welfare	3	0	0	3
HSL782	Perspectives on Development in India	3	0	0	3
HSL801	Law, Technology and Citizenship	3	0	0	3
HSL877	Industry and Society	3	0	0	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, credits)						Lecture courses	Contact h/week				Credits	
								L	T	P	Total		
I	Programme Core												
	HSL511 Microeconomics-1 (3-1-0) 4	HSL512 Macroeconomics-1 (3-1-0) 4	HSL513 Probability and Statistics for Economics (3-0-2) 4	HSL516 Mathematical Economics (3-1-0) 4	HSL514 Issues in Development* (3-0-0) 3		5	15	3	2	20	19	
II	Programme Core					Programme Elective							
	HSL611 Microeconomics-2 (3-1-0) 4	HSL612 Macroeconomics-2 (3-1-0) 4	HSL613 Econometrics (3-0-2) 4	HSL614 Development Economics (3-0-0) 3	HSP511 Economics Lab (0-0-8) 4	OE-1 (3-0-0) 3	5	15	2	10	27	22	
Summer		Summer Internship/Project (optional and non-credit)											
Programme Core					Programme Elective								
III	HSL515 Indian Economy* (3-0-0) 3	HSP612 Research Seminar (1-0-4) 3	HSP611 Advanced Economics Lab (0-0-8) 4	HSP520 Project-I (0-0-6) 3	PE-1 (3-0-0) 3	PE-2 (3-0-0) 3	PE-3 (3-0-0) 3	5	13	0	18	31	22
Programme Core					Programme Elective								
IV	HSP620 Project-II (0-0-18) 9				OE-2 (3-0-0) 3			1	3	0	18	21	12

*Issues in Development and India Economy will be offered in alternate years so that both first year and second year students attend one compulsory course together.

Total = 75

Master of Science in Mathematics

Department of Mathematics

The overall credits structure

Category	PC	PE	OC	Total
Credits	57	12	6	75

Program Core

MTD701	Project-I	0	0	10	5	MTL745	Advanced Matrix Theory	3	0	0	3
MTL501	Algebra	3	1	0	4	MTL746	Methods of Applied Mathematics	3	0	0	3
MTL502	Linear Algebra	3	1	0	4	MTL747	Mathematical Logic	3	0	0	3
MTL503	Real Analysis	3	1	0	4	MTL751	Symbolic Dynamics	3	0	0	3
MTL504	Ordinary Differential Equations	3	1	0	4	MTL752	Data Structures for Applied Mathematics	3	0	2	4
MTL505	Computer Programming	3	0	2	4	MTL755	Algebraic Geometry	3	0	0	3
MTL506	Complex Analysis	3	1	0	4	MTL756	Lie Algebras and Lie Groups	3	0	0	3
MTL507	Topology	3	1	0	4	MTL757	Introduction to Algebraic Topology	3	0	0	3
MTL508	Mathematical Programming	3	1	0	4	MTL760	Advanced Algorithms	3	0	0	3
MTL509	Numerical Analysis	3	1	0	4	MTL761	Basic Ergodic Theory	3	0	0	3
MTL510	Measure and Integration	3	1	0	4	MTL762	Probability Theory	3	0	0	3
MTL601	Probability and Statistics	3	1	0	4	MTL763	Introduction to Game Theory	3	0	0	3
MTL602	Functional Analysis	3	1	0	4	MTL766	Multivariate Statistical Methods	3	0	0	3
MTL603	Partial Differential Equations	3	1	0	4	MTL768	Graph Theory	3	0	0	3
Total Credits					57	MTL773	Wavelets and Applications	3	0	0	3
						MTL776	Graph Algorithms	3	0	0	3
						MTL780	Parameterized Algorithms for NP-hard Problems	3	0	0	3

Program Electives

MTD702	Project-II	0	0	12	6	MTL781	Finite Element Theory and Applications	3	0	0	3
MTL625	Principles of Optimization Theory	3	0	0	3	MTL782	Data Mining	3	0	2	4
MTL704	Numerical Optimization	3	0	0	3	MTL785	Natural Language Processing	3	0	0	3
MTL712	Computational Methods for Differential Equations	3	0	2	4	MTL792	Modern Methods in Partial Differential equations	3	0	0	3
MTL717	Fuzzy Sets and Applications	3	0	0	3	MTL793	Numerical Methods for Hyperbolic PDEs	3	0	0	3
MTL720	Neurocomputing and Applications	3	0	0	3	MTL794	Advanced Probability Theory	3	0	0	3
MTL725	Stochastic Processes and its Applications	3	0	0	3	MTL795	Numerical Method for Partial Differential Equations	3	1	0	4
MTL728	Category Theory	3	0	0	3	MTL799	Mathematical Analysis in Learning Theory	3	0	0	3
MTL729	Computational Algebra and its Applications	3	0	0	3	MTV791	Special Module in Dynamical System	1	0	0	1
MTL730	Cryptography	3	0	0	3	MTL843	Mathematical Modeling of Credit Risk	3	0	0	3
MTL731	Introduction to Chaotic Dynamical Systems	3	0	0	3	MTL851	Applied Numerical Analysis	3	0	0	3
MTL732	Financial Mathematics	3	0	0	3	MTL854	Interpolation and Approximation	3	0	0	3
MTL733	Stochastic of Finance	3	0	0	3	MTL855	Multiple Decision Procedures in Ranking and Selection	3	0	0	3
MTL735	Advanced Number Theory	3	0	0	3	MTL860	Linear Algebra	3	0	0	3
MTL736	Analytic Number Theory	3	0	0	3	MTL863	Algebraic Number Theory	3	0	0	3
MTL737	Differential Geometry	3	0	0	3	MTV874	Analysis	3	0	0	3
MTL738	Commutative Algebra	3	0	0	3	MTL882	Applied Analysis	3	0	0	3
MTL739	Representation of Finite Groups	3	0	0	3	MTL883	Physical Fluid Mechanics	3	0	0	3
MTL741	Fractal Geometry	3	0	0	3	MTL888	Boundary Elements Methods with Computer Implementation	3	0	0	3
MTL742	Operator Theory	3	0	0	3						
MTL743	Fourier Analysis	3	0	0	3						
MTL744	Mathematical Theory of Coding	3	0	0	3						

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)						Lecture courses	Contact h/week				Credits
								L	T	P	Total	
I	MTL501 Algebra (3-1-0) 4	MTL502 Linear Algebra (3-1-0) 4	MTL503 Real Analysis (3-1-0) 4	MTL504 Ordinary Differential Equations (3-1-0) 4	MTL505 Computer Programming (3-0-2) 4		5	15	5	0	20	20
II	MTL506 Complex Analysis (3-1-0) 4	MTL507 Topology (3-1-0) 4	MTL508 Mathematical Programming (3-1-0) 4	MTL509 Numerical Analysis (3-1-0) 4	MTL510 Measure and Integration (3-1-0) 4		5	15	5	0	20	20
Summer												
III	MTL601 Probability and Statistics (3-1-0) 4	MTL602 Functional Analysis (3-1-0) 4	MTL603 Partial Differential Equations (3-1-0) 4	DE-1	MAD701 Project-I (0-0-10) 5		4	12	3	10	25	20
IV	DE-2	DE-3	DE-4	OC-1	OC-2		5	15	0	0	15	15

Total = 75

Master of Science in Physics

Department of Physics

The overall credits structure

Category	PC	PE	OE	Total
Credits	62	12	6	80

Optional Departmental specialization : Additional 6 credits : Total Credits : 86 with specialization

Program Core

PYD561	Project-I	0	0	6	3
PYD562	Project-II	0	0	12	6
PYL551	Classical Mechanics	3	1	0	4
PYL552	Electrodynamics	3	1	0	4
PYL553	Mathematical Physics	3	1	0	4
PYL555	Quantum Mechanics-I	3	1	0	4
PYL556	Quantum Mechanics-II	3	0	0	3
PYL557	Electronics	3	1	0	4
PYL558	Statistical Mechanics	3	1	0	4
PYL560	Applied Optics	3	1	0	4
PYL563	Solid State Physics	3	1	0	4
PYL567	Atomic and Molecular Physics	3	0	0	3
PYL569	Nuclear and Particle Physics	3	0	0	3
PYP561	Laboratory-I	0	0	8	4
PYP562	Laboratory-II	0	0	8	4
PYP563	Advanced Laboratory	0	0	8	4
Total Credits					62

PYL749	Quantum Information and Computation	3	0	0	3
PYL760	Biomedical optics and Bio-photonics	3	0	0	3
PYL761	Liquid Crystals	3	0	0	3
PYL762	Statistical Optics and Optical Coherence Theory	3	0	0	3
PYL770	Ultra-fast optics and applications	3	0	0	3
PYL793	Photonic Devices	3	0	0	3
PYL892	Guided Wave Photonic Sensors	3	0	0	3

Specialization in Condensed Matter Physics Min. 12 credits

PYL651	Advanced Solid State Physics	3	0	0	3
PYL652	Magnetism and Spintronics	3	0	0	3
PYL702	Physics of Semiconductor Devices	3	0	0	3
PYL704	Science and Technology of Thin Films	3	0	0	3
PYL707	Characterization Techniques for Materials	3	0	0	3
PYL727	Energy Materials and Devices	3	0	0	3
PYL728	Quantum Heterostructures	2	0	0	2
PYL739	Computational Techniques for Solid State Materials	3	0	0	3
PYL740	Advanced Condensed Matter Theory	3	0	0	3

Program Electives

PYD658	Mini Project	0	0	6	3
PYL653	Semiconductor Electronics	3	0	0	3
PYL656	Microwaves	3	0	0	3
PYL705	Nanostructured Materials	3	0	0	3
PYL711	Introduction to Nonlinear Dynamics	3	1	0	4
PYL723	Vacuum Science and Cryogenics	3	0	0	3
PYL725	Physics of Amorphous Materials	3	0	0	3
PYL792	Optical Electronics	3	0	0	3

Specialization in Theoretical Physics Min. 12 credits

PYL657	Plasma Physics	3	0	0	3
PYL658	Advanced Plasma Physics	3	0	0	3
PYL730	Plasma Theory and Simulations	3	0	0	3
PYL740	Advanced Condensed Matter Theory	3	0	0	3
PYL741	Field Theory and Quantum Electrodynamics	3	0	0	3
PYL742	General Relativity and Introductory Astrophysics	3	0	0	3
PYL743	Group Theory and its Applications	3	0	0	3
PYL744	High Energy Physics	3	0	0	3
PYL745	Advanced Statistical Mechanics	3	0	0	3
PYL746	Non-equilibrium Statistical Mechanics with Interdisciplinary Applications	3	0	0	3
PYL748	Quantum Optics	3	0	0	3
PYL749	Quantum Information and Computation	3	0	0	3

Specialization in Photonics Min. 12 credits

PYL650	Fiber and Integrated Optics	3	0	0	3
PYL655	Laser Physics	3	0	0	3
PYL659	Laser Spectroscopy	3	0	0	3
PYL747	Non-linear Optics	3	0	0	3
PYL748	Quantum Optics	3	0	0	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)								Lecture courses	Contact h/week				Credits
										L	T	P	Total	
I	PYL551 Classical Mechanics (3-1-0) 4	PYL553 Mathematical Physics (3-1-0) 4	PYL555 Quantum Mechanics (3-1-0) 4	PYL557 Electronics (3-1-0) 4	PYP561 Laboratory-I (0-0-8) 4				4	12	4	8	24	20
II	PYL552 Electro- dynamics (3-1-0) 4	PYL556 Quantum Mechanics-II (3-0-0) 3	PYL558 Statistical Mechanics (3-1-0) 4	PYL560 Applied Optics (3-1-0) 4	PYP562 Laboratory-II (0-0-8) 4	PYL563 Solid State Physics (3-1-0) 4			5	15	3	8	26	22
Summer														
III	PYD561 Project-I (0-0-6) 3	PYL567 Atomic and Molecular Physics (3-0-0) 3	PYL569 Nuclear and Particle Physics (3-0-0) 3	PYP563 Advanced Laboratory (0-0-8) 4	PE-1 (3-0-0) 3	PE-2 (3-0-0) 3	OE-1 (3-0-0) 3	DS-1 (3-0-0) 3	5-6	15-18	1	14	30-33	23-26
IV	PYD562 Project-II (0-0-12) 6	PE-3 (3-0-0) 3	PE-4 (3-0-0) 3	OE-2 (3-0-0) 3	DS-2 (3-0-0) 3				3-4	9-12	0	12	21-24	15-18

Total = 75-81

Master of Science in Biological Sciences

Department of Biological Sciences

The overall credits structure

Category	PC	PE/OE	NGU	Total
Credits	64	12	1	76

Program Core

SBL500	Molecular Cell Biology & Genetics	3	0	2	4
SBL510	Mathematical and Statistical Biology	3	0	2	4
SBL520	Biochemistry	3	0	2	4
SBL530	Biology of Microbes	3	0	2	4
SBL540	Experimental Methods in Biology	3	0	2	4
SBL600	Computational Biology and Data Analyses	1	0	6	4
SBV601	Intellectual Property Rights in Biosciences	1	0	0	1
SBQ601	Seminars in Biological Sciences	0	1	0	1
				NGU	
SBL610	Genetic Engineering and Biotechnology	3	0	2	4
SBL620	Immunobiology	3	0	2	4

SBL630	Advanced Cellular Biology	3	0	2	4
SBL640	Biological Pathways	3	0	2	4
SBD601	Minor Project	0	0	12	6
SBD602	Project / Internship	0	0	34	17

Total Credits

65

All 700/800 courses from KSBS and other academic units at IIT Delhi may be considered as electives, as long as the pre-requisites are fulfilled. No distinction has been made between open electives (OE) and program electives (PE) to ensure and strengthen the interdisciplinary nature of the program.

Program/Open Electives

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)						Lecture courses	Contact h/week				Credits
								L	T	P	Total	
I	SBL500 Molecular Cell Biology & Genetics (3-0-2) 4	SBL510 Mathematical and Statistical Biology (3-0-2) 4	SBL520 Biochemistry (3-0-2) 4	SBL530 Biology of Microbes (3-0-2) 4	SBL540 Experimental Methods in Biology (3-0-2) 4			15	0	5	25	20
II	SBL600 Computational Biology and Data Analyses (1-0-6) 4	SBL610 Genetic Engineering and Biotechnology (3-0-2) 4	SBL620 Immunobiology (3-0-2) 4	SBL630 Advanced Cellular Biology (3-0-2) 4	SBL640 Biological Pathways (3-0-2) 4	SVL601 Intellectual Property Rights in Biosciences (1-0-0) 1		14	0	7	28	21
III	Elective I (3-0-0) 3	Elective II (3-0-0) 3	Elective III (3-0-0) 3	Elective IV (3-0-0) 3	SBD601 Minor Project (0-0-12) 6	SBQ601 Seminars in Biological Sciences (0-1-0) 1 NGU		12	1 NGU	6	24	18 (+1 NGU)
IV	SBD602 Project/Internship (0-0-34) 17							0	0	17	34	17
	Overall Credits											76 (+1 NGU)

Total = 76

Master of Design in Industrial Design

Department of Design

The overall credits structure

Category	PC	PE	OE	Total
Credits	39	9	3	51

Program Core

DDD792	Design Project-I	0	0	6	3
DDD891	Design Project-II	0	0	12	6
DDD892	Industry/Research Design Project	0	0	18	9
DDL710	Framework of Design	2	0	0	2
DDL732	Adv. Mat. Processes & Die Design	2	0	2	3
DDL751	Form and Aesthetics	2	0	2	3
DDP711	Computer Aided Product Detailing	1	0	4	3
DDP721	Design and Innovation Methods	1	0	4	3
DDP722	Applied Ergonomics	1	0	2	2
DDP731	Communication and Presentation Skills	1	0	4	3
DDP741	Product Interface & Design	1	0	2	2
DDR761	Social Immersion (Non-credit)	0	0	2	0
DDR801	Summer Internship (Non-credit)	0	0	4	0
Total Credits				39	

Program Electives

DDL725	Information Design & Data Visualization	2	0	2	3
DDL768	Design Research Methodology	2	0	2	3
DDL782	Design for Usability	2	0	2	3
DDL810	Special Topics in Design-I	3	0	0	3
DDL820	Special Topics in Design-II	3	0	0	3
DDL841	Design Management and Professional Practice	3	0	0	3
DDP712	Exhibitions and Environmental Design	2	0	2	3
DDR762	Vehicle Design	2	0	2	3
DDR772	Transportation Design	2	0	2	3
DDR812	Media Studies	2	0	2	3
DDR822	Design for Sustainability	2	0	2	3
DDD830	Self-initiated Design Project	0	0	6	3
DDR832	Design for User Experience	3	0	0	3
DDR852	Strategic Design Management	2	0	2	3
DDR862	Design in Indian Context	3	0	0	3
DDV820	Special Modules in Design	1	0	0	1

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)						Lecture courses	Contact h/week				Credits
								L	T	P	Total	
I	DDL710 Framework of Design (2-0-0) 2	DDP721 Design and Innovation Methods (1-0-4) 3	DDP731 Communication and Presentation Skills (1-0-4) 3	DDP741 Product Interface & Design (1-0-2) 2	DDL751 Form and Aesthetics (2-0-2) 3		2	7	0	12	19	13
Winter	DDR761 Social Immersion (Non-credit core)											
II	DDP711 Computer Aided Product Detailing (1-0-4) 3	DDP722 Applied Ergonomics (1-0-2) 2	DDL732 Adv. Mat. Processes & Die Design (2-0-2) 3	DDD792 Design Project-I (0-0-6) 3	PE-1 (2-0-2/3-0-0) 3		2	6-7	0	14-16	21-22	14
Summer	DDR801 Summer Internship (Non-credit core)											
III	DDD891 Design Project-II (0-0-12) 6	PE-2 (2-0-2/3-0-0) 3	PE-3 (2-0-2/3-0-0) 3				2	4-6	0	6-10	12-14	12
IV	DDD892 Industry/ Research Design Project (0-0-18) 9	OE (3-0-0) 3					1	3	0	18	21	12

Total = 51

Master of Business Administration

Department of Management Studies

The overall credits structure

Category	Programme Core PC (Total 33 Credits)		Streamed Electives SE (Total 12 credits)		Non-credit Core NC	Programme Electives PE	Total
	Common Core CC	Unique Core UC	Analytical Skills Stream AS	People Skills Stream PS			
Credits	30	3	6	6	3	27	72

Program Core

Programme Core consists of Common Core (CC) courses and Unique Core (UC) courses. The total credits of Programme Core would be 33.

MSL705* HRM Systems	1.5	0	0	1.5
MSL706** Business Laws	3	0	0	3
MSL707* Management Accounting	3	0	0	3
MSL708* Financial Management	3	0	0	3
MSL709* Business Research Methods	1.5	0	0	1.5
MSL711* Strategic Management	3	0	0	3
MSL712* Ethics & Values Based Leadership	1.5	0	0	1.5
MSL713* Information Systems Management	3	0	0	3
MSL720* Macroeconomic Environment of Business	3	0	0	3
MSL745 Operations Management	3	0	0	3
MSL760 Marketing Management	3	0	0	3
MSL780* Managerial Economics	1.5	0	0	1.5
MDD801 MBA Project (Unique Core)	0	0	6	3

Notes:

The UC will include the MBA project which would focus on a research driven application of skills acquired in a particular functional area, through the programme.

* These are new courses which have been designed and/or modified as a part of the curriculum review.

** MSL706 was initially an elective, MSL887. This course's content is the same, only the number has been changed to now reflect a core course.

Total Credits 33

Streamed Electives (SE)

Streamed Electives consist of Analytical Skills (AS) Stream and People Skills (PS) Stream. The total credits of Streamed Electives would be 12 – 6 from AS and 6 from PS.

a) Analytical Skills (AS) Stream

MSL719* Statistics for Management	3	0	0	3
MSL721* Econometrics	3	0	0	3
MSL740 Quantitative Methods in Management	3	0	0	3
MTL732 Financial Mathematics	3	1	0	4

* These are new courses which have been designed and/or modified as a part of the curriculum review.

b) People Skills (PS) Stream

MSL710 Creative Problem Solving	3	0	0	3
MSL724* Business Communication	1.5	0	0	1.5
MSL725* Business Negotiations	1.5	0	0	1.5
MSL727* Interpersonal Behavior & Team Dynamics	1.5	0	0	1.5
MSL729* Individual Behavior in Organization	1.5	0	0	1.5
MSL730* Managing With Power	1.5	0	0	1.5
MSL731* Developing Self Awareness	1.5	0	0	1.5
MSL733* Organization Theory	1.5	0	0	1.5

* These are new courses which have been designed and/or modified as a part of the curriculum review.

Non-credit Core (NC)

MST893 Corporate Sector Attachment	0	0	4	2
MST894* Social Sector Attachment	0	0	2	1

* This is a new course which has been designed as a part of the curriculum review.

Program Electives (PE)

MDL800 Management of Blockchain Technologies	1.5	0	0	1.5
MDL801 Managing Digital Transformation	1.5	0	0	1.5
MDL802 Entrepreneurial Finance	3	0	0	3
MDL803 Fixed Income Securities	3	0	0	3

MDL804 Behavioral Finance	1.5	0	0	1.5
MDL805 Financial Technology	1.5	0	0	1.5
MDL809 Career Management Strategies	3	0	0	3
MDL810 Consumer Behavior	3	0	0	3
MSL716 Fundamentals of Management Systems	3	0	0	3
MSL717* Business Systems Analysis & Design	3	0	0	3
MSL811 Management Control Systems	3	0	0	3
MSL802 Management of Intellectual Property Rights	3	0	0	3
MSL835 Labor Legislation and Industrial Relations	3	0	0	3
MSL704 Science & Technology Policy Systems	3	0	0	3
MSL801 Technology Forecasting & Assessment	1	0	0	3
MSV801 Selected Topics in OB & HR Management	1	0	0	1
MSV802 Selected Topics in Finance	1	0	0	1
MSL803 Technical Entrepreneurship	3	0	0	3
MSV803 Selected Topics in Information Technology Mgmt.	1	0	0	1
MSV804 Selected Topics in Operations Management	1	0	0	1
MSV805 Selected Topics in Economics	1	0	0	1
MSL806* Mergers & Acquisitions	3	0	0	3
MSV806 Selected Topics in Marketing Management	1	0	0	1
MSL807* Selected Topics in Strategic Management	1	0	0	1
MSL808* Systems Thinking	3	0	0	3
MSL809* Cyber Security: Managing Risks	3	0	0	3
MSL810* Advanced Data Mining for Business Decisions	1.50	0	0	1.5
MSL812 Flexible Systems Management	3	0	0	3
MSL813 Systems Methodology for Management	3	0	0	3
MSL814* Data Visualization	1.50	0	0	1.5
MSL815 Decision Support and Expert Systems	3	0	0	3
MSL817 Systems Waste & Sustainability	3	0	0	3
MSL819 Business Process Re-engineering	3	0	0	3
MSL820 Global Business Environment	3	0	0	3
MSL821* Strategy Execution Excellence	3	0	0	3
MSL822 International Business	3	0	0	3
MSL823 Strategic Change & Flexibility	3	0	0	3
MSL824 Policy Dynamics & Learning Organization	3	0	0	3
MSL825 Strategies in Functional Management	3	0	0	3
MSL826 Business Ethics	3	0	0	3
MSL827 International Competitiveness	3	0	0	3
MSL828 Global Strategic Management	3	0	0	3
MSL829 Current & Emerging Issues in Strategic Management	3	0	0	3
MSL851* Strategic Alliance	1.50	0	0	1.5
MSL714 Organizational Dynamics and Environment	3	0	0	3
MSL830 Organizational Structure and Processes	3	0	0	3
MSL831 Management of Change	3	0	0	3
MSL832 Managing Innovation for Organizational Effectiveness	3	0	0	3
MSL833 Organizational Development	3	0	0	3
MSL834* Managing Diversity at Workplace	1.50	0	0	1.5
MSL836* International Human Resources Management	1.50	0	0	1.5
MSL839 Current & Emerging Issues in Organizational Management	3	0	0	3
MSL804* Procurement Management	3	0	0	3
MSL805* Services Operations Management	3	0	0	3
MSL715 Quality and Environment Management Systems	3	0	0	3
MSL816 Total Quality Management	3	0	0	3
MSL818 Industrial Waste Management	3	0	0	3
MSL840 Manufacturing Strategy	3	0	0	3
MSL841* Supply Chain Analytics	3	0	0	3
MSL842* Supply Chain Modeling	3	0	0	3
MSL843 Supply Chain Logistics Management	3	0	0	3
MSL844 Systems Reliability, Safety and Maintenance Management	3	0	0	3
MSL845 Total Project Systems Management	3	0	0	3

Master of Business Administration (Telecommunication Systems Management)

Department of Management Studies

The overall credits structure

Category	Programme Core PC (Total 33 Credits)		Streamed Electives SE (Total 12 credits)		Focus Electives FE	Non-credit Core NC	Programme Electives PE	Total
	Common Core CC	Unique Core UC	Analytical Skills Stream AS	People Skills Stream PS				
Credits	30	3	6	6	6	3	21	72

Program Core (PC)

Programme Core consists of Common Core (CC) courses and Unique Core (UC) courses. The total credits of Programme Core would be 33.

MSL705* HRM Systems	1.5	0	0	1.5
MSL706** Business Laws	3	0	0	3
MSL707* Management Accounting	3	0	0	3
MSL708* Financial Management	3	0	0	3
MSL709* Business Research Methods	1.5	0	0	1.5
MSL711* Strategic Management	3	0	0	3
MSL712* Ethics & Values Based Leadership	1.5	0	0	1.5
MSL713* Information Systems Management	3	0	0	3
MSL720* Macroeconomic Environment of Business	3	0	0	3
MSL745 Operations Management	3	0	0	3
MSL760 Marketing Management	3	0	0	3
MSL780* Managerial Economics	1.5	0	0	1.5
MDD802 MBA Project (Unique Core)	0	0	6	3

Notes:

The UC will include the MBA project which would focus on a research driven application of skills acquired in a particular functional area, through the programme.

* These are new courses which have been designed and/or modified as a part of the curriculum review.

** MSL706 was initially an elective, MSL887. This course's content is the same, only the number has been changed to now reflect a core course.

Total Credits 33

Streamed Electives (SE)

Streamed Electives consist of Analytical Skills (AS) Stream and People Skills (PS) Stream. The total credits of Streamed Electives would be 12 – 6 from AS and 6 from PS.

a) Analytical Skills (AS) Stream

MSL719* Statistics for Management	3	0	0	3
MSL721* Econometrics	3	0	0	3
MSL740 Quantitative Methods in Management	3	0	0	3
MTL732 Financial Mathematics	3	1	0	4

* These are new courses which have been designed and/or modified as a part of the curriculum review.

b) People Skills (PS) Stream

MSL710 Creative Problem Solving	3	0	0	3
MSL724* Business Communication	1.5	0	0	1.5
MSL725* Business Negotiations	1.5	0	0	1.5
MSL727* Interpersonal Behavior & Team Dynamics	1.5	0	0	1.5
MSL729* Individual Behavior in Organization	1.5	0	0	1.5
MSL730* Managing With Power	1.5	0	0	1.5
MSL731* Developing Self Awareness	1.5	0	0	1.5
MSL733* Organization Theory	1.5	0	0	1.5

* These are new courses which have been designed and/or modified as a part of the curriculum review.

Focus Electives (FE)

MSL723 Telecommunication Systems	3	0	0	3
MSL726 Telecom System Analysis, Planning & Design	3	0	0	3
MSL728 International Telecommunication Management	3	0	0	3
EEL767 Telecom Systems	3	0	0	3

Non-credit Core (NC)

MST893 Corporate Sector Attachment	0	0	4	2
MST894* Social Sector Attachment	0	0	2	1

* This is a new course which has been designed as a part of the curriculum review.

Program Electives (PE)

MDL800 Management of Blockchain Technologies	1.5	0	0	1.5
MDL801 Managing Digital Transformation	1.5	0	0	1.5
MDL803 Fixed Income Securities	3	0	0	3
MDL804 Behavioral Finance	1.5	0	0	1.5
MDL805 Financial Technology	1.5	0	0	1.5
MDL809 Career Management Strategies	3	0	0	3
MDL810 Consumer Behavior	3	0	0	3
MSL716 Fundamentals of Management Systems	3	0	0	3
MSL717* Business Systems Analysis & Design	3	0	0	3
MSL811 Management Control Systems	3	0	0	3
MSL802 Management of Intellectual Property Rights	3	0	0	3
MSL835 Labor Legislation and Industrial Relations	3	0	0	3
MSL704 Science & Technology Policy Systems	3	0	0	3
MSL801 Technology Forecasting & Assessment	3	0	0	3
MSL803 Technical Entrepreneurship	3	0	0	3
MSL806* Mergers & Acquisitions	3	0	0	3
MSL807* Selected Topics in Strategic Management	1	0	0	1
MSL808* Systems Thinking	3	0	0	3
MSL812 Flexible Systems Management	3	0	0	3
MSL813 Systems Methodology for Management	3	0	0	3
MSL817 Systems Waste & Sustainability	3	0	0	3
MSL819 Business Process Re-engineering	3	0	0	3
MSL820 Global Business Environment	3	0	0	3
MSL821* Strategy Execution Excellence	3	0	0	3
MSL822 International Business	3	0	0	3
MSL823 Strategic Change & Flexibility	3	0	0	3
MSL824 Policy Dynamics & Learning Organization	3	0	0	3
MSL825 Strategies in Functional Management	3	0	0	3
MSL826 Business Ethics	3	0	0	3
MSL827 International Competitiveness	3	0	0	3
MSL828 Global Strategic Management	3	0	0	3
MSL829 Current & Emerging Issues in Strategic Mgmt.	3	0	0	3
MSL851* Strategic Alliance	1.5	0	0	1.5
MSL714 Organizational Dynamics and Environment	3	0	0	3
MSL809* Cyber Security: Managing Risks	3	0	0	3
MSL810* Advanced Data Mining for Business Decisions	1.5	0	0	1.5
MSL814* Data Visualization	1.5	0	0	1.5
MSL815 Decision Support and Expert Systems	3	0	0	3
MSL830 Organizational Structure and Processes	3	0	0	3
MSL831 Management of Change	3	0	0	3
MSL832 Managing Innovation for Organizational Effectiveness	3	0	0	3
MSL833 Organizational Development	3	0	0	3
MSL834* Managing Diversity at Workplace	1.5	0	0	1.5
MSL836* International Human Resources Management	1.5	0	0	1.5
MSL839 Current & Emerging Issues in Organizational Management	3	0	0	3
MSL804* Procurement Management	3	0	0	3
MSL805* Services Operations Management	3	0	0	3
MSL715 Quality and Environment Management Systems	3	0	0	3
MSL816 Total Quality Management	3	0	0	3
MSL818 Industrial Waste Management	3	0	0	3
MSL840 Manufacturing Strategy	3	0	0	3
MSL841* Supply Chain Analytics	3	0	0	3
MSL842* Supply Chain Modeling	3	0	0	3
MSL843 Supply Chain Logistics Management	3	0	0	3
MSL844 Systems Reliability, Safety and Maintenance Management	3	0	0	3
MSL845 Total Project Systems Management	3	0	0	3
MSL846 Total Productivity Management	3	0	0	3
MSL848* Applied Operations Research	3	0	0	3

MSL849	Current & Emerging Issues in Manufacturing Management	3	0	0	3	MSL880	Selected Topics in Management Methodology	3	0	0	3
MSL850	Management of Information Technology	3	0	0	3	MSL881	Mgmt. of Public Sector Enterprises in India	3	0	0	3
MSL852	Network System: Applications and Management	3	0	0	3	MSL889	Current & Emerging Issues in Public Sector Mgmt.	3	0	0	3
MSL853*	Software Project Management	3	0	0	3	MSL897	Consultancy Process & Skills	3	0	0	3
MSL854*	Big Data Analytics & Data Science	1.5	0	0	1.5	MSL898	Consultancy Professional Practice	3	0	0	3
MSL855*	Electronic Commerce	3	0	0	3	MSL899	Current & Emerging Issues in Consultancy Management	3	0	0	3
MSL856*	Business Intelligence	3	0	0	3	MSL895	Advanced Data Analysis for Management	3	0	0	3
MSL858*	Business Process Management with IT	1.5	0	0	1.5	MSL896	International Economic Policy	3	0	0	3
MSL859	Current and Emerging Issues in IT Mgmt.	3	0	0	3	MSV801	Selected Topics in OB & HR Management	1	0	0	1
MSL868*	Digital Research Methods	1.5	0	0	1.5	MSV802	Selected Topics in Finance	1	0	0	1
MSL876*	Economics of Digital Business	1.5	0	0	1.5	MSV803	Selected Topics in IT Management	1	0	0	1
MSL877*	Electronic Government	1.5	0	0	1.5	MSV804	Selected Topics in Operations Management	1	0	0	1
MSL878*	Electronic Payments	1.5	0	0	1.5	MSV805	Selected Topics in Economics	1	0	0	1
MSL882*	Enterprise Cloud Computing	1.5	0	0	1.5	MSV806	Selected Topics in Marketing Management	1	0	0	1
MSL883*	ICTs, Development and Business	1.5	0	0	1.5	MSV815	Case Study Teaching and Writing	1	0	0	1
MSL884*	Information System Strategy	3	0	0	3	MSL799	Theories in IS research	3	0	0	3
MSL885*	Digital Marketing-Analytics & Optimization	3	0	0	3	MSV816	Contemporary Issue in Management	1	0	0	1
MSL886*	IT Consulting & Practice	3	0	0	3	MSV817	Art of Scholarship in Management Research	1	0	0	1
MSL887*	Mobile Commerce	3	0	0	3	MSP801	Interdisciplinary Research in Management	0	0	2	1
MSL888*	Data Warehousing for Business Decision	1.5	0	0	1.5	MSV818	Contemporary Issues in OB & HR Mgmt.	1	0	0	1
MSL891*	Data Analytics using SPSS	1.5	0	0	1.5	MSV819	Contemporary Issues in Finance	1	0	0	1
MSL892*	Predictive Analytics	1.5	0	0	1.5	MSV820	Contemporary Issues in IT Management	1	0	0	1
MSL861	Market Research	3	0	0	3	MSV821	Contemporary Issues in Operations Mgmt.	1	0	0	1
MSL862	Product Management	3	0	0	3	MSV822	Contemporary Issues in Economic Policy	1	0	0	1
MSL863	Advertising and Sales Promotion Management	3	0	0	3	MSV824	Contemporary Issues in Strategic Mgmt.	1	0	0	1
MSL864*	Corporate Communication	3	0	0	3	MSV826	Frontiers in OB & HR Management	1	0	0	1
MSL865	Sales Management	3	0	0	3	MSV827	Frontiers in Finance	1	0	0	1
MSL866	International Marketing	3	0	0	3	MSV828	Frontiers in Information Systems Mgmt.	1	0	0	1
MSL867	Industrial Marketing Management	3	0	0	3	MSV832	Frontiers in Strategic Management	1	0	0	1
MSL869	Current & Emerging Issues in Marketing	3	0	0	3	MSL735	Qualitative Methods in Management	3	0	0	3
MSL870*	Corporate Governance	1.5	0	0	1.5	MSL781	Macroeconomic Dynamics	3	0	0	3
MSL871*	Banking and Financial Services	1.5	0	0	1.5	MDL806	Derivatives	3	0	0	3
MSL872	Working Capital Management	3	0	0	3	MSL890	Financial Engineering	3	0	0	3
MSL873	Security Analysis & Portfolio Management	3	0	0	3	MSL310	Financial Institutions and Markets	3	0	0	3
MSL874*	Indian Financial System	1.5	0	0	1.5	MSL718	Management of Blockchain Technology	1.5	0	0	1.5
MSL875	International Financial Management	3	0	0	3	MSL722	Managing Enterprise AI/ML Systems	1.5	0	0	1.5
MSL879	Current & Emerging Issues in Finance	3	0	0	3	MSL782	Business Cycles and Global Economy	1.5	0	0	1.5
MSL734	Management of Small & Medium Scale Industrial Enterprises	3	0	0	3	MSL783	Global Economic Development	1.5	0	0	1.5
MSL847	Advanced Methods for Management Research	3	0	0	3	MSL784	Sovereign Debt and Default	3	0	0	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, credits)									Lecture courses	Contact h/week				Credits
											L	T	P	Total	
I	MSL707 Mgmt. Accounting (3-0-0) 3	MSL709 Business Research Methods (1.5-0-0) 1.5	MSL712 Ethics & Values Based Leadership (1.5-0-0) 1.5	MSL760 Marketing Mgmt. (3-0-0) 3	MSL780 Managerial Economics (1.5-0-0) 1.5	SE AS-1 (3-0-0) 3	SE AS-2 (3-0-0) 3	SE PS-1 (1.5-0-0) 1.5	SE PS-2 (1.5-0-0) 1.5	9	19.5	0	0	19.5	19.5
Winter	MST894 Social Sector Attachment														
II	MSL705 HRM Systems (1.5-0-0) 1.5	MSL708 Financial Mgmt. (3-0-0) 3	MSL711 Strategic Mgmt. (3-0-0) 3	MSL713 Infor- mation Systems Mgmt. (3-0-0) 3	MSL720 Macroeconomic Environment of Business (3-0-0) 3	MSL745 Operations Mgmt. (3-0-0) 3	SE PS-3 (1.5-0-0) 1.5	SE PS-4 (1.5-0-0) 1.5	7/8	19.5	0	0	19.5	19.5	
						SE PS-5 (3-0-0) 3									
Summer	MST893 Corporate Sector Attachment														
III	PE (Credits 15-18)									6/7	15-18	0	0	15-18	15-18
IV	MDD802 MBA Project (0-0-6) 3	MSL706 Business Laws (3-0-0) 3	FE-1 (3-0-0) 3	FE-2 (3-0-0) 3	PE (Credits 3-6)				4/6	12-15	0	6	18-21	15-18	

SE = Streamed Electives, AS = Analytical Skills Stream, PS = People Skills Stream, PE = Programme Electives

Total = 72

Master of Business Administration (Executive)

Department of Management Studies

The overall credits structure

Category	Programme Core PC (Total 36 Credits)		Streamed Electives SE (Total 12 credits)		Non-credit Core NC	Programme Electives PE	Total
	Common Core CC	Unique Core UC	Analytical Skills Stream AS	People Skills Stream PS			
Credits	24	3	6	4.5	3	16.5	54

Program Core (PC)

Programme Core consists of Common Core (CC) courses and Unique Core (UC) courses. The total credits of Programme Core would be 33.

MSL705* HRM Systems	1.5	0	0	1.5
MSL707* Management Accounting	3	0	0	3
MSL709* Business Research Methods	1.5	0	0	1.5
MSL711* Strategic Management	3	0	0	3
MSL712* Ethics & Values Based Leadership	1.5	0	0	1.5
MSL713* Information Systems Management	3	0	0	3
MSL719* Statistics for Management	3	0	0	3
MSL720* Macroeconomic Environment of Business	3	0	0	3
MSL724* Business Communication	1.5	0	0	1.5
MSL727* Interpersonal Behavior & Team Dynamics	1.5	0	0	1.5
MSL729* Individual Behavior in Organization	1.5	0	0	1.5
MSL740 Quantitative Methods in Management	3	0	0	3
MSL745 Operations Management	3	0	0	3
MSL760 Marketing Management	3	0	0	3
MSL780* Managerial Economics	1.5	0	0	1.5
MDD803 MBA Project (Unique Core)	0	0	6	3
Total Credits			33	

Notes:

The UC will include MBA project which would focus on a research driven application of skills acquired in a particular functional area, through the programme.

* These are new courses which have been designed and/or modified as a part of the curriculum review.

** MSL706 was initially an elective, MSL887. This course's content is the same, only the number has been changed to now reflect a core course.

Streamed Electives (SE)

Streamed Electives consist of Analytical Skills (AS) Stream and People Skills (PS) Stream. The total credits of Streamed Electives would be 12 – 6 from AS and 6 from PS.

a) Analytical Skills (AS) Stream

MSL721* Econometrics	3	0	0	3
MTL732 Financial Mathematics	3	1	0	4

* These are new courses which have been designed and/or modified as a part of the curriculum review.

b) People Skills (PS) Stream

MSL710 Creative Problem Solving	3	0	0	3
MSL725* Business Negotiations	1.5	0	0	1.5
MSL730* Managing With Power	1.5	0	0	1.5
MSL731* Developing Self Awareness	1.5	0	0	1.5
MSL733* Organization Theory	1.5	0	0	1.5

* These are new courses which have been designed and/or modified as a part of the curriculum review.

Focus Electives (FE)

MSL700 Fundamentals of Management of Technology	3	0	0	3
MSL701 Strategic Technology Management	3	0	0	3
MSL702 Management of Innovation and R&D	3	0	0	3
MSL703 Mgmt. of Technology Transfer and Absorption	3	0	0	3

Non-credit Core (NC)

MSC894* Seminar	0	0	6	3
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* This is a new course which has been designed as a part of the curriculum review.

Program Electives (PE)

MSL706** Business Laws	3	0	0	3
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MSL708* Financial Management	3	0	0	3
MDL800 Management of Blockchain Technologies	1.5	0	0	1.5
MDL801 Managing Digital Transformation	1.5	0	0	1.5
MDL802 Entrepreneurial Finance	3	0	0	3
MDL803 Fixed Income Securities	3	0	0	3
MDL804 Behavioral Finance	1.5	0	0	1.5
MDL805 Financial Technology	1.5	0	0	1.5
MDL809 Career Management Strategies	3	0	0	3
MDL810 Consumer Behavior	3	0	0	3
MSL716 Fundamentals of Management Systems	3	0	0	3
MSL717* Business Systems Analysis & Design	3	0	0	3
MSL811 Management Control Systems	3	0	0	3
MSL802 Management of Intellectual Property Rights	3	0	0	3
MSL835 Labor Legislation and Industrial Relations	3	0	0	3
MSL704 Science & Technology Policy Systems	3	0	0	3
MSL801 Technology Forecasting & Assessment	3	0	0	3
MSL803 Technical Entrepreneurship	3	0	0	3
MSL806* Mergers & Acquisitions	3	0	0	3
MSL807* Selected Topics in Strategic Management	1	0	0	1
MSL808* Systems Thinking	3	0	0	3
MSL812 Flexible Systems Management	3	0	0	3
MSL813 Systems Methodology for Management	3	0	0	3
MSL817 Systems Waste & Sustainability	3	0	0	3
MSL819 Business Process Re-engineering	3	0	0	3
MSL820 Global Business Environment	3	0	0	3
MSL821* Strategy Execution Excellence	3	0	0	3
MSL822 International Business	3	0	0	3
MSL823 Strategic Change & Flexibility	3	0	0	3
MSL824 Policy Dynamics & Learning Organization	3	0	0	3
MSL825 Strategies in Functional Management	3	0	0	3
MSL826 Business Ethics	3	0	0	3
MSL827 International Competitiveness	3	0	0	3
MSL828 Global Strategic Management	3	0	0	3
MSL829 Current & Emerging Issues in Strategic Mgmt.	3	0	0	3
MSL851* Strategic Alliance	1.5	0	0	1.5
MSL714 Organizational Dynamics and Environment	3	0	0	3
MSL830 Organizational Structure and Processes	3	0	0	3
MSL831 Management of Change	3	0	0	3
MSL832 Managing Innovation for Organizational Effectiveness	3	0	0	3
MSL833 Organizational Development	3	0	0	3
MSL834* Managing Diversity at Workplace	1.5	0	0	1.5
MSL836* International Human Resources Management	1.5	0	0	1.5
MSL839 Current & Emerging Issues in Organizational Management	3	0	0	3
MSL804* Procurement Management	3	0	0	3
MSL805* Services Operations Management	3	0	0	3
MSL715 Quality and Environment Management Systems	3	0	0	3
MSL814* Data Visualization	1.5	0	0	1.5
MSL815 Decision Support and Expert Systems	3	0	0	3
MSL816 Total Quality Management	3	0	0	3
MSL818 Industrial Waste Management	3	0	0	3
MSL840 Manufacturing Strategy	3	0	0	3
MSL841* Supply Chain Analytics	3	0	0	3
MSL842* Supply Chain Modeling	3	0	0	3
MSL843 Supply Chain Logistics Management	3	0	0	3
MSL844 Systems Reliability, Safety and Maintenance Mgmt.	3	0	0	3
MSL845 Total Project Systems Management	3	0	0	3
MSL846 Total Productivity Management	3	0	0	3
MSL848* Applied Operations Research	3	0	0	3
MSL849 Current & Emerging Issues in Manufacturing Mgmt.	3	0	0	3
MSL809* Cyber Security: Managing Risks	3	0	0	3
MSL810* Advanced Data Mining for Business Decisions	1.5	0	0	1.5

Master of Arts in Culture, Society, Thought

Department of Humanities and Social Sciences

The overall credits structure

Category	PC	PE	OE	Total
Credits	53	15	6	74

Program Core

HSN500	Interdisciplinary Reading Bootcamp: Culture, Society, Thought	0	1	0	1
HSN590	Writing Workshop	1	0	6	4
XXX	Dept Seminars	0	0	0	4
HSL531	Literary Theory	3	0	0	3
HSL551	Philosophical Thinking	3	0	0	3
HSL582	Theory & Methods in Humanities and Social Sciences	3	0	0	3
HSL571	Contemporary Social Theory	3	0	0	3
HSL584	Political Thought	3	0	0	3
HSL583	Introduction to Research Methods	3	0	0	3
HSL572	Programming for Digital Humanities*	1	0	2	2
HSL575	Fieldwork Methods*	2	0	0	2
HSV735	Narrative Matters*	2	0	0	2
HSL581	Archival Methods*	1	0	2	2
HSV734	Dimensions of Language*	2	0	0	2
HSL585	Introduction to Digitization*	1	0	2	2
HSD690	Research Project Part I	0	0	6	3
HSD691	Research Project Part II	0	0	8	4
HSD692	Research Project Part III	0	0	26	13

Total Credits **53**

(*any three have to be taken)

Program Electives

HSL631	Study of an Author	3	0	0	3
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HSL686	Foundations in Digital Humanities	3	0	0	3
HSL675	Education and Society	3	0	0	3
HSL676	Religion and Society	3	0	0	3
HSL682	Introduction to Anti-Caste Thought and Literature	3	0	0	3
HSL688	Agrarian Societies and Transformation	3	0	0	3
HSL683	Feminist Thought	3	0	0	3
HSL657	Machine Minds: Philosophy of Artificial Intelligence	3	0	0	3
HSL632	The World Novel	3	0	0	3
HSL633	The Lyric Poet in Modernity	3	0	0	3
HSL634	Political Theatre	3	0	0	3
HSL652	Philosophy and Its Histories	3	0	0	3
HSL653	Principles of Rationalism and Empiricism	3	0	0	3
HSL654	Ideas of Freedom	3	0	0	3
HSL655	The Idea of Democracy	3	0	0	3
HSL656	Critical Philosophy of Race	3	0	0	3
HSL671	Reading Ethnographic Texts	3	0	0	3
HSL672	Making Sense of the Everyday: Theories, Debates and Critiques	3	0	0	3
HSL673	Family, Marriage and Kinship	3	0	0	3
HSL674	Urban Ethnography	3	0	0	3
HSL677	Comparative Literature: History, Theory and Method	3	0	0	3
HSL678	Cinemas of India	3	0	0	3
HSL681	Print Culture and History	3	0	0	3
HSL684	Queering Gender and Sexuality: Debates from India	3	0	0	3
HSL685	Culture and Politics	3	0	0	3
HSL687	Introduction to Medical Humanities	3	0	0	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, credits)						Lecture course	Contact h/week				Credits	NGU	Total	
	L	T	P	Total											
I	Programme Core				Programme Electives		Colloquium								
	HSN500 Reading Bootcamp (0-1-0) 1	HSL531 Literary Theory (3-0-0) 3	HSL551 Philosophical Thinking (3-0-0) 3	HSL582 Theory & Methods in HSS (3-0-0) 3	PE-1 (3-0-0) 3		NGU-1 Department Seminar Series		4	12	0	0	12	12	2
II	Programme Core			Research	Programme Electives		Colloquium								
	HSL571 Contemporary Social Theory (3-0-0) 3	HSL584 Political Thought (3-0-0) 3	HSL583 Introduction to Research Methods (3-0-0) 3	HSD690 Research Project Part-1 (0-0-6) 3	PE-2 (3-0-0) 3	PE-3 (3-0-0) 3	NGU-1 Department Seminar Series		5	15	15	0	15	15+3 (Re- search Proj- ect)=18	1
Summer	HSN900 Writing Workshop													4	4
III	Programme Core		Research	Programme Electives		Open	Colloquium								
	HSL572 Programming for DH	HSL574 Dimensions of Language	HSD691 Research Project Part II (0-0-8) 4	PE-4 (3-0-0) 3	PE-5 (3-0-0) 3	OE-1 (3-0-0) 3	NGU-1 Department Seminar Series	5	13	0	0	13	13+4 (Re- search Project) = 17	1	18
	HSL573 Narrative Matters	HSL575 Fieldwork Methods													
	HSL581 Archival Methods	HSL585 Intro to Digitisation													
	Any Two form this list														
IV	Programme Core		Research		Open		Colloquium								
	HSL572 Programming for DH	HSL574 Dimensions of Language	HSD692 Research Project Part III (0-0-26) 13		OE-2 (3-0-0) 3	NGU-1 Department Seminar Series	2	5	0	0	5	5+13 (Re- search Project) = 18	1	19	
	HSL573 Narrative Matters	HSL575 Fieldwork Methods													
	HSL581 Archival Methods	HSL585 Intro to Digitisation													
	Any One form this list														
Note: HSI 572, HSI 573, HSI 575, HSI 581 and HSI 585 are basket of courses on methods. Students compulsorily have to choose three out of the combination of courses that are offered.															

Note: HSL572, HSL573, HSL575, HSL581 and HSL585 are basket of courses on methods. Students compulsorily have to choose three out of the combination of courses that are offered.

Total = 74

Master of Technology in Engineering Analysis and Design

Department of Applied Mechanics

The overall credits structure

Category	PC	PE	OC	Total
Credits	34	12	6	52

Program Core (PC)

APL701	Continuum Mechanics	3	0	0	3
APL702	Experimental Methods for Solids and Fluids	2	0	2	3
APL703	Engineering Mathematic and Computation	3	0	2	4
APL734	Advanced Dynamics	3	0	0	3
APL775	Design Methods	3	0	0	3
AMP811	Project-I	0	0	12	6
AMP812	Project-II	0	0	24	12

Product Design (Program Electives)

APL710	Computer Aided Design	3	0	2	4
APL737	Advanced Design of Machine Elements	3	0	0	3
APL767	Engineering Failure Analysis and Prevention	3	0	0	3
APL771	Design Optimization and Decision Theory	3	0	0	3
APL774	Modeling & Analysis of Mechanical Systems	3	0	0	3
APL776	Product Design and Feasibility Study (Stream Core)	2	0	4	4
APL787	Fatigue Failure and Design	3	0	0	3
APL871	Product Reliability	3	0	0	3
MCL741	Control Engineering	3	0	2	4
MCL749	Mechatronics Product Design	3	0	2	4

Engineering Mechanics (Program Electives)

APL705	Finite Element Method	3	0	2	4
APL711	Advanced Fluid Mechanics	3	0	0	3
APL713	Turbulence and its Modeling	3	0	0	3
APL715	Physics of Turbulent Flows	3	0	0	3
APL716	Fluid Transportation Systems	3	0	0	3
APL720	Computational Fluid Dynamics	3	0	2	4

AML833	Applied Plasticity	3	0	0	3
APL737	Advanced Design of Machine Elements	3	0	0	3
APL736	Multiscale Modelling of Crystalline Materials	3	0	2	4
APL742	Advanced Biomechanics	3	0	0	3
APL744	Probabilistic Machine Learning for Mechanics	3	0	2	4
APL745	Deep Learning for Mechanics	3	0	2	4
APL746	Environmental Fluid Dynamics	3	0	0	3
APL747	Uncertainty Quantification and Propagation	3	0	0	3
APL765	Fracture Mechanics	3	0	0	3
APL787	Fatigue Failure and Design	3	0	0	3
APL796	Advanced Solid Mechanics	3	0	0	3
APL805	Advanced Finite Element Method	3	0	0	3
AML811	Advanced CFD	3	0	0	3
APL815	Hydrodynamic Stability	3	0	0	3
AML816	Compressible Fluid Flow and Gas Dynamics	3	0	0	3
APL831	Theory of Plates and Shells	3	0	0	3
APL835	Mechanics of Composite Materials	3	0	0	3

Materials (Program Electives)

APL750	Modern Engineering Materials	3	0	0	3
APL756	Microstructural Characterization of Materials	3	0	2	4
APL759	Phase Transformations	3	0	0	3
APL763	Micro & Nanoscale Mechanical Behaviour of Materials	3	0	2	4
APL764	Mechanical Behaviour of Biomaterials	3	0	0	3
APL765	Fracture Mechanics	3	0	0	3
APL767	Engineering Failure Analysis and Prevention	3	0	0	3
APL	Selected Topics in Material Engineering	3	0	0	3

Semester wise course breakup for two streams

Sem.	Courses (Number, Abbreviated Title, L-T-P, credits)							Lecture courses	Contact h/week				Credits
									L	T	P	Total	
I	APL775 Design Methods (3-0-0) 3	APL734 Advanced Dynamics (3-0-0) 3	APL703 Engineering Mathematics & Computation (3-0-2) 4	APL701 Continuum Mechanics (3-0-0) 3	APL702 Experimental Methods for Solids & Fluids (2-0-2) 3			5	14	0	4	18	16
II	PE-1	PE-2	PE-3	OE-1				4	12	0	0	12	12
Summer													
III	OE2	AMP811	PE-4					2	6	0	12	18	12
IV	AMP812							0	0	0	24	24	12

Total = 52

Master of Technology in Biomolecular and Bioprocess Engineering

Department of Biochemical Engineering and Biotechnology

The overall credits structure

Category	PC	PE	OE	Total
Credits	41	12	-	53

Program Core

BBL771	Microbial Biochemistry and Molecular Biology	3	0	2	4	BBL736	Dynamics of Microbial Systems	3	0	0	3
BBL772	Data Analytics and Informatics in Biotechnology	2	0	2	3	BBL737	Instrumentation and Analytical Methods in Bioengineering	2	0	2	3
BBL773	Applied Mathematics for Biochemical Engg.	3	0	0	3	BBL740	Plant Cell Technology	3	0	2	4
BBL774	Biomolecular Engineering	3	0	2	4	BBL741	Protein Science & Engineering	3	0	0	3
BBL775	Bioreaction Engineering	3	0	3	4.5	BBL742	Biological Waste Treatment	3	0	2	4
BBL731	Bioseparation Engineering	3	0	3	4.5	BBL745	Combinatorial Biotechnology	3	0	0	3
BBD855	MTP PART-I	0	0	12	6	BBL747	Bionanotechnology	3	0	0	3
BBD856	MTP PART-II	0	0	24	12	BBL749	Cancer Cell Biology	3	0	3	4.5
Total Credits		41				BBL750	Genome Engineering	2	0	2	3

Program Electives

BBL734	Metabolic Regulation & Engineering	3	0	0	3	BBL752	Microbial Ecology	3	0	0	3
BBL735	Genomics and Proteomics	2	0	2	3	BBL754	Optics with Life Sciences	3	0	0	3
						BBL757	Electromicrobiology and Bioelectrochemical Systems	3	0	0	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)				Lecture courses	Contact h/week				Credits
						L	T	P	Total	
I	BBL771 Microbial Biochemistry and Molecular Biology (3-0-2) 4	BBL772 Data Analytics and Informatics in Biotechnology (2-0-2) 3	BBL773 Applied Mathematics for Biochemical Engineering (3-0-0) 3	PE-1	4	--	--	--		13
II	BBL731 Bioseparation Engineering (3-0-3) 4.5	BBL774 Biomolecular Engineering (3-0-2) 4	BBL775 Bioreaction Engineering (3-0-3) 4.5	PE-2	4	--	--	--		16
III	BBD855 Major Project-I (0-0-12) 6		PE-3	PE-4	2	--	--	--		12
IV	BBD856 Major Project-II (0-0-24) 12				0	0	0	24	24	12

Total = 53

Master of Technology in Chemical Engineering

Department of Chemical Engineering

The overall credits structure

Category	PC	PE	OE	Total
Credits	37	12	3	52

Program Core

CLD771	Minor Project	0	0	6	3	CLL743	Petrochemicals Technology	3	0	0	3
CLD781	Major Project Part-I	0	0	16	8	CLL761	Chemical Engineering Mathematics	3	0	0	3
CLD782	Major Project Part-II	0	0	24	12	CLL762	Advanced Computational Techniques in Chemical Engineering	2	0	2	3
CLL701	Modelling of Transport Processes	2	0	0	2	CLL766	Interfacial Engineering	3	0	0	3
CLL702	Principles of Thermodynamics, Reaction Kinetics and Reactors	2	0	0	2	CLL767	Structures and Properties of Polymers	3	0	0	3
CLL703	Process Engineering	3	0	0	3	CLL768	Fundamentals of Computational Fluid Dynamics	2	0	2	3
CLP704	Technical Communication for Chemical Engineers	0	0	2	1	CLL769	Applications of Computational Fluid Dynamics	2	0	2	3
CLL731	Advanced Transport Phenomena	3	0	0	3	CLL770	Introduction to Microfluidics and Microfabrication	3	0	0	3
CLL733	Industrial Multiphase Reactors	3	0	0	3	CLL771	Introduction to Complex Fluids	3	0	0	3
Total Credits		37				CLL772	Transport Phenomena in Complex Fluids	3	0	0	3

Program Electives

CLL704	Natural Gas Processing	3	0	0	3	CLL776	Granular Materials	3	0	0	3
CLL705	Petroleum Reservoir Engineering	3	0	0	3	CLL777	Complex Fluids Technology	3	0	0	3
CLL706	Petroleum Production Engineering	3	0	0	3	CLL778	Interfacial Behaviour and Transport of Biomolecules	3	0	0	3
CLL707	Population Balance Modeling	3	0	0	3	CLL779	Molecular Biotechnology and in-vitro Diagnostics	3	0	0	3
CLL720	Principles of Electrochemical Engineering	3	0	0	3	CLL780	Bioprocessing and Bioseparations	3	0	0	3
CLL721	Electrochemical Methods	3	0	0	3	CLL781	Process Operations Scheduling	3	0	0	3
CLL722	Electrochemical Conversion and Storage Devices	3	0	0	3	CLL782	Process Optimization	3	0	0	3
CLL723	Hydrogen Energy and Fuel Cell Technology	3	0	0	3	CLL783	Advanced Process Control	3	0	0	3
CLL724	Environmental Engineering and Waste Management	3	0	0	3	CLL784	Process Modeling and Simulation	3	0	0	3
CLL725	Air Pollution Control Engineering	3	0	0	3	CLL785	Evolutionary Optimization	3	0	0	3
CLL726	Molecular Modeling of Catalytic Reactions	3	0	0	3	CLL786	Fine Chemicals Technology	3	0	0	3
CLL727	Heterogeneous Catalysis and Catalytic Reactors	3	0	0	3	CLL787	Statistical Methods for Chemical Engg.	3	0	0	3
CLL728	Biomass Conversion and Utilization	3	0	0	3	CLL788	Process Data Analytics	3	0	0	3
CLL730	Structure, Transport and Reactions in BioNano Systems	3	0	0	3	CLL791	Chemical Product and Process Integration	3	0	0	3
CLL732	Advanced Chemical Engineering Thermodynamics	3	0	0	3	CLL792	Chemical Product Development and Commercialization	3	0	0	3
CLL734	Process Intensification and Novel Reactors	3	0	0	3	CLL793	Membrane Science and Engineering	3	0	0	3
CLL735	Design of Multicomponent Separation Processes	3	0	0	3	CLL794	Petroleum Refinery Engineering	3	0	0	3
CLL736	Experimental Characterization of Multiphase Reactors	3	0	0	3	CLL798	Selected Topics in Chemical Engineering-I	3	0	0	3
CLL742	Experimental Characterization of BioMacromolecules	3	0	0	3	CLL799	Selected Topics in Chemical Engineering-II	3	0	0	3
						CLV796	Current Topics in Chemical Engineering	1	0	0	1
						CLV797	Recent Advances in Chemical Engineering	2	0	0	2

Sem.	Courses (Number, Abbreviated Title, L-T-P, credits)					Lecture courses	Contact h/week				Credits
							L	T	P	Total	
I	CLL701 Modelling of Transport Processes (2-0-0) 2	CLL702 Principles of Thermodynamics, Reaction Kinetics and Reactors (2-0-0) 2	CLL703 Process Engineering (3-0-0) 3	PE-1 (3-0-0) 3	PE-2 (3-0-0) 3	5	13	0	0	13	13
II	CLL731 Advanced Transport Phenomena (3-0-0) 3	CLL733 Industrial Multiphase Reactors (3-0-0) 3	CLD771 Minor Project (0-0-6) 3	PE-3 (3-0-0) 3	CLP704 Tech. Commu. Chem. Engineers (0-0-2) 1	3	9	0	8	17	13
Summer											
III			CLD781 Major Project Part-I (0-0-16) 8	PE-4 (3-0-0) 3	OE-1 (3-0-0) 3	2	6	0	16	22	14
IV			CLD782 Major Project Part-II (0-0-24) 12			0	0	0	24	24	12

Total = 52

Master of Technology in Molecular Engineering : Chemical Synthesis and Analysis

Department of Chemistry

The overall credits structure

Category	PC	PE/OE	Total
Credits	42	12	54

Program Core

CMD806	Major Project Part-I	0	0	18	9
CMD807	Major Project Part-II	0	0	18	9
CML721	Design and Synthesis of Organic Molecules	3	0	0	3
CML724	Synthesis of Industrially Important Inorganic Materials	3	0	0	3
CML726	Cheminformatics and Molecular Modelling	3	0	0	3
CMP728	Instrumentation Laboratory	0	0	6	3
CML729	Material Characterization	3	0	0	3
CML731	Chemical Separation and Electroanalytical Methods	3	0	0	3
CML737	Applied Spectroscopy	3	0	0	3
CMP722	Synthesis of Organic and Inorganic Compounds	0	0	6	3
Total Credits		42			

Program Electives

CMD799	Minor Project	0	0	6	3
CML723	Principles and Practice of NMR and Optical Spectroscopy	3	0	0	3
CML733	Chemistry of Industrial Catalysts	3	0	0	3
CML735	Biosynthetic Approach Towards Natural Products	3	0	0	3
CML734	Chemistry of Nanostructured Materials	3	0	0	3
CML738	Applications of P-block Elements and their Compounds	3	0	0	3
CML739	Applied Biocatalysis	3	0	0	3
CML740	Chemistry of Heterocyclic Compounds	3	0	0	3
CML741	Organo and Organometallic Catalysis	3	0	0	3
CML742	Reagents in Synthetic Transformations	3	0	0	3
CML743	Physical Organic Chemistry Advanced	3	0	0	3
CML801	Molecular Modelling and Simulations: Concepts and Techniques	3	0	0	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, credits)							Lecture courses	Contact h/week				Credits
									L	T	P	Total	
I	CML721 Design & Synthesis (3-0-0) 3	CML726 Cheminformatics (3-0-0) 3	CML731 Separation & Electroanalytical (3-0-0) 3	CMP722 Lab on Synthesis (0-0-6) 3	PE-1 (3-0-0) 3			4	12	0	6	18	15
II	CML724 Inorganic Materials (3-0-0) 3	CML729 Material Characterization (3-0-0) 3	CML737 Applied Spectroscopy (3-0-0) 3	CMP728 Instru. Lab. (0-0-6) 3	PE-2 (3-0-0) 3			4	12	0	6	18	15
III	CMD805 Major Project Part-I (0-0-12) 6	PE-3 (3-0-0) 3	PE-4 (3-0-0) 3					2	6	0	18	24	12
IV	CMD807 Major Project Part-II (0-0-24) 12							0	0	0	18	18	12

Total = 54

Master of Technology in Construction Technology and Management

Department of Civil Engineering

The overall credits structure

Category	PC	PE	OE	Total
Credits	37.5	15	0	52.5

Program Core

CVC771	Seminar In Construction Technology and Management-I	0	0	2	0
CVC772	Seminar In Construction Technology and Management-II	0	0	2	0
CVD772	Major Project Part-I (CEC)	0	0	18	9
CVD773	Major Project Part-II (CEC)	0	0	24	12
CVL772	Construction Project Management	3	0	0	3
CVL773	Quantitative Methods in Construction Management	3	0	0	3
CVL774	Construction Contract Management	3	0	0	3
CVL775	Construction Economics and Finance	3	0	0	3
CVL776	Construction Practices and Equipment	3	0	0	3
CVP772	Computational Laboratory for Construction Management	0	0	3	1.5
Total Credits		37.5			

Program Electives for All Background

CVD771	Minor Project (CEC)	0	0	6	3
CVS771	Independent Study (CEC)	0	3	0	3
MCL754	Operations Planning and Control	3	0	0	3
MCL756	Supply Chain Management	3	0	0	3
MCL757	Logistics	3	0	0	3
MCL771	Value Engineering and Life Cycle Costing	3	0	0	3
MSL705	HRM Systems	2	0	0	1.5
MSL804	Procurement Management	3	0	0	3
MSL822	International Business	3	0	0	3
MSL846	Total Productivity Management	3	0	0	3
MCL772	Reliability Engineering	3	0	0	3

Program Electives for Civil Engineering Background

EEL747	Electrical Systems for Construction Industries	3	0	2	4
CVL702	Ground Improvement and Geosynthetics	3	0	0	3
CVL714	Field Exploration and Geotechnical Processes	3	0	0	3
CVL715	Excavation Methods and Underground Space Technology	3	0	0	3
CVL727	Environmental risk assessment	3	0	0	3
CVL747	Transportation Safety and Environment	3	0	0	3
CVL750	Intelligent Transportation Systems	3	0	0	3
CVL765	Concrete Mechanics	3	0	0	3
CVL771	Advanced Concrete Technology	3	0	0	3
CVL777	Building Science	3	0	0	3
CVL778	Building Services and Maintenance Management	3	0	0	3
CVL779	Formwork for Concrete Structures	3	0	0	3
CVL820	Environmental Impact Assessment	3	0	0	3
CVL838	Geographic Information Systems	2	0	2	3
CVL840	Planning and Design of Sustainable Transport Systems	3	0	0	3
CVL871	Durability and Repair of Concrete Structures	3	0	0	3
CVL872	Infrastructure Development and Management	3	0	0	3
CVL873	Fire Engineering and Design	3	0	0	3
CVL874	Quality and Safety in Construction	3	0	0	3
CVL875	Sustainable Materials and Green Buildings	3	0	0	3

Program Electives for Electrical Engineering Background

ELL700	Linear Systems Theory	3	0	0	3
ELL712	Digital Communications	3	0	0	3
ELL750	Modelling of Electrical Machines	3	0	0	3
ELL751	Power Electronic Converters	3	0	0	3

ELL752	Electric Drive System	3	0	0	3
ELL753	Physical Phenomena in Electrical Machines	3	0	0	3
ELL754	Permanent Magnet Machines	3	0	0	3
ELL755	Variable Reluctance Machines	3	0	0	3
ELL756	Special Electrical Machines	3	0	0	3
ELL757	Energy Efficient Motors	3	0	0	3
ELL758	Power Quality	3	0	0	3
ELL759	Power Electronic Converters for Renewable Energy Systems	3	0	0	3
ELL760	Switched Mode Power Conversion	3	0	0	3
ELL761	Power Electronics for Utility Interface	3	0	0	3
ELL762	Intelligent Motor Controllers	3	0	0	3
ELL763	Advanced Electric Drives	3	0	0	3
ELL764	Electric Vehicles	3	0	0	3
ELL765	Smart Grid Technology	3	0	0	3
ELL766	Appliance Systems	3	0	0	3
ELL767	Mechatronics	3	0	0	3
ELL770	Power System Analysis	3	0	0	3
ELL771	Advanced Power System Protection	3	0	0	3
ELL772	Planning and Operation of a Smart Grid	3	0	0	3
ELL773	High Voltage DC Transmission	3	0	0	3
ELL774	Flexible AC Transmission system	3	0	0	3
ELL775	Power System Dynamics	3	0	0	3
ELL776	Advanced Power System Optimization	3	0	0	3
ELL777	Power System operation and control	3	0	0	3
ELL778	Dynamic Modelling And Control of Sustainable Energy Systems	3	0	0	3
ELL850	Digital Control of Power Electronics and Drive Systems	3	0	0	3
ELL851	Computer Aided Design of Electrical Machines	3	0	0	3
ELL852	Condition Monitoring of Electrical Machines	3	0	0	3
ELL853	Advanced Topics in Electrical Machines	3	0	0	3
ELL854	Selected Topics in Electrical Machines	3	0	0	3
ELL855	High Power Converters	3	0	0	3
ELL856	Advanced Topics in Power Electronics	3	0	0	3
ELL857	Selected Topics in Power Electronics	3	0	0	3
ELL858	Advanced Topics in Electric Drives	3	0	0	3
ELL859	Selected Topics in Electric Drives	3	0	0	3
ELL870	Restructured Power System	3	0	0	3
ELL871	Distribution System Operation and Planning	3	0	0	3
ELL872	Selected Topics in Power System	3	0	0	3
ELL873	Power System Transient	3	0	0	3
ELL874	Power System Reliability	3	0	0	3
ELP850	Electrical Machines Laboratory	0	0	3	1.5
ELP851	Power Electronics Laboratory	0	0	3	1.5
ELP852	Electrical Drives Laboratory	0	0	3	1.5
ELP853	DSP Based Control of Power Electronics and Drives Laboratory	0	0	3	1.5
ELP854	Electrical Machines CAD Laboratory	0	1	4	3
ELP855	Smart Grids Laboratory	0	1	4	3
ELP870	Power System Lab I	0	1	4	3
ELP871	Power System Lab II	0	1	4	3
ESL718	Power Generation, Transmission and Distribution	3	0	0	3
ESL732	Bioconversion and Processing of Waste	3	0	0	3
ESL734	Nuclear Energy	3	0	0	3
ESL740	Non-conventional Sources of Energy	3	0	0	3
ESL746	Hydrogen Energy	3	0	0	3
ESL768	Wind Energy and Hydro Power Systems	3	0	0	3
ESL770	Solar Energy Utilization	3	0	0	3
ESL870	Fusion Energy	3	0	0	3

Master of Technology in Geotechnical and Geoenvironmental Engineering

Department of Civil Engineering

The overall credits structure

Category	PC	PE	OE	Total
Credits	36	12	0	48

Program Core

CVD800	Major Project Part-I	0	0	12	6
CVD801	Major Project Part-II	0	0	24	12
CVL700	Engineering Behaviour of Soils	3	0	0	3
CVL701	Site Investigation and Foundation Design	3	0	0	3
CVL702	Ground Improvement and Geosynthetics	3	0	0	3
CVL703	Geoenvironmental Engineering	3	0	0	3
CVP700	Soil Engineering Lab	0	0	6	3
CVP800	Geoenvironmental and Geotechnical Engineering Lab	0	0	6	3
Total Credits		36			

Program Electives

CVD700*	Minor Project	0	0	6	3
CVL704	Finite Element Method in Geotechnical Engg.	3	0	0	3
CVL705	Slopes and Retaining Structures	3	0	0	3
CVL706	Soil Dynamics and Earthquake Geotechnical Engineering	3	0	0	3
CVL707	Soil-Structure Interaction Analysis	3	0	0	3
CVL708	Geotechnology of Waste Disposal Facilities	3	0	0	3
CVL709	Offshore Geotechnical Engineering	3	0	0	3
CVL800	Emerging Topics in Geotechnical Engineering	3	0	0	3
CVL801	Constitutive Modelling in Geotechnics	3	0	0	3
CVS800	Independent Study	0	3	0	3

* This course is only for Part-Time students in lieu of CVP800 and DAAD students.

Sem.	Courses (Number, Abbreviated Title, L-T-P, credits)					Lecture courses	Contact h/week				Credits
							L	T	P	Total	
I	CVL700 Engineering Behaviour of Soils (3-0-0) 3	CVL701 Site Investigation and Foundation Design (3-0-0) 3	CVP700 Soil Engineering Lab (0-0-6) 3	PE-1 (3-0-0) 3		3	9	0	6	15	12
II	CVL702 Ground Improvement and Geosynthetics (3-0-0) 3	CVL703 Geoenvironmental Engineering (3-0-0) 3	CVP800 Geoenvironmental and Geotechnical Engg. Lab/ CVD700 Minor Project (for Part Time Students) (0-0-6) 3	PE-2 (3-0-0) 3	PE-3 (3-0-0) 3	4	12	0	6	18	15
Summer											
III	CVD800 Major Project Part-I (0-0-12) 6	PE-4 (3-0-0) 3				1	3	0	12	15	9
IV	CVD801 Major Project Part-II (0-0-24) 12					0	0	0	24	24	12

Total = 48

Master of Technology in Transportation Engineering

Department of Civil Engineering

The overall credits structure

Category	PC	PE	OC	Total
Credits	36	18	0	54

Including 6 Credits of Restricted Electives

Program Core

CVD853	Major Project Part-I	0	0	18	9
CVD854	Major Project Part-II	0	0	24	12
CVL740	Pavement Materials and Design of Pavements	3	0	2	4
CVL741	Urban and Regional Transportation Planning	3	0	2	4
CVL742	Traffic Engineering	3	0	2	4
CVS852	Advanced Topics in Transportation Engineering	0	0	6	3
Total Credits					36

Restricted Electives (6 Credits)

CVL763	Analytical & Numerical Methods in Structural Engineering	3	0	0	3
CVL729	Environmental Statistics and Experimental Design	2	0	2	3
MCL761	Probability and Statistics	3	0	0	3
CVL731	Optimization Techniques in Water Resources	3	0	0	3
CVL748	Data Analysis for Transportation Engineering	3	0	0	3
CVS753	Minor Project in Transportation Engineering	0	0	6	3

Program Electives

CVL743	Airport Planning and Design	3	0	0	3
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CVL744	Transportation Infrastructure Design	2	0	2	3
CVL745	Modeling of Pavement Materials	2	0	2	3
CVL746	Public Transportation Systems	3	0	0	3
CVL747	Transportation Safety and Environment	3	0	0	3
CVL749	Planning and Design of Bus Transportation System	3	0	0	3
CVL750	Intelligent Transportation Systems	3	0	0	3
CVL840	Planning and Design of Sustainable Transport Systems	3	0	0	3
CVL841	Advanced Transportation Modelling	2	0	2	3
CVL842	Geometric Design of Roads	2	0	2	3
CVL844	Transportation Infrastructure Management	3	0	0	3
CVL845	Viscoelastic Behavior of Bituminous Materials	3	0	0	3
CVL846	Transportation System Management	3	0	0	3
CVL847	Transportation Economics	3	0	0	3
CVL848	Discrete Choice Methods for Travel Demand Analysis	2	0	2	3
CVL849	Traffic Flow Modelling	3	0	0	3
CVL850	Transportation Logistics	3	0	0	3
CVL851	Special Topics in Transportation Engineering	3	0	0	3
CVS754	Independent Study	0	3	0	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, credits)					Lecture courses	Contact h/week				Credits
							L	T	P	Total	
I	CVL741 Urban & Regional Transport Planning (3-0-2) 4	CVL740 Pavement Materials and Design of Pavements (3-0-2) 4	CVL742 Traffic Engineering (3-0-2) 4	RE-1* (3-0-0 or 2-0-2) 3			Min. 11 Max. 12	0	Min. 6 Max. 8	18	15
II	PE-1 (2-0-2) 3	PE-2 (2-0-2) 3	PE-3 (2-0-2) 3	RE-2# (3-0-0 or 2-0-2) 3			Min. 8 Max. 9		Min. 6 Max. 8	12	12
Summer	CVS852 Advanced Topics in Transportation Engineering (0-0-6) 3										3
III	PE-4 (2-0-2) 3	CVD853 Major Project Part-I (0-0-18) 9					0	0	6	6	12
IV	CVD854 Major Project Part-II (0-0-24) 12						3	0	18	24	12

* Should be listed in restricted elective course category.

Any course (relevant to research area) offered in that semester with consent of thesis supervisor. Alternatively minor project can be opted.

Total = 54

Master of Technology in Structural Engineering

Department of Civil Engineering

The overall credits structure

Category	PC	PE	OC	Total
Credits	42	12	0	54

Program Core

CVD757	Major Project Part-I (CES)	0	0	18	9	CVL769	Design of Tall Buildings	3	0	0	3
CVD758	Major Project Part-II (CES)	0	0	18	9	CVL770	Prestressed and Composite Structures	2	0	2	3
CVL756	Advanced Structural Analysis	3	0	0	3	CVL856	Strengthening and Retrofitting of Structures	3	0	0	3
CVL757	Finite Element Methods in Structural Engineering	2	0	2	3	CVL857	Structural Safety and Reliability	3	0	0	3
CVL758	Solid Mechanics in Structural Engineering	3	0	0	3	CVL858	Theory of Plates and Shells	3	0	0	3
CVL759	Structural Dynamics	3	0	0	3	CVL859	Theory of Structural Stability	3	0	0	3
CVL760	Theory of Concrete Structures	3	0	0	3	CVL860	Advanced Finite Element Method and Programming	2	0	2	3
CVL761	Theory of Steel Structures	3	0	0	3	CVL861	Analysis and Design of Machine Foundations	2	0	2	3
CVL762	Earthquake Analysis and Design	3	0	0	3	CVL862	Design of Offshore Structures	3	0	0	3
CVP756	Structural Engineering Laboratory	0	0	6	3	CVL863	General Continuum Mechanics	3	0	0	3
Total Credits					42	CVL864	Structural Health Monitoring	2	0	2	3
						CVL865	Structural Vibration Control	3	0	0	3

Program Electives

CVD756	Minor Project in Structural Engineering	0	0	6	3	CVL866	Wind Resistant Design of Structures	3	0	0	3
CVL763	Analytical and Numerical Methods for Structural Engineering	3	0	0	3	CVS756	Independent Study (CES)	0	3	0	3
CVL764	Blast Resistant Design of Structures	2	0	2	3	CVL771	Advanced Concrete Technology	3	0	0	3
CVL765	Concrete Mechanics	3	0	0	3	CVL873	Fire Engineering and Design	3	0	0	3
CVL766	Design of Bridge Structures	3	0	0	3	CVL779	Formwork for Concrete Structures	3	0	0	3
CVL767	Design of Fiber Reinforced Composite Structures	3	0	0	3	CVL871	Durability and Repair of Concrete Structures	3	0	0	3
CVL768	Design of Masonry Structures	3	0	0	3	CVL868	Waves through Periodic Structures and metamaterials	3	0	0	3
						CVL867	Atomistic & Multiscale Modelling of Materials	3	0	0	3
						CVL869	Probabilistic Structural Dynamics	3	0	0	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, credits)						Lecture courses	Contact h/week				Credits
								L	T	P	Total	
I	CVL756 Advanced Structural Analysis (3-0-0) 3	CVL759 Structural Dynamics (3-0-0) 3	CVL757 Finite Element Methods in Structural Engineering (2-0-2) 3	CVL758 Solid Mechanics in Structural Engineering (3-0-0) 3	PE-1 (3-0-0) 3 or (2-0-2) 3		5	(13, 14)	0	(2,4)	(16, 17)	15
II	CVP756 Structural Engineering Laboratory (0-0-6) 3	CVL762 Earthquake Analysis and Design (3-0-0) 3	CVL760 Theory of Concrete Structures (3-0-0) 3	CVL761 Theory of Steel Structures (3-0-0) 3	PE-2 (3-0-0) 3 or (2-0-2) 3		4	(11, 12)	0	(6,8)	(18, 19)	15
Summer												
III	CVD757 Major Project Part-I (CES) (0-0-18) 9	PE-3 (3-0-0) 3 or (2-0-2) 3	PE-4 (3-0-0) 3 or (2-0-2) 3				2	(4,6)	0	(18, 22)	(24, 26)	15
IV	CVD758 Major Project Part-I (CES) (0-0-18) 9						0	0	0	18	18	9

Total = 54

Master of Technology in Construction Engineering and Management

Department of Civil Engineering

The overall credits structure

Category	PC	PE	OC	Total
Credits	42	12	0	54

Program Core

CVD777	Major Project Part-I (CET)	0	0	18	9
CVD778	Major Project Part-II (CET)	0	0	24	12
CVL771	Advanced Concrete Technology	3	0	0	3
CVL772	Construction Project Management	3	0	0	3
CVL773	Quantitative Methods in Construction Management	3	0	0	3
CVL774	Construction Contract Management	3	0	0	3
CVL775	Construction Economics and Finance	3	0	0	3
CVL776	Construction Practices and Equipment	3	0	0	3
CVP771	Construction Technology Laboratory	0	0	3	1.5
CVP772	Computational Laboratory for Construction Management	0	0	3	1.5
Total Credits		42			

Program Electives

CVD776	Minor Project (CET)	0	0	6	3
CVL765	Concrete Mechanics	3	0	0	3
CVL777	Building Science	3	0	0	3
CVL778	Building Services and Maintenance Management	3	0	0	3
CVL779	Formwork for Concrete Structures	3	0	0	3
CVL871	Durability and Repair of Concrete Structures	3	0	0	3
CVL872	Infrastructure Development and Management	3	0	0	3
CVL873	Fire Engineering and Design	3	0	0	3
CVL874	Quality and Safety in Construction	3	0	0	3
CVL875	Sustainable Materials and Green Buildings	3	0	0	3
CVL876	Digital Design and Construction	2	0	2	3
CVS776	Independent Study (CET)	0	3	0	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)						Lecture courses	Contact h/week				Credits
								L	T	P	Total	
I	CVL772 Construction Project Management (3-0-0) 3	CVL773 Quantitative Methods in Construction Management (3-0-0) 3	CVL771 Advanced Concrete Technology (3-0-0) 3	CVP772 Computational Laboratory for Construction Management (0-0-3) 1.5	PE-1 (3-0-0) 3		4	12	0	3	15	13.5
II	CVL775 Construction Economics and Finance (3-0-0) 3	CVL776 Construction Practices and Equipment (3-0-0) 3	CVL774 Construction Contract Management (3-0-0) 3	CVP771 Construction Technology Laboratory (0-0-3) 1.5	PE-2 (3-0-0) 3		4	12	0	3	15	13.5
Summer												
III	CVD777 Major Project Part-I (CET) (0-0-18) 9	PE-3 (3-0-0) 3	PE-4 (3-0-0) 3				2	6	0	18	24	15
IV	CVD778 Major Project Part-II (CET) (0-0-24) 12						0	0	0	24	24	12

Total = 54

Master of Technology in Rock Engineering and Underground Structures

Department of Civil Engineering

The overall credits structure

Category	PC	PE	OC	Total
Credits	36	12	0	48

Program Core

CVD810	Major Project Part-I	0	0	12	6
CVD811	Major Project Part-II	0	0	24	12
CVL710	Engineering Properties of Rocks and Rock Masses	3	0	0	3
CVL711	Structural Geology	3	0	0	3
CVL712	Slopes and Foundations	3	0	0	3
CVL713	Analysis and Design of Underground Structures	3	0	0	3
CVP710	Rock Mechanics Laboratory-I	0	0	6	3
CVP810	Rock Mechanics Laboratory-II	0	0	6	3
Total Credits					36

Program Electives

CVD710*	Minor Project	0	0	6	3
CVL704	Finite Element Method in Geotechnical Engg.	3	0	0	3
CVL714	Field Exploration and Geotechnical Processes	3	0	0	3
CVL715	Excavation Methods and Underground Space Technology	3	0	0	3
CVL716	Environmental Rock Engineering	3	0	0	3
CVL810	Emerging Topics in Rock Engineering and Underground Structures	3	0	0	3
CVL811	Numerical and Computer Methods in Geomechanics	3	0	0	3
CVS810	Independent Study	0	0	6	3

* This course is only for Part-Time students in lieu of CVP800 and DAAD students.

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)						Lecture courses	Contact h/week				Credits
								L	T	P	Total	
I	CVL710 Engineering Properties of Rocks and Rock Masses (3-0-0) 3	CVL711 Structural Geology (3-0-0) 3	CVP710 Rock Mechanics Laboratory-I (0-0-6) 3	PE-1 (3-0-0) 3			3	9	0	6	15	12
II	CVL712 Slopes and Foundations (3-0-0) 3	CVL713 Analysis and Design of Underground Structures (3-0-0) 3	CVP810 Rock Mechanics Laboratory-II / CVD710 Minor Project (0-0-6) 3	PE-2 (3-0-0) 3	PE-3 (3-0-0) 3		4	12	0	6	18	15
Summer												
III	CVD810 Major Project Part-I (0-0-12) 6	PE-4 (3-0-0) 3					1	3	0	12	15	9
IV	CVD811 Major Project Part-II (0-0-24) 12						0	0	0	24	24	12

Total = 48

Master of Technology in Environmental Engineering and Management

Department of Civil Engineering

The overall credits structure

Category	PC	PE	OC	Total
Credits	39	9	6	54

Program Core

CVD720	Major Thesis Part-I	0	0	12	6
CVD721	Major Thesis Part-II	0	0	24	12
CVD726	Minor Project	0	0	6	3
CVL720	Air Pollution and Control	3	0	0	3
CVL721	Solid Waste Engineering	3	0	0	3
CVL722	Water Engineering	3	0	0	3
CVL723	Wastewater Engineering	3	0	0	3
CVL724	Environmental Systems Analysis	3	0	0	3
CVL725	Environmental Chemistry and Microbiology	1	0	4	3
Total Credits		39			

Program Electives

CVL727	Environmental risk assessment	3	0	0	3
CVL728	Environmental Quality Modeling	3	0	0	3
CVL729	Environmental Statistics and Experimental Design	2	0	2	3
CVL820	Environmental Impact Assessment	3	0	0	3

CVL821	Industrial Waste Management and Audit	3	0	0	3
CVL822	Emerging Technologies for Environmental Management	3	0	0	3
CVL823	Thermal Techniques for Waste Management	3	0	0	3
CVL824	Life Cycle Analysis and Design for Environment	3	0	0	3
CVL825	Fundamental of Aerosol: Health and Climate Change	3	0	0	3
CVL826	Quantitative Microbial Risk Assessment	1	0	0	1
CVL827	Environmental Implications of Engineered Nanomaterials	2	0	0	2
CVL828	Water Distribution and Sewerage Network Design	3	0	0	3
CVL829	Indoor Environmental Quality: Theory and Practice	2	0	2	3
CVP820	Advanced Air Pollution Laboratory	1	0	4	3
CVP821	Advanced Water and Wastewater Laboratory	1	0	4	3
CVS720	Independent Study	0	3	0	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)					Lecture courses	Contact h/week				Credits
							L	T	P	Total	
I	CVL725 Environmental Chemistry and Microbiology (1-0-4) 3	CVL722 Water Engineering (3-0-0) 3	CVL720 Air pollution and control (3-0-0) 3	PE-1 (3-0-0) 3 or (2-0-2) 3 or (1-0-4)		4	(8, 10)	0	(4,8)	(14, 16)	12
II	CVL721 Solid Waste Engineering (3-0-0) 3	CVL724 Environmental Systems Analysis (3-0-0) 3	CVL723 Wastewater Engineering (3-0-0) 3	PE-2 (3-0-0) 3 or (2-0-2) 3 or (1-0-4)	OE-1 (3-0-0) 3	5	(13, 15)	0	(0,4)	(15, 17)	15
Summer	CVD726 Minor project (0-0-6) 3										3
III	CVD720 Major Thesis Part-I (0-0-12) 6			PE-3 (3-0-0) 3 or (2-0-2) 3 or (1-0-4)	OE-2 (3-0-0) 3	2	(4,6)	0	(12, 16)	(18, 20)	12
IV	CVD721 Major Thesis Part-II (0-0-24) 12					0	0	0	24	24	12

Total = 54

Master of Technology in Water Resources Engineering

Department of Civil Engineering

The overall credits structure

Category	PC	PE	OC	Total
Credits	39	15	0	54

Program Core

CVD831	Major Project Part-I	0	0	12	6
CVD832	Major Project Part-II	0	0	24	12
CVL730	Hydrologic Processes and Modeling	3	0	0	3
CVL731	Optimization Techniques in Water Resources	3	0	0	3
CVL732	Groundwater Hydrology	3	0	0	3
CVL733	Stochastic Hydrology	2	0	2	3
CVL734	Advanced Hydraulics	3	0	0	3
CVL735	Finite Element in Water Resources	3	0	0	3
CVP730	Simulation Laboratory-I	0	0	3	1.5
CVP731	Simulation Laboratory-II	0	0	3	1.5
Total Credits		39			

Program Electives

CVL736	Soft Computing Techniques in Water Resources	2	0	2	3
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CVL737	Environmental Dynamics and Management	3	0	0	3
CVL738	Economic Aspects of Water Resources Development	3	0	0	3
CVL830	Groundwater Flow and Pollution Modeling	3	0	0	3
CVL831	Surface Water Quality Modeling and Control	3	0	0	3
CVL832	Hydroelectric Engineering	3	0	0	3
CVL833	Water Resources Systems	3	0	0	3
CVL834	Urban Water Infrastructure	3	0	0	3
CVL835	Eco-hydraulics and Hydrology	3	0	0	3
CVL836	Advanced Hydrologic Land Surface Processes	3	0	0	3
CVL837	Mechanics of Sediment Transport	2	0	2	3
CVL838	Geographic Information Systems	2	0	2	3
CVL839	Hydrologic Applications of Remote Sensing	2	0	2	3
CVS730	Minor Project	0	0	6	3
CVS830	Independent Study	0	3	0	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)						Lecture courses	Contact h/week				Credits
								L	T	P	Total	
I	CVL730 Hyd. Process (3-0-0) 3	CVL731 Opt. Tech. (3-0-0) 3	CVL732 GW Hyd. (3-0-0) 3	CVL735 Stochastic Hyd. (2-0-2) 3	PE-1 (3-0-0) 3 or (2-0-2) 3		5	14/13	0	2/4	16/17	15
II	CVL733 Adv. Hydraulics (3-0-0) 3	CVL734 Finite Element (3-0-0) 3	CVP730 Sim. Lab-I (0-0-3) 1.5	CVP731 Sim. Lab-II (0-0-3) 1.5	PE-2 (3-0-0) 3 or (2-0-2) 3	PE-3 (3-0-0) 3 or (2-0-2) 3	4	12-10	0	6-10	18-20	15
Summer	Major Project Part I (CEW)											0
III	CVD831 Major Project Part-I (0-0-12) 6			PE-4 (3-0-0) 3 or (2-0-2) 3	PE-5 (3-0-0) 3 or (2-0-2) 3		2	6-4	0	12-16	18-20	12
IV	CVD832 Major Project Part-II (0-0-24) 12						0	0	0	24	24	12

Total = 54

Master of Technology in Computer Science and Engineering

Department of Computer Science and Engineering

The overall credits structure

Category	PC	PE	OC	Total
Credits	21	27-33	Maximum of 4 Cr. in lieu of a PE (track B)	48-54

Program Core

COD891	Minor Project	0	0	6	3
COD892	M.Tech. Project Part-I	0	0	14	7
COL702	Advanced Data Structures	3	0	2	4
COL765	Logic & Functional Programming	3	0	2	4
COP701	Software Systems Laboratory	0	0	6	3

Total Credits **21**

Bridge Courses - Min. 6 credits, this may be waived in exceptional cases on the recommendation by DRC. Please note that bridge course credits do not count towards the 48-54 credit requirement at in the credit structure table above Bridge Course credits should be considered an additional requirement over and above the 48-54 credit requirement.

COL632	Introduction to Data Base Systems	3	0	2	4
COL633	Resources Management in Computer Systems	3	0	2	4
COL671	Artificial Intelligence	3	0	2	4
COL672	Computer Networks	3	0	2	4

Program Electives (PE)

COD893	Major Project Part-II	0	0	28	14
COP820	Processor Design Lab	0	0	8	4
COS799	Independent Study	0	3	0	3

Specialization Streams - At least 6 credits from PE; Project work relevant to specialization

1. Architecture & Embedded Systems (AES)

COL718	Architecture of High Performance Computers	3	0	2	4
COL719	Synthesis of Digital Systems	3	0	2	4
COL720	Real Time Systems	3	0	2	4
COL788	Embedded Computing	3	0	0	3
COL812	System Level Design and Modelling	3	0	0	3
COL818	Principles of Multiprocessor Systems	3	0	2	4
COL821	Reconfigurable Computing	3	0	0	3
COL861	Special Topics in Hardware Systems	3	0	0	3
COP745	Digital System Design Laboratory	0	0	6	3
COP820	Processor Design Lab	0	0	8	4
COV881	Special Module in Hardware Systems	1	0	0	1

2. Graphics & Vision (GV)

COL726	Numerical Algorithms	3	0	2	4
COL780	Computer Vision	3	0	2	4
COL781	Computer Graphics	3	0	3	4.5
COL783	Digital Image Analysis	3	0	3	4.5
COL785	Virtual and Augmented Reality	3	0	2	4
COL828	Advanced Computer Vision	3	0	2	4
COL829	Advanced Computer Graphics	3	0	2	4
COV877	Special Module on Visual Computing	1	0	0	1
SIL801	Special Topics in Multimedia Systems	3	0	0	3

3. Software Systems (SS)

COL720	Real Time Systems	3	0	2	4
COL724	Advanced Computer Networks	3	0	2	4
COL728	Compiler Design	3	0	3	4.5
COL729	Compiler Optimization	3	0	3	4.5
COL730	Parallel Programming	3	0	2	4
COL731	Advanced Compiler Techniques for Optimization, Safety and Security	3	0	2	4
COL732	Virtualization and Cloud Computing	3	0	2	4
COL733	Cloud Computing Technology Fundamentals	3	0	2	4
COL740	Software Engineering	3	0	2	4
COL750	Foundations of Automatic Verification	3	0	2	4
COL768	Wireless Networks	3	0	2	4
COL818	Principles of Multiprocessor Systems	3	0	2	4
COL819	Advanced Distributed Systems	3	0	2	4
COL851	Special Topics in Operating Systems	3	0	0	3
COL852	Special Topics in Compilers	3	0	0	3

COL860	Special Topics in Parallel Computation	3	0	0	3
COL862	Special Topics in Software Systems	3	0	0	3
COL867	Special Topics in High Speed Networks	3	0	0	3
COL869	Special Topics in Concurrency	3	0	0	3
COL871	Special Topics in Programming Languages	3	0	0	3
COL874	Special Topics in Compilers and Language Implementation	3	0	0	3
COL876	Special Topics in Formal Methods	3	0	0	3
COL886	Special Topics in Operating Systems	3	0	0	3
COV880	Special Module in Parallel Computation	1	0	0	1
COV882	Special Module in Software Systems	1	0	0	1
COV887	Special Module in High Speed Networks	1	0	0	1
COV889	Special Module in Concurrency	1	0	0	1
SIL765	Networks & System Security	3	0	2	4
SIL769	Internet Traffic - Measurement, Modeling & Analysis	3	0	2	4

4. Theoretical Computer Science (TH)

COL703	Logic for CS (LCS)	3	0	2	4
COL726	Numerical Algorithms	3	0	2	4
COL727	Rapid Mixing in Markov Chains	3	0	0	3
COL749	Computational Social Choice	3	0	0	3
COL750	Foundations of Automatic Verification	3	0	2	4
COL751	Algorithmic Graph Theory	3	0	0	3
COL752	Geometric Algorithms	3	0	2	4
COL753	Complexity Theory	3	0	0	3
COL754	Approximation Algorithms	3	0	0	3
COL755	Algorithmic Game Theory	3	0	0	3
COL756	Mathematical Programming	3	0	0	3
COL757	Model Centric Algorithm Design	3	0	2	4
COL758	Advanced Algorithms	3	0	2	4
COL759	Cryptography & Computer Security	3	0	0	3
COL787	Online Algorithms and Competitive Analysis	3	0	0	3
COL830	Distributed Computing	3	0	0	3
COL831	Semantics of Programming Languages	3	0	0	3
COL832	Proofs and Types	3	0	0	3
COL860	Special Topics in Parallel Computation	3	0	0	3
COL863	Special Topics in Theoretical Computer Science	3	0	0	3
COL866	Special Topics in Algorithms	3	0	0	3
COL869	Special Topics in Concurrency	3	0	0	3
COL871	Special Topics in Programming Languages	3	0	0	3
COL872	Special Topics in Cryptography	3	0	0	3
COL874	Special Topics in Compilers and Language Implementation	3	0	0	3
COL876	Special Topics in Formal Methods	3	0	0	3
COV879	Special Module in Financial Algorithms	2	0	0	2
COV883	Special Module in Theoretical Computer Science	1	0	0	1
COV886	Special Module in Algorithms	1	0	0	1
COV889	Special Module in Concurrency	1	0	0	1

5. Data Analytics & AI (DAAI)

COL726	Numerical Algorithms	3	0	2	4
COL760	Advanced Data Management	3	0	2	4
COL761	Data Mining	3	0	2	4
COL762	Database Implementation	3	0	2	4
COL764	Information Retrieval and Web Search	3	0	2	4
COL770	Advanced Artificial Intelligence	3	0	2	4
COL772	Natural Language Processing	3	0	2	4
COL774	Machine Learning	3	0	2	4
COL775	Deep Learning	3	0	2	4
COL776	Learning Probabilistic Graphical Models	3	0	2	4
COL777	Deep Reinforcement Learning	3	0	2	4
COL778	Principles of Autonomous Systems	3	0	2	4
COL786	Advanced Functional Brain Imaging	3	0	2	4
COL864	Special Topics in Artificial Intelligence	3	0	0	3
COL868	Special Topics in Database Systems	3	0	0	3

Master of Technology in Control and Automation

Department of Electrical Engineering

The overall credits structure

Category	PC	PE	OE	Total
Credits	24	18	6	48

Program Core

ELD801	Major Project Part-I	0	0	12	6
ELL700	Linear Systems Theory	3	0	0	3
ELL701	Mathematical Methods in Control	3	0	0	3
ELL702	Nonlinear Systems	3	0	0	3
ELL703	Optimal Control Theory	3	0	0	3
ELL705	Stochastic Filtering and Identification	3	0	0	3
ELP800	Control Systems Laboratory	0	0	2	1
ELP801	Advanced Control Laboratory	0	0	4	2
Total Credits		24			

Program Electives

ELD800	Minor Project (EEA)	0	0	6	3
ELD802	Major Project Part-II	0	0	24	12
ELL704	Advanced Robotics	3	0	0	3
MTL704	Numerical Optimization	3	0	0	3
ELL707	Systems Biology	3	0	0	3
ELL708	Selected Topics in Systems and Control	3	0	0	3
ELL709	Design Aspects in Control	3	0	0	3
DSL711	Sensors & Transducers	3	0	0	3
ELL714	Basic Information Theory	3	0	0	3
ELL720	Advanced Digital Signal Processing	3	0	0	3
MTL731	Introduction to Chaotic Dynamical System	3	0	0	3
ELL762	Intelligent Motor Controllers	3	0	0	3
ELL765	Smart Grid Technology	3	0	0	3
ELL767	Mechatronics	3	0	0	3
ELL775	Power System Dynamics	3	0	0	3
ELL778	Dynamic Modelling And Control of Sustainable Energy Systems	3	0	0	3

MCL783	Automation Manufacturing	3	0	2	4
ELL784	Introduction to Machine Learning	3	0	0	3
ELL787	Embedded Systems and Applications	3	0	0	3
ELL789	Intelligent Systems	3	0	0	3
ELL791	Neural Systems and Learning Machines	3	0	2	4
ELL793	Computer Vision	3	0	0	3
ELL795	Swarm Intelligence	3	0	0	3
ELL796	Signals and Systems in Biology	3	0	0	3
ELL800	Numerical Linear Algebra and Optimization in Engineering	3	0	0	3
ELL801	Nonlinear Control	3	0	0	3
ELL802	Adaptive and Learning Control	3	0	0	3
ELL803	Model Reduction in Control	3	0	0	3
ELL804	Robust Control	3	0	0	3
ELL805	Networked and Multi-Agent Control Systems	3	0	0	3
ELL806	Modeling and Control of Distributed Parameter Systems	3	0	0	3
ELL807	Stochastic Control	3	0	0	3
ELL808	Advanced Topics in Systems and Control	3	0	0	3
MCL845	Advanced Robotics	2	0	2	3
ELL850	Digital Control of Power Electronics and Drive Systems	3	0	0	3
ELL883	Embedded Intelligence	3	0	0	3
ELL888	Advanced Machine Learning	3	0	0	3
ELL890	Computational Neuroscience	3	0	0	3
ELL891	Advances in Deep Learning	3	0	0	3
ELL893	Cyber-Physical Systems	3	0	0	3
ELV700	Special Module in Systems and Control	1	0	0	1

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)					Lecture courses	Contact h/week				Credits
							L	T	P	Total	
I	ELL700 Linear Systems Theory (3-0-0)	ELL701 Mathematical Methods in Control (3-0-0)	ELL702 Nonlinear Systems (3-0-0)	ELP800 Control Systems Lab (0-0-2)	OE (3-0-0)	4	12	0	2	14	13
II	ELL703 Optimal Control Theory (3-0-0)	ELL705 Stochastic Filtering and Identification (3-0-0)	ELP801 Advanced Control Lab (0-0-4)	PE (3-0-0)		3	9	0	4	13	11
Summer											
III (Project based) OR	ELD801 Major Project Part-I (0-0-12)		PE (3-0-0)	OE (3-0-0)		2	6	0	12	18	12
III (Course based)	PE (3-0-0)	PE (3-0-0)	PE (3-0-0)	OE (3-0-0)		4	12	0	0	12	12
IV (Project based) OR	ELD802 Major Project Part-II (0-0-24)					0	0	0	24	24	12
IV (Course based)	ELD801 Major Project Part-I (0-0-12)		PE (3-0-0)	PE (3-0-0)		2	6	0	12	18	12

Total = 48

Master of Technology in Communication Engineering

Department of Electrical Engineering

The overall credits structure

Category	PC	PE	OE	Total
Credits	24	18	6	48

Program Core

ELD811	Major Project Part-I (Communication Engineering)	0	0	12	6
ELL711	Signal Theory	3	0	0	3
ELL712	Digital Communications	3	0	0	3
ELL713	Microwave Theory and Techniques	3	0	0	3
ELL719	Detection and Estimation Theory	3	0	0	3
ELP719	Microwave Laboratory	0	1	4	3
ELP725	Wireless Communication Laboratory	0	1	4	3
Total Credits					24

Program Electives

BSP710	Communication & Signal Processing Tech. Lab.	0	1	4	3
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Streamed Electives (EEE) in (Communication Systems)

BSP710	Communication & Signal Processing Tech. Lab.	0	1	4	3
ELD810	Minor Project (Communication Engineering)	0	0	6	3
ELD812	Major Project Part-II	0	0	24	12
ELL701	Mathematical Methods in Control	3	0	0	3
ELL710	Coding Theory	3	0	0	3
ELL714	Basic Information Theory	3	0	0	3
ELL716	Telecommunication Switching and Transmission	3	0	0	3
ELL717	Optical Communication Systems	3	0	0	3
ELL720	Advanced Digital Signal Processing	3	0	0	3
ELL722	Antenna Theory and Techniques	3	0	0	3
ELL723	Broadband Communication Systems	3	0	0	3
ELL724	Multichannel Signal Processing	3	0	0	3
ELL725	Wireless Communications	3	0	0	3
ELL730	I.C. Technology	3	0	0	3
ELL732	Micro and Nanoelectronics	3	0	0	3
ELL734	MOS VLSI design	3	0	0	3
ELL735	Analog Integrated Circuits	3	0	0	3
ELL785	Computer Communication Networks	3	0	0	3
ELL810	Cyber Security and Information Assurance	3	0	0	3
ELL812	Microwave Propagation and Systems	3	0	0	3
ELL813	Advanced Information Theory	3	0	0	3
ELL814	Wireless Optical Communications	3	0	0	3
ELL815	MIMO Wireless Communications	3	0	0	3
ELL816	Satellite Communication	3	0	0	3
ELL818	Telecommunication Technologies	3	0	0	3

ELL821	Selected Topics in Communication Systems and Networking-I	3	0	0	3
ELL822	Selected Topics in Communication Systems and Networking-II	3	0	0	3
ELL833	CMOS RF IC Design	3	0	0	3
ELL894	Network Performance Modeling and Analysis	3	0	0	3
ELP718	Telecommunication Software Laboratory	0	1	4	3
ELP721	Embedded Telecommunication Systems Laboratory	0	1	4	3
ELV710	Special Module in Cyber Security	1	0	0	1
ELV720	Special Module in Communication Systems and Networking-I	1	0	0	1
ELV821	Special Module in Communication Systems and Networking-II	1	0	0	1
CRL708	Sonar Systems Engineering	3	0	0	3
CRL709	Underwater Electronic Systems	3	0	0	3
CRL712	RF and Microwave Active Circuits	3	0	0	3
CRL715	Radiating Systems for RF Communication	3	0	0	3

Streamed Electives (EEE) in (Information Processing)

ELD810	Minor Project (Communication Engineering)	0	0	6	3
ELD812	Major Project Part-II	0	0	24	12
ELL701	Mathematical Methods in Control	3	0	0	3
ELL714	Basic Information Theory	3	0	0	3
ELL715	Digital Image Processing	3	0	2	4
ELL718	Statistical Signal Processing	3	0	0	3
ELL720	Advanced Digital Signal Processing	3	0	0	3
ELL724	Multichannel Signal Processing	3	0	0	3
ELL784	Introduction to Machine Learning	3	0	0	3
ELL786	Multimedia Systems	3	0	0	3
ELL792	Computer Graphics	3	0	0	3
ELL793	Computer Vision	3	0	0	3
ELL794	Human-Computer Interface	3	0	0	3
ELL823	Selected Topics in Information Processing-I	3	0	0	3
ELL824	Selected Topics in Information Processing-II	3	0	0	3
ELV781	Special Modules in Information Processing-I	1	0	0	1
ELV823	Special Modules in Information Processing-II	1	0	0	1
CRL704	Sensor Array Signal Processing	3	0	0	3
CRL707	Human & Machine Speech Communication	3	0	0	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, credits)					Lecture courses	Contact h/week				Credits
							L	T	P	Total	
I	ELL 711 Signal theory (3-0-0)	ELL712 Digital Comm. (3-0-0)	ELL 713 Microwave Theory and Techniques (3-0-0)	ELP 719 Microwave Lab. (0-1-4)		3	9	1	4	14	12
II	ELL719 Detection and Estimation Theory (3-0-0)	ELP725 Wireless Comm. Lab. (0-1-4)	PE-1 (3-0-0)	PE-2 (3-0-0)		3	9	1	4	14	12
Summer											
III	ELD811 Major Project Part-I (0-0-12) 6		OE-1 (3-0-0)	OE-2 (3-0-0)		2	6	0	12	18	12
IV (Project based) OR	ELD812 Major Project Part-II (0-0-24) 12					0	0	0	24	24	12
IV (Course based)	PE-3 (3-0-0)	PE-4 (3-0-0)	PE-5 (3-0-0)	PE-6 (3-0-0)		4	12	0	0	12	12

Total = 48

Master of Technology in Integrated Electronics and Circuits

Department of Electrical Engineering

The overall credits structure

Category	PC	PE	OC	Total
Credits	24	18	6	48

Program Core

ELD831	Major Project Part-I (Integrated Electronic Circuits)	0	0	12	6
ELL730	I.C. Technology	3	0	0	3
ELL732	Micro and Nanoelectronics	3	0	0	3
ELL734	MOS VLSI design	3	0	0	3
ELL735	Analog Integrated Circuits	3	0	0	3
ELP831	IEC Laboratory-I	0	0	6	3
ELP832	IEC Laboratory-II	0	0	6	3
Total Credits		24			

Program Electives

BSP710	Communication & Signal Processing Tech. Lab.	0	1	4	3
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Streamed Electives (EEN) in (VLSI Design)

COL719	Synthesis of Digital Systems	3	0	2	4
ELD830	Minor Project	0	0	6	3
ELD832	Major Project Part-II	0	0	24	12
ELL720	Advanced Digital Signal Processing	3	0	0	3
ELL731	Mixed Signal Circuit Design	3	0	0	3
ELL733	Digital ASIC Design	3	0	2	4
ELL736	Solid State Imaging Sensors	3	0	0	3
ELL737	Flexible Electronics	3	0	0	3
ELL740	Compact Modeling of Semiconductor Devices	3	0	0	3
ELL741	Neuromorphic Engineering	3	0	0	3
ELL747	Active and Passive Filter Design	3	0	0	3
ELL748	System-on-Chip Design and Test	3	0	0	3
ELL749	Semiconductor Memory Design	3	0	0	3
ELL782	Computer Architecture	3	0	0	3
ELL791	Neural Systems and Learning Machines	3	0	2	4
ELL830	Issues in Deep Submicron VLSI Design	3	0	0	3
ELL831	CAD for VLSI, MEMS, and Nanoassembly	3	0	0	3
ELL832	Selected Topics in IEC-I	3	0	0	3
ELL833	CMOS RF IC Design	3	0	0	3
ELL834	Selected Topics in IEC-II	3	0	0	3
ELP830	Semiconductor Processing Laboratory	0	0	6	3
ELV734	Special Module in Scientific Writing for Research	1	0	0	1
ELV830	Special Module in Low Power IC Design	1	0	0	1
ELV831	Special Module in VLSI Testing	1	0	0	1
ELV832	Special Module in Machine Learning	1	0	0	1

Streamed Electives (EEN) in (Nanoelectronics and Photonics)

ELD830	Minor Project	0	0	6	3
ELD832	Major Project Part-II	0	0	24	12
ELL737	Flexible Electronics	3	0	0	3

ELL738	Micro and Nano Photonics	3	0	0	3
ELL739	Advanced Semiconductor Devices	3	0	0	3
ELL740	Compact Modeling of Semiconductor Devices	3	0	0	3
ELL741	Neuromorphic Engineering	3	0	0	3
ELL742	Introduction to MEMS Design	3	0	0	3
ELL743	Photovoltaics	3	0	0	3
ELL744	Electronic and Photonic Nanomaterials	3	0	0	3
ELL745	Quantum Electronics	3	0	0	3
ELL746	Biomedical Electronics	3	0	0	3
ELL749	Semiconductor Memory Design	3	0	0	3
ELL791	Neural Systems and Learning Machines	3	0	2	4
ELL830	Issues in Deep Submicron VLSI Design	3	0	0	3
ELL832	Selected Topics in IEC-I	3	0	0	3
ELL834	Selected Topics in IEC-II	3	0	0	3
ELP830	Semiconductor Processing Laboratory	0	0	6	3
ELP833	Device and Materials Characterization Lab.	0	0	6	3
ELV734	Special Module in Scientific Writing for Research	1	0	0	1
ELV833	Special Module in Semiconductor Business Management	1	0	0	1
ELV834	Special Module in Nanoelectronics	1	0	0	1

Streamed Electives (EEN) in (Embedded Intelligent Systems)

COL719	Synthesis of Digital Systems	3	0	2	4
COL788	Advanced Topics in Embedded Computing	3	0	0	3
ELD830	Minor Project	0	0	6	3
ELD832	Major Project Part-II	0	0	24	12
ELL720	Advanced Digital Signal Processing	3	0	0	3
ELL731	Mixed Signal Circuit Design	3	0	0	3
ELL733	Digital ASIC Design	3	0	2	4
ELL736	Solid State Imaging Sensors	3	0	0	3
ELL748	System-on-Chip Design and Test	3	0	0	3
ELL782	Computer Architecture	3	0	0	3
ELL784	Introduction to Machine Learning	3	0	0	3
ELL787	Embedded Systems and Applications	3	0	0	3
ELL789	Intelligent Systems	3	0	0	3
ELL791	Neural Systems and Learning Machines	3	0	2	4
ELL830	Issues in Deep Submicron VLSI Design	3	0	0	3
ELL831	CAD for VLSI, MEMS, and Nanoassembly	3	0	0	3
ELL832	Selected Topics in IEC-I	3	0	0	3
ELL834	Selected Topics in IEC-II	3	0	0	3
ELL883	Embedded Intelligence	3	0	0	3
ELV734	Special Module in Scientific Writing for Research	1	0	0	1
ELV831	Special Module in VLSI Testing	1	0	0	1
ELV832	Special Module in Machine Learning	1	0	0	1

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)					Lecture courses	Contact h/week				Credits
							L	T	P	Total	
I	ELL732 Micro and Nanoelectronics (3-0-0)	ELL735 Analog Integrated Circuits (3-0-0)	ELL734 MOS VLSI Design (3-0-0)	ELP831 IEC Lab-I (0-0-6)		3	9	0	6	15	12
II	PE (3-0-0)	ELP832 IEC Lab-II (0-0-6)	ELL730 I.C. Technology (3-0-0)	PE/OE (3-0-0)		3	9	0	6	15	12
Summer											
III	ELD831 Major Project Part-I (0-0-12)		PE/OE (3-0-0)	PE/OE (3-0-0)		2	6	0	12	18	12
IV (Project based) OR	ELD832 Major Project Part-II (0-0-24)					0	0	0	24	24	12
IV (Course based)	PE/OE (3-0-0)	PE/OE (3-0-0)	PE/OE (3-0-0)	PE/OE (3-0-0)		4	12	0	0	12	12

Total = 48

Master of Technology in Power Electronics, Electrical Machines and Drives

Department of Electrical Engineering

The overall credits structure

Category	PC	PE	OC	Total
Credits	24	18	6	48

Program Core

ELD851	Major Project Part-I	0	0	12	6	ELL758	Power Quality	3	0	0	3
ELL750	Modelling of Electrical Machines	3	0	0	3	ELL759	Power Electronic Converters for Renewable Energy Systems	3	0	0	3
ELL751	Power Electronic Converters	3	0	0	3	ELL760	Switched Mode Power Conversion	3	0	0	3
ELL752	Electric Drive System	3	0	0	3	ELL761	Power Electronics for Utility Interface	3	0	0	3
ELL850	Digital Control of Power Electronics and Drive Systems	3	0	0	3	ELL762	Intelligent Motor Controllers	3	0	0	3
ELP850	Electrical Machines Laboratory	0	0	3	1.5	ELL763	Advanced Electric Drives	3	0	0	3
ELP851	Power Electronics Laboratory	0	0	3	1.5	ELL764	Electric Vehicles	3	0	0	3
ELP852	Electrical Drives Laboratory	0	0	3	1.5	ELL765	Smart Grid Technology	3	0	0	3
ELP853	DSP Based Control of Power Electronics and Drives Laboratory	0	0	3	1.5	ELL766	Appliance Systems	3	0	0	3
Total Credits					24	ELL767	Mechatronics	3	0	0	3
						ELL768	Computer Aided Design of Power Electronic Systems	3	0	0	3

Program Electives

ELD850	Minor Project	0	0	6	3	ELL787	Embedded Systems and Applications	3	0	0	3
ELD852	Major Project Part-II	0	0	24	12	ELL791	Neural Systems and Learning Machines	3	0	2	4
ELL700	Linear Systems Theory	3	0	0	3	ELL851	Computer Aided Design of Electrical Machines	3	0	0	3
ELL703	Optimal Control Theory	3	0	0	3	ELL852	Condition Monitoring of Electrical Machines	3	0	0	3
ELL704	Advanced Robotics	3	0	0	3	ELL853	Advanced Topics in Electrical Machines	3	0	0	3
ELL706	Digital Control	3	0	0	3	ELL854	Selected Topics in Electrical Machines	3	0	0	3
ELL720	Advanced Digital Signal Processing	3	0	0	3	ELL855	High Power Converters	3	0	0	3
ELL753	Physical Phenomena in Electrical Machines	3	0	0	3	ELL856	Advanced Topics in Power Electronics	3	0	0	3
ELL754	Permanent Magnet Machines	3	0	0	3	ELL857	Selected Topics in Power Electronics	3	0	0	3
ELL755	Variable Reluctance Machines	3	0	0	3	ELL858	Advanced Topics in Electric Drives	3	0	0	3
ELL756	Special Electrical Machines	3	0	0	3	ELL859	Selected Topics in Electric Drives	3	0	0	3
ELL757	Energy Efficient Motors	3	0	0	3	ELP854	Electrical Machines CAD Laboratory	0	1	4	3
						ELP855	Smart Grids Laboratory	0	1	4	3
						ELT850	Industrial Training and Seminar	0	0	6	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)					Lecture courses	Contact h/week				Credits
	L	T	P	Total							
I	ELL750 Modelling of Electrical Machines (3-0-0)	ELL751 Power Electronic Converters (3-0-0)	ELP850 Electrical Machines Laboratory (0-0-3)	ELP851 Power Electronics Laboratory (0-0-3)	PE/OE (3-0-0)*	3	9	0	6	15	12
II	ELL752 Electric Drive System (3-0-0)	ELL850 Digital Control of Power Electronics and Drive Systems (3-0-0)	ELP852 Electrical Drives Laboratory (0-0-3)	ELP853 DSP Based Control of Power Electronics and Drives Laboratory (0-0-3)	PE/OE (3-0-0)*	3	9	0	6	15	12
Project Based											
III	ELD851 Major Project Part-I (0-0-12)	PE/OE (3-0-0)*	PE/OE (3-0-0)*			2	6	0	12	18	12
IV	ELD852 Major Project Part-II (0-0-24)					0	0	0	24	24	12
(OR) Course Based											
III	PE/OE (3-0-0)	PE/OE (3-0-0)	PE/OE (3-0-0)	PE/OE (3-0-0)		4	12	0	0	12	12
IV	ELD851 Major Project Part-I (0-0-12)	PE/OE (3-0-0)	PE/OE (3-0-0)			2	6	0	12	18	12

Total = 48

Master of Technology in Power Systems

Department of Electrical Engineering

The overall credits structure

Category	PC	PE	OC	Total
Credits	24	18	6	48

Program Core

ELD871	Major Project Part-I	0	0	12	6	ELL758	Power Quality	3	0	0	3
ELL770	Power System Analysis	3	0	0	3	ELL759	Power Electronic Converters for Renewable Energy Systems	3	0	0	3
ELL771	Advanced Power System Protection	3	0	0	3	ELL772	Planning and Operation of a Smart Grid	3	0	0	3
ELL775	Power System Dynamics	3	0	0	3	ELL773	High Voltage DC Transmission	3	0	0	3
ELL776	Advanced Power System Optimization	3	0	0	3	ELL774	Flexible AC Transmission System	3	0	0	3
ELP870	Power System Lab-I	0	1	4	3	ELL777	Power System operation and control	3	0	0	3
ELP871	Power System Lab-II	0	1	4	3	ELL778	Dynamic Modelling And Control of Sustainable Energy Systems	3	0	0	3
Total Credits		24				ELL779	Forecasting Techniques for Power System	3	0	0	3

Program Electives

ELD870	Minor Project-I	0	0	6	3	ELL870	Restructured Power System	3	0	0	3
ELD872	Major Project Part-II	0	0	24	12	ELL871	Distribution System Operation and Planning	3	0	0	3
ELL700	Linear Systems Theory	3	0	0	3	ELL872	Selected Topics in Power System	3	0	0	3
ELL712	Digital Communications	3	0	0	3	ELL873	Power System Transient	3	0	0	3
						ELL874	Power System Reliability	3	0	0	3

Sem.	Courses (number, Abbreviated Title, L-T-P, Credits)					Lecture courses	Contact h/week				Credits
							L	T	P	Total	
I	ELL 770 Power System Analysis (3-0-0)	ELL771 Advanced Power System Protection (3-0-0)	ELL775 Power System Dynamics (3-0-0)	ELP870 Power System Lab-I (0-1-4)		3	9	1	4	14	12
II	ELL776 Advanced Power System Optimization (3-0-0)	ELP871 Power System Lab-II (0-1-4)	PE/OE (3-0-0)	PE/OE (3-0-0)		3	9	1	4	14	12
Summer											
III	ELD871 Major Project Part-I (0-0-12)		PE/OE (3-0-0)	PE/OE (3-0-0)		2	6	0	12	18	12
IV (Project based) OR	ELD871 Major Project Part-II (0-0-24)					0	0	0	24	24	12
IV (Course based)	PE/OE (3-0-0)	PE/OE (3-0-0)	PE/OE (3-0-0)	PE/OE (3-0-0)		4	12	0	0	12	12

Total = 48

Master of Technology in Computer Technology

Department of Electrical Engineering

The overall credits structure

Category	PC	PE	OC	Total
Credits	21	24/27	3/6	51

Program Core

ELD780	Minor Project	0	0	4	2
ELD880	Major Project Part-I	0	0	12	6
ELL780	Mathematical Foundations of Computer Technology	3	0	0	3
ELL781	Software Fundamentals for Computer Technology	3	0	0	3
ELL782	Computer Architecture	3	0	0	3
ELL783	Operating Systems	3	0	2	4
Total Credits					21

Program Electives

BSP710	Communication & Signal Processing Tech. Lab.	0	1	4	3
ELD881	Major Project Part-II	0	0	24	12
ELL880	Special Topics in Computers-I	3	0	0	3
ELL881	Special Topics in Computers-II	3	0	0	3
ELV752	Special Modules in EET-I	1	0	0	1
ELV780	Special Module in Computers	1	0	0	1

Streamed Electives (EET) in (Cognitive and Intelligent Systems)

Required Electives

ELL784	Introduction to Machine Learning	3	0	0	3
ELL786	Multimedia Systems	3	0	0	3

Other Electives

ELL704	Advanced Robotics	3	0	0	3
ELL707	Systems Biology	3	0	0	3
ELL715	Digital Image Processing	3	0	2	4
ELL724	Multichannel Signal Processing	3	0	0	3
ELL741	Neuromorphic Engineering	3	0	0	3
ELL785	Computer Communication Networks	3	0	0	3
ELL787	Embedded Systems and Applications	3	0	0	3
ELL788	Computational Perception and Cognition	3	0	0	3
ELL789	Intelligent Systems	3	0	0	3
ELL791	Neural Systems and Learning Machines	3	0	2	4
ELL793	Computer Vision	3	0	0	3
ELL794	Human-Computer Interface	3	0	0	3
ELL795	Swarm Intelligence	3	0	0	3
ELL796	Signals and Systems in Biology	3	0	0	3
ELL798	Agent Technologies	3	0	0	3
ELL799	Natural Computing	3	0	0	3
ELL882	Large-Scale Machine Learning	3	0	0	3
ELL883	Embedded Intelligence	3	0	0	3
ELL884	Deep Learning for Natural Language Processing	3	0	0	3
ELL885	Machine Learning for Computational Finance	3	0	0	3
ELL886	Big Data Systems	3	0	0	3
ELL887	Cloud Computing	3	0	0	3
ELL888	Advanced Machine Learning	3	0	0	3
ELL890	Computational Neuroscience	3	0	0	3
ELL891	Advances in Deep Learning	3	0	0	3
ELL893	Cyber-Physical Systems	3	0	0	3

Streamed Electives (EET) in (Embedded Intelligent Systems)

Required Electives

ELL784	Introduction to Machine Learning	3	0	0	3
ELL787	Embedded Systems and Applications	3	0	0	3

Other Electives

COL719	Synthesis of Digital Systems	3	0	2	4
COL812	System Level Design and Modelling	3	0	0	3
ELL704	Advanced Robotics	3	0	0	3
ELL710	Coding Theory	3	0	0	3
ELL720	Advanced Digital Signal Processing	3	0	0	3
ELL728	Optoelectronic Instrumentation	3	0	0	3
ELL731	Mixed Signal Circuit Design	3	0	0	3
ELL733	Digital ASIC Design	3	0	2	4

ELL734	MOS VLSI design	3	0	0	3
ELL735	Analog Integrated Circuits	3	0	0	3
ELL748	System-on-Chip Design and Test	3	0	0	3
ELL766	Appliance Systems	3	0	0	3
ELL767	Mechatronics	3	0	0	3
ELL785	Computer Communication Networks	3	0	0	3
ELL786	Multimedia Systems	3	0	0	3
ELL790	Digital Hardware Design	3	0	0	3
ELL791	Neural Systems and Learning Machines	3	0	2	4
ELL797	Energy-Efficient Computing	3	0	0	3
ELL802	Adaptive and Learning Control	3	0	0	3
ELL883	Embedded Intelligence	3	0	0	3
ELL887	Cloud Computing	3	0	0	3
ELL898	Pervasive Computing	3	0	0	3
ELL899	Testing and Fault Tolerance	3	0	0	3
ELP780	Software Lab	0	1	4	3
ELP781	Digital Systems Lab	0	1	4	3
ELP831	IEC Laboratory-I	0	0	6	3

Streamed Electives (EET) in (Computer Communication and Networks)

Required Electives

ELL785	Computer Communication Networks	3	0	0	3
ELL786	Multimedia Systems	3	0	0	3

Other Electives

ELL710	Coding Theory	3	0	0	3
ELL711	Signal Theory	3	0	0	3
ELL712	Digital Communications	3	0	0	3
ELL714	Basic Information Theory	3	0	0	3
ELL716	Telecommunication Switching and Transmission	3	0	0	3
ELL717	Optical Communication Systems	3	0	0	3
ELL723	Broadband Communication Systems	3	0	0	3
ELL725	Wireless Communications	3	0	0	3
ELL784	Introduction to Machine Learning	3	0	0	3
ELL787	Embedded Systems and Applications	3	0	0	3
ELL797	Energy-Efficient Computing	3	0	0	3
ELL813	Advanced Information Theory	3	0	0	3
ELL816	Satellite Communication	3	0	0	3
ELL817	Access Networks	3	0	0	3
ELL818	Telecommunication Technologies	3	0	0	3
ELL820	Photonic Switching and Networking	3	0	0	3
ELL887	Cloud Computing	3	0	0	3
ELL889	Protocol Engineering	3	0	0	3
ELL892	Internet Technologies	3	0	0	3
ELL894	Network Performance Modeling and Analysis	3	0	0	3
ELL895	Network Security	3	0	0	3
ELL896	Mobile Computing	3	0	0	3
ELL897	Network Management	3	0	0	3
ELL898	Pervasive Computing	3	0	0	3
ELP720	Telecommunication Networks Laboratory	0	1	4	3
ELP780	Software Lab	0	1	4	3
ELP781	Digital Systems Lab	0	1	4	3
ELP782	Computer Networks Lab	0	1	4	3
ELP821	Advanced Telecommunication Networks Laboratory	0	1	4	3
ELP822	Network Software Laboratory	0	1	4	3

Streamed Electives (EET) in (Multimedia Information Processing)

Required Electives

ELL786	Multimedia Systems	3	0	0	3
ELL787	Embedded Systems and Applications	3	0	0	3

Master of Technology in Materials Engineering

Department of Materials Science and Engineering

The overall credits structure

Category	PC	PE	OE/PE	RE	Total
Credits	32	06	06	04	48

Program Core

MLL701	Structure and Characterization of Materials	3	0	0	3	MLL714	Fracture Mechanics	3	0	0	3
MLL702	Thermodynamics of Materials	3	0	0	3	MLL715	Advanced Engineering Materials	3	0	0	3
MLL703	Mathematical and Computational Methods in Materials	3	0	2	4	MLL716/					
MLP704	Materials Processing and Characterization Lab	1	1	4	4	APL767	Engineering Failure Analysis and Prevention	3	0	0	3
MLD801	M.Tech. Project-I		0	0	18	MLL717	Engineering and Specialty Polymers	3	0	0	3
MLD802	M.Tech. Project-II		0	0	18	MLL718	Polymeric Nanomaterials and Nanocomposites	3	0	0	3
Total Credits					32	MLL727	Smart Polymeric Materials	3	0	0	3
						MLL730	Diffusion and Kinetics	3	0	0	3
						MLL732	Porous Materials	3	0	0	3
						MLL734	Texture and Grain Boundary Engineering in Metals and Alloys	3	0	0	3

Non-graded Core (NG)

MLN710	Research Seminar		0	0	2	1	MLL736	Tribology and Surface Engineering of Materials	3	0	0	3
							MLL738	Electronic Devices and Characterization	3	0	0	3

Restricted Elective (RE) or Stream Core

MLV705	Special Topics in Materials	1	0	0	1	MLL740	Nanostructures and Nanomaterials	3	0	0	3
MLL711	Mechanical Behaviour of Materials	3	0	0	3	MLL742	Micro and Nano Fabrication in Materials Engg.	3	0	0	3

Program Electives (PE)

MLL712/						MLL748	Solid State Diffusion and Kinetics	3	0	0	3
APL753	Materials Design and Selection	3	0	0	3	MLL744	Materials for Additive Manufacturing	3	0	0	3
MLL713	Phase Transformations	3	0	0	3	MLL746	Crystals, Symmetry and Tensors	3	2	0	5
						MLL750	Imperfections in Materials and Applications	3	0	0	3
						MLL752	Creep and Superplasticity of Materials	3	0	0	3
						MLL760	Materials Simulation Methods Using High Performance Computing	2	0	2	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, credits)						Lecture courses	Contact h/week				Credits
								L	T	P	Total	
I	MLL701 Structure and Characterization of Materials (3-0-0) 3	MLL702 Thermodynamics of materials (3-0-0) 3	MLL703 Mathematical and Computational methods in Materials (3-0-2) 4	MLL704 Materials Processing and Characterization Lab (1-1-4) 4	MLV705 (RE-1) Special topics in Materials (1-0-0) 1		5	11	1	6	18	15
II	MLN710 (NG) Research Seminar (0-0-2) 1	MLL711 (RE) 2 Mechanical Behaviour of Materials (3-0-0) 3	PE-1 (3-0-0) 3	PE-2 (3-0-0) 3	OE-1 / PE-3 (3-0-0) 3	OE-2 / PE-4 (3-0-0) 3	6	15	0	2	17	15
Summer												
III	MLD801 (0-0-18) 9						0	0	0	0	18	9
IV	MLD802 (0-0-18) 9						0	0	0	0	18	9

Total = 48

Master of Technology in Polymer Science and Technology

Department of Materials Science and Engineering

The overall credits structure

Category	PC	PE	OE	Total
Credits	42	12	0	54

Program Core

MLL719	Polymer Chemistry	3	0	0	3
MLL720	Polymer Processing	3	0	0	3
MLL721	Polymer Physics	3	0	0	3
MLL722	Polymer Technology	3	0	0	3
MLL723	Polymer Characterization	3	0	0	3
MLL724	Polymer Engineering and Rheology	3	0	0	3
MLL725	Polymer Testing and Properties	3	0	0	3
MLP726	Polymer Synthesis and Characterization Laboratory	0	0	4	2
MLP728	Polymer Rheology and Processing Lab	0	0	2	1
MLD811	Major Project Part-I	0	0	12	6
MLD812	Major Project Part-II	0	0	24	12
Total Credits		42			

Program Electives

MLL717	Engineering and Specialty Polymers	3	0	0	3
MLL718	Polymeric Nanomaterials and Nanocomposites	3	0	0	3
MLL729	Polymer Blends and Composites	3	0	0	3
MLL731	Rubber Technology	3	0	0	3
MLL733	Polymer Reaction Engineering	3	0	0	3
MLL735	Polymer Product & Mould Design	3	0	0	3
MLL737	Polymer Degradation and Stabilization	3	0	0	3
MLL739	Polymer Coatings	3	0	0	3
MLL741	Biodegradable Polymeric Materials	3	0	0	3
MLS800	Independent Study	0	3	0	3
MLV700	Special Lectures in Polymers	1	0	0	1
MLD799	Minor Project	0	0	6	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)						Lecture courses	Contact h/week				Credits
								L	T	P	Total	
I	MLL719 Polymer Chemistry (3-0-0) 3	MLL721 Polymer Physics (3-0-0) 3	MLL723 Polymer Characterization (3-0-0) 3	MLL724 Polymer Engineering and Rheology (3-0-0) 3	MLP726 Polymer Synthesis and Characterization Laboratory (0-0-4) 2		4	12	0	4	16	14
II	MLL720 Polymer Processing (3-0-0) 3	MLL722 Polymer Technology (3-0-0) 3	MLP728 Polymer Rheology and Processing Lab (0-0-2) 1	MLL725 Polymer Testing and Properties (3-0-0) 3	PE-1 (3-0-0) 3		4	12	0	2	14	13
Summer												
III	PE-2 (3-0-0) 3	PE-3 (3-0-0) 3	PE-4 (3-0-0) 3	MLD811 Major Project Part-I (0-0-12) 6			3	9	0	12	21	15
IV	MLD812 Major Project Part-II (0-0-24) 12						0	0	0	24	24	12

Total = 54

Master of Technology in Industrial Engineering

Department of Mechanical Engineering

The overall credits structure

Category	PC	PE	OE	Total
Credits	36	12	0	48

Program Core

MCD861	M.Tech. Project Part-I	0	0	24	12
MCD862	M.Tech. Project Part-II	0	0	24	12
MCL751	Industrial Engineering Systems	1	0	4	3
MCL754	Operations Planning and Control	3	0	0	3
MCL761	Probability and Statistics	3	0	0	3
MCL765	Operations Research	3	0	0	3
Total Credits					36

Streamed Electives (MEE) in (Analytics and Optimization)

CTL729	Automotive Reliability and Life Testing	3	0	0	3
CTL732	Advanced Vehicle Propulsion	3	0	0	3
MCL753	Manufacturing Informatics	3	0	2	4
MCL758	Optimization	3	0	0	3
MCL770	Stochastic Modeling and Simulation	3	0	0	3
MCL865	Advanced Operations Research	3	0	0	3
COL702	Advanced Data Structures	3	0	2	4
COL770	Advanced Artificial Intelligence	3	0	2	4
COL772	Natural Language Processing or	3	0	2	4
MTL785	Natural Language Processing	3	0	0	3
COL774	Machine Learning or	3	0	2	4
ELL784	Introduction to Machine Learning	3	0	2	4
ELL791	Neural Systems and Learning Machines	3	0	0	3
MSL713	Information Systems Management	3	0	0	3
MSL717	Business Systems Analysis & Design	3	0	0	3
MTL763	Introduction to Game Theory	3	0	0	3

Streamed Electives (MEE) in (Product Life Cycle Management)

CTL729	Automotive Reliability and Life Testing	3	0	0	3
CTL732	Advanced Vehicle Propulsion	3	0	0	3
MCL771	Value Engineering and Life Cycle Costing	3	0	0	3
MCL772	Reliability Engineering	3	0	0	3
MCL773	Quality Systems	3	0	0	3
MSL841	Supply Chain Analytics	3	0	0	3
ITL709	Maintenance Planning and Control	3	0	0	3
ITL711	Reliability, Availability and Maintainability (RAM) Engineering	3	0	0	3
ITL702	Diagnostic Maintenance and Condition Monitoring	3	0	2	4

Streamed Electives (MEE) in (Operations Management)

CTL729	Automotive Reliability and Life Testing	3	0	0	3
CTL732	Advanced Vehicle Propulsion	3	0	0	3
MCL763	Network Models for Public Systems	3	0	0	3
MCL755	Service System Design	3	0	0	3
MCL756	Supply Chain Management	3	0	0	3
MCL757	Logistics	3	0	0	3
MCL759	Entrepreneurship	3	0	0	3
MCL760	Project Management	3	0	0	3
MCL775	Special Topics in IE	3	0	0	3
MCL866	Maintenance management	3	0	0	3
CVL746	Public Transportation Systems	3	0	0	3
CVL750	Intelligent Transportation Systems	3	0	0	3
MSL805	Services Operations Management	3	0	0	3
MSL704	Science & Technology Policy Systems	3	0	0	3
MSL877	Electronic Government	1.5	0	0	1.5

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)				Lecture courses	Contact h/week				Credits
						L	T	P	Total	
I	MCL761 Probability and Statistics (3-0-0) 3	MCL765 Operations Research (3-0-0) 3	MCL751 Industrial Engineering Systems (1-0-4) 3	PE-1 (from PLM Stream) (3-0-0) 3	4	10	0	4	14	12
II	MCL754 Operations Planning and Control (3-0-0) 3	PE-2 (3-0-0) 3	PE-3 (3-0-0) 3	PE-4 (3-0-0) 3	4	12	0	0	12	12
Professional Project Activity In Summer Vacation										
III	MCD861 Major Project Part-I (0-0-24) 12				1	3	0	24	27	15
IV	MCD862 Major Project Part-II (0-0-24) 12				0	0	0	24	24	12

Total = 48

Master of Technology in Mechanical Design

Department of Mechanical Engineering

The overall credits structure

Category	PC	PE	OE	Total
Credits	32	22	0	54

Program Core

APL701	Continuum Mechanics	3	0	0	3
MCD831	Major Project Part-I	0	0	12	6
MCD832	Major Project Part-II	0	0	24	12
MCL731	Analytical Dynamics	3	0	0	3
MCL735	CAD and Finite Element Analysis	3	0	2	4
MCL742	Design & Optimization	3	0	2	4
Total Credits		32			

Streamed Electives MEM - (A1) (atleast 12 credits)

MCL730	Designing with advanced materials	3	0	2	4
MCL733	Vibration and Noise	3	0	2	4
MCL736	Automotive Design	3	0	2	4
MCL741	Control Engineering	3	0	2	4
MCL745	Robotics	3	0	2	4
MCL748	Tribological Systems Design	3	0	2	4

Streamed Electives MEM - (A2) (atleast 10 credits)

MCL711	Fracture Mechanics in Design	2	0	2	3
MCL712	Engineering Acoustics	3	0	0	3
MCL713	Active Noise Control	3	0	0	3
MCL714	Orthopedic Biomechanics and Implant Design	2	0	2	3
MCL715	Design for Noise, Vibration and Harshness	2	0	2	3

MCL716	Mechatronics Product Design	2	0	2	3
MCL717	Machine Tool Design	2	0	2	3
MCL718	Design for Manufacture and Assembly	2	0	2	3
MCL723	Vehicle Dynamics	2	0	2	3
MCL728	Nanotribology	3	0	0	3
MCL729	Nanomechanics	2	0	2	3
MCL737	Biomechanics of Trauma and Automotive Design	3	0	0	3
MCL738	Dynamics of Multibody Systems	2	0	2	3
MCL740	Lubrication	3	0	0	3
MCL743	Plant Equipment Design	3	0	0	3
MCL747	Design of Precision Machines	2	0	2	3
MCL797	Freedom and Constraints in Design	3	0	0	3
MCL798	Medical Robotics	2	0	2	3
MCL834	Vibroacoustics	2	0	2	3
MCL837	Advanced Mechanisms	2	0	2	3
MCL839	Rotor Dynamics	2	0	2	3
MCL840	Experimental Modal Analysis and Dynamic Design	2	0	2	3
MCL845	Advanced Robotics	2	0	2	3
MCL848	Special topics in Systems Design-I	2	0	0	2
MCL849	Special topics in Systems Design-II	3	0	0	3
MCS830	Independent Study	0	3	0	3
MCV849	Special Module in Systems Design	1	0	0	1

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)							Lecture courses	Contact h/week				Credits
									L	T	P	Total	
I	APL701 Continuum Mechanics (3-0-0) 3	MCL731 Analytical Dynamics (3-0-0) 3	MCL735 CAD and Finite Element Analysis (3-0-2) 4	MCL742 Design & Optimization (3-0-2) 4				4	12	0	4	16	14
II	PE-1 (3-0-2) 4	PE-2 (3-0-2) 4	PE-3 (3-0-2) 4	PE-4 (3-0-0) 3				4	12	0	6	18	15
Summer													
III	MCD831 Major Project Part-I (0-0-12) 6	PE-5 (3-0-2) 4	PE-6 (3-0-0) 3					2	6	0	14	0	13
IV	MCD832 Major Project Part-II (0-0-24) 12							0	0	0	24	24	12

Total = 54

Master of Technology in Production Engineering

Department of Mechanical Engineering

The overall credits structure

Category	PC	PE	OE	Total
Credits	31	18	0	49

Program Core

MCD881	Major Project Part-I	0	0	12	6	MCL751	Industrial Engineering Systems	1	0	4	3
MCL705	Experimental Methods	3	0	2	4	MCL753	Manufacturing Informatics	3	0	2	4
MCL769	Metal Forming Analysis	3	0	2	4	MCL754	Operations Planning and Control	3	0	0	3
MCL781	Machining Processes and Analysis	3	0	2	4	MCL773	Quality Systems	3	0	0	3
MCL782	Computational Methods	2	0	0	2	MCL776	Advances in Metal Forming	3	0	0	3
MCL784	Computer Aided Manufacturing	3	0	2	4	MCL777	Machine Tool Design	3	0	2	4
MCL786	Metrology	2	0	2	3	MCL778	Design and Metallurgy of Welded Joints	3	0	2	4
MCL787	Welding Science and Technology	3	0	2	4	MCL780	Casting Technology	3	0	2	4
Total Credits		31				MCL783	Automation in Manufacturing	3	0	2	4
						MCL785	Advanced Machining Processes	3	0	0	3
						MCL788	Surface Engineering	3	0	2	4
						MCL791	Processing and Mechanics of Composite Materials	3	0	2	4

Program Electives

MCD882	Major Project Part-II	0	0	24	12	MCL792	Injection Molding and Mold Design	2	0	2	3
MCL718	Design for Manufacture and Assembly	2	0	2	3	MCL795	Laser Processing of Materials	3	0	2	4
MCL729	Nanomechanics	3	0	0	3	MCL796	Additive Manufacturing	3	0	2	4
MCL749	Mechatronics Product Design	3	0	2	4	MCP790	Process Engineering	2	0	4	4
MCL750	Product Design and Manufacturing	1	0	4	3						

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)							Lecture courses	Contact h/week				Credits
									L	T	P	Total	
I	MCL781 Machining Processes and Analysis (3-0-2) 4	MCL787 Welding Science and Technology (3-0-2) 4	MCL769 Metal Forming Analysis (3-0-2) 4					3	9	0	6	15	12
II	MCL705 Experimental Methods (3-0-2) 4	MCL784 CAM (3-0-2) 4	MCL786 Metrology (2-0-2) 3	MCL782 Computational Methods (2-0-0) 3				4	10	0	6	16	13
Professional Project Activity In Summer Vacation													
III	MCD881 Major Project Part-I (Core) (0-0-12) 6	PE-1 (3-0-0) 3	PE-2 (3-0-0) 3					2	6	0	12	18	12
IV	MCD882 Major Project Part-II (PE) (0-0-24) 12							0	0	0	24	24	12

Total = 49

Master of Technology in Thermal Engineering

Department of Mechanical Engineering

The overall credits structure

Category	PC	PE	OE	Total
Credits	36	12	3	51

Non-graded Core

MCD800 Professional Project Activity	0	0	6	3
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Program Core

MCD811 Major Project Part-I (Thermal Engineering)	0	0	16	8
MCD812 Major Project Part-II (Thermal Engineering)	0	0	24	12
MCL701 Advanced Thermodynamics	3	0	0	3
MCL702 Advanced Fluid Mechanics	3	0	0	3
MCL703 Advanced Heat and Mass Transfer	3	0	0	3
MCL704 Applied Mathematics for Thermofluids	3	0	0	3
MCL705 Experimental Methods	3	0	2	4
Total Credits				36

Program Electives

MCL707 Thermal Turbomachines	3	0	0	3
MCL732 Air Pollution: Sources and Apportionment	3	0	0	3

MCL811 Advanced Power Generation Systems	3	0	0	3
MCL812 Combustion	3	0	0	3
MCL813 Computational Heat Transfer	3	0	2	4
MCL814 Convective Heat Transfer	3	0	0	3
MCL815 Fire Dynamics and Engineering	2	0	4	4
MCL816 Gas Dynamics	3	0	2	4
MCL817 Heat Exchangers	3	0	0	3
MCL818 Heating, Ventilating and Air-conditioning	3	0	2	3
MCL819 Lattice Boltzmann method	3	0	0	3
MCL820 Micro/nano Scale Heat Transfer	3	0	2	4
MCL821 Radiative Heat Transfer	3	0	0	3
MCL822 Steam and Gas Turbines	3	0	2	4
MCL823 Thermal Design	3	0	2	4
MCL824 Turbocompressors	3	0	0	3
MCL825 Design of Wind Power Farms	3	0	2	4

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)						Lecture courses	Contact h/week				Credits
								L	T	P	Total	
I	MCL701 Adv. Thermodynamics (3-0-0) 3	MCL702 Adv. Fluid Mechanics (3-0-0) 3	MCL703 Adv. Heat & Mass Transfer (3-0-0) 3	MCL704 Applied Math. (3-0-0) 3			4	12	0	0	12	12
II	MCL705 Exptl Methods (3-0-2) 4	PE-1 (3-0-0) 3	PE-2 (3-0-0) 3	PE-3 (3-0-0) 3			4	12	0	2	14	13
Summer	Professional Project Activity (compulsory audit)						0					0
III	MED811 Major Project Part-I (MET) (0-0-16) 8	PE-4 (3-0-0) 3	OE-1 (3-0-0) 3				2	6	0	12	18	14
IV	MED812 Major Project Part-II (MET) (0-0-24) 12						0	0	0	24	24	12

Total = 51

Master of Technology in Applied Optics

Department of Physics

The overall credits structure

Category	PC	PE	OE	Total
Credits	39	9	3	51

Program Core

PYD851	Major Project Part-I	0	0	12	6	PYL770	Ultra-fast Optics and Applications	3	0	0	3
PYD852	Major Project Part-II	0	0	24	12	PYL771	Green Photonics	3	0	0	3
PYL751	Optical Sources, Photometry and Metrology	3	0	0	3	PYL772	Plasmonic Sensors	3	0	0	3
PYL752	Laser Systems and Applications	3	0	0	3	PYL780	Diffractive and Micro Optics	3	0	0	3
PYL753	Optical Systems Design	3	0	0	3	PYL791	Fiber Optics	3	0	0	3
PYL755	Basic Optics and Optical Instrumentation	3	0	0	3	PYL792	Optical Electronics	3	0	0	3
PYL756	Fourier Optics and Holography	3	0	0	3	PYL795	Optics and Lasers	3	0	0	3
PYP761	Optical Fabrication and Metrology Laboratory	0	0	6	3	PYL858	Advanced Holographic Techniques	3	0	0	3
PYP762	Advanced Optics Laboratory	0	0	6	3	PYL879	Selected Topics in Applied Optics	3	0	0	3
Total Credits					39	PYL881	Selected Topics – I	1	0	0	1
						PYL882	Selected Topics – II	1	0	0	1
						PYL883	Minor Project	0	0	6	3
						PYL892	Guided Wave Photonic Sensors	3	0	0	3

Program Electives

PYL757	Statistical and Quantum Optics	3	0	0	3	PYP763	Computational Optics Laboratory	0	0	6	3
PYL758	Advanced Quantum Optics and Applications	3	0	0	3	PYP764	Advanced Optical Workshop	0	0	6	3
PYL759	Computational Optical Imaging	3	0	0	3	PYS855	Independent Study	0	3	0	3
PYL760	Biomedical Optics and Bio-photonics	3	0	0	3	PYL747	Non-linear Optics	3	0	0	3
						PYL774	Polarised Light and its Application	3	0	0	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)						Lecture courses	Contact h/week				Credits
								L	T	P	Total	
I	PYL755 Basic Optics and Optical Instrumentation (3-0-0) 3	PYL751 Optical Sources, Photometry and Metrology (3-0-0) 3	PYL753 Optical Systems Design (3-0-0) 3	PYP761 Optical Fabrication and Metrology Laboratory (0-0-6) 3	PE-1 (3-0-0) 3		4	12	0	6	18	15
II	PYL752 Laser Systems and Application (3-0-0) 3	PYL756 Fourier Optics and Holography (3-0-0) 3	PYP762 Advanced Optics Laboratory (0-0-6) 3	PE-2 (3-0-0) 3	PE-3 (3-0-0) 3		4	12	0	6	18	15
Summer												
III	OE-1 (3-0-0) 3	PYD851 Maj. Proj. Part-I (0-0-12) 6					1	3	0	12	15	9
IV	PYD852 Maj. Proj. Part-II (0-0-24) 12						0	0	0	24	24	12

Total = 51

Master of Technology in Solid State Materials

Department of Physics

The overall credits structure

Category	PC	PE	OE	Total
Credits	39	9	3	51

Program Core

PYD801	Major Project Part-I	0	0	12	6
PYD802	Major Project Part-II	0	0	24	12
PYL701	Physical Foundations of Materials Science	3	0	0	3
PYL702	Physics of Semiconductor Devices	3	0	0	3
PYL703	Electronic Properties of Materials	3	0	0	3
PYL704	Science and Technology of Thin Films	3	0	0	3
PYL705	Nanostructured Materials	3	0	0	3
PYP701	Solid State Materials Laboratory-I	0	0	6	3
PYP702	Solid State Materials Laboratory-II	0	0	6	3
Total Credits		39			

Program Electives

PYL707	Characterization Techniques for Materials	3	0	0	3
PYL723	Vacuum Science and Cryogenics	3	0	0	3
PYL724	Advances in Spintronics	3	0	0	3
PYL725	Surface Physics and Analysis	3	0	0	3
PYL726	Semiconductor Device Technology	3	0	0	3
PYL727	Energy Materials and Devices	3	0	0	3
PYL728	Quantum Heterostructures	2	0	0	2
PYL729	Nanoprobe Techniques	1	0	0	1
PYL750	Topology in Condensed Matter Physics	3	0	0	3
PYV759	Selected Topics in Solid State Materials	1	0	0	1

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)						Lecture courses	Contact h/week				Credits
								L	T	P	Total	
I	PYL701 Physical Foundations of Materials Science (3-0-0) 3	PYL702 Physics of Semiconductor Devices (3-0-0) 3	PYL703 Electronic Properties of Materials (3-0-0) 3	PYP701 Solid State Materials Laboratory-I (0-0-6) 3	PE-1 (3-0-0) 3		4	12	0	6	18	15
II	PYL704 Science and Technology of Thin Films (3-0-0) 3	PYL705 Nanostructured Materials (3-0-0) 3	PYP702 Solid State Materials Laboratory-II (0-0-6) 3	PE-2 (3-0-0) 3	PE-3 (3-0-0) 3		4	12	0	6	18	15
Summer												
III	OE-1 (3-0-0) 3	PYD801 Maj. Proj. Part-I (0-0-12) 6					1	3	0	12	15	9
IV	PYD802 Maj. Proj. Part-II + Report (0-0-24) 12						0	0	0	24	24	12

Total = 51

Master of Technology in Textile Engineering

Department of Textile and Fibre Engineering

The overall credits structure

Category	PC	PE	OE	Total
Credits	42	12	0	54

Program Core

TXD801	Major Project Part-I (TXE)	0	0	12	6	TXL751	Apparel Engineering and Quality Control	2	0	2	3
TXD803	Major Project Part-II (TXE)	0	0	24	12	TXL752	Design of Functional Clothing	3	0	0	3
TXL721	Theory of Yarn Structure	3	0	0	3	TXL766	Design and Manuf. of Textile Structural Composites	3	0	0	3
TXL722	Mechanics of Spinning Processes	3	0	0	3	TXL771	Electronics and Controls for Textile Industry	3	0	2	4
TXL725	Mechanics of Spinning Machines	3	0	0	3	TXL772	Computational Methods for Textiles	2	0	2	3
TXL731	Theory of Fabric Structure	3	0	0	3	TXL773	Medical Textiles	3	0	0	3
TXL732	Advanced Fabric Manufacturing Systems	3	0	0	3	TXL774	Process Control in Yarn & Fabric Manufacturing	3	0	0	3
TXL775	Technical Textiles	3	0	0	3	TXL777	Product Design and Development	3	0	0	3
TXL783	Design of Experiments and Statistical Techniques	3	0	0	3	TXL780	Principles of Characterization of Functional and Technical Textiles	3	0	0	3
TXP725	Mechanics of Textile Machines Laboratory	0	0	2	1	TXL781	Project Appraisal and Finance	3	0	0	3
TXP761	Evaluation of Textile Materials	0	0	4	2	TXL782	Production and Operations Management in Textile Industry	3	0	0	3
Total Credits		42									

Program Electives

TXD809	Mini Project (Textile Engineering)	0	0	8	4	TXL784	Supply Chain Management in Textile Industry	3	0	0	3
TXL700	Modelling and Simulation in Fibrous Assemblies	2	0	2	3	TXL785	Heat and Mass Transport in Fibrous Materials	3	0	0	3
TXL710	High Performance and Specialty Fiber	3	0	0	3	TXL786	Technology of Textile Coating and Lamination	2	0	2	3
TXL712	Polymer and Fibre Physics	3	0	0	3	TXL807	Seminar (Textile Engineering)	0	2	0	2
TXL719	Functional & Smart Textiles	3	0	0	3	TXS805	Independent Study (Textile Engineering)	0	3	0	3
TXL724	Textured Yarn Technology	3	0	0	3	TXV702	Management of Textile Business	1	0	0	1
TXL734	Nonwoven Processes and Products	3	0	0	3	TXV703	Special Module in Textile Product Mgmt.	1	0	0	1
TXL740	Science & App. of Nanotechnology in Textiles	3	0	0	3	TXV704	Special Module in Yarn Manufacture	1	0	0	1
TXL750	Science of Clothing Comfort	3	0	0	3	TXV705	Special Module in Fabric Manufacture	1	0	0	1

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)						Lecture courses	Contact h/week				Credits
								L	T	P	Total	
I	TXL721 Theory of Yarn Structure (3-0-0) 3	TXL722 Mechanics of Spinning Processes (3-0-0) 3	TXL731 Theory of Fabric Structure (3-0-0) 3	TXL732 Advanced Fabric Manufacturing Systems (3-0-0) 3		PE-1 (3-0-0) 3	5	15	0	0	15	15
II	TXL775 Technical Textiles (3-0-0) 3	TXL725 Mechanics of Spinning Machines (3-0-0) 3	TXL783 Design of Expt. and Stat. Tech. (3-0-0) 3	TXP761 Evaluation of Textile Materials (0-0-4) 2	TXP725 Mechanics of Spinning Machines Lab. (0-0-2) 1	PE-2 (3-0-0) 3	4	12	0	6	18	15
Summer												
III	TXD801 Major Project Part-I (TXE) (0-0-12) 6	PE-3 (3-0-0) 3	PE-4 (3-0-0) 3				2	6	0	12	18	12
IV	TXD803 Major Project Part-II (TXE) (0-0-24) 12						0	0	0	24	24	12

Total = 54

Master of Technology in Textile Chemical Processing

Department of Textile and Fibre Engineering

The overall credits structure

Category	PC	PE	OE	Total
Credits	42	12	0	54

Program Core

TXD805	Major Project Part-I (TXC)	0	0	12	6	TXL724	Textured Yarn Technology	3	0	0	3
TXD806	Major Project Part-II (TXC)	0	0	24	12	TXL740	Science and Application of Nanotechnology in Textiles	3	0	0	3
TXL712	Polymer and Fibre Physics	3	0	0	3	TXL750	Science of Clothing Comfort	3	0	0	3
TXL747	Colour Science	3	0	0	3	TXL751	Apparel Engineering and Quality Control	2	0	2	3
TXL748	Advances in Finishing of Textiles	3	0	0	3	TXL752	Design of Functional Clothing	3	0	0	3
TXL749	Theory and Practice of Dyeing	3	0	0	3	TXL755	Textile Wet Processing Machines: Automation and Control*	3	0	0	3
TXL753	Advanced Textile Printing Technology	2	0	0	2	TXL756	Textile Auxiliaries*	3	0	0	3
TXL754	Sustainable Chemical Processing of Textiles	2	0	0	2	TXL766	Design and Manuf. of Textile Structural Composites	3	0	0	3
TXL783	Design of Experiments and Statistical Techniques	3	0	0	3	TXL773	Medical Textiles	3	0	0	3
TXP748	Textile Preparation and Finishing Lab	0	0	2	1	TXL775	Technical Textiles	3	0	0	3
TXP749	Textile Coloration Lab	0	0	2	1	TXL777	Product Design and Development	3	0	0	3
TXP751	Characterization and Evaluation of Dyed and Finished Textiles Lab	0	0	2	1	TXL780	Principles of Characterization of Functional and Technical Textiles	3	0	0	3
TXS751	Research Seminar	0	0	2	1	TXL781	Project Appraisal and Finance	2	1	0	3
TXR752	Professional Practices	0	0	2	1	TXL782	Production and Operations Management in Textile Industry	3	0	0	3
TXR800	Industrial Summer Training	Non credit				TXL784	Supply Chain Mgmt. in Textile Industry	3	0	0	3
Total Credits				42		TXL785	Heat and Mass Transport in Fibrous Materials	3	0	0	3

Program Electives

MSL760	Marketing Management	2	0	2	3	TXL786	Technology of Textile Coating and Lamination	2	0	2	3
MSL802	Management of Intellectual Property Rights	3	0	0	3	TXP711	Polymer and Fibre Chemistry Laboratory	0	2	0	1
MSL816	Total Quality Management	2	0	2	3	TXP712	Polymer and Fibre Physics Laboratory	0	0	2	1
TXD812	Mini Projects (TCP)	0	0	6	3	TXP716	Fibre Production and Post Spinning Operation Laboratory	0	0	4	2
TXL711	Polymer and Fibre Chemistry	3	0	0	3	TXP761	Evaluation of Textile Materials	0	0	4	2
TXL713	Technology of Melt Spun Fibres	3	1	0	4	TXS811	Independent Study	0	3	0	3
TXL714	Advanced Materials Characterization Techniques	1	0	0	1	TXV703	Special Module in Textile Product Mgmt.	1	0	0	1
TXL715	Technology of Solution Spun Fibres	3	0	0	3	TXV707	Special Module-Textile Chemical Processing-1	1	0	0	1
TXL719	Functional and Smart Textiles	3	0	0	3	* TCP PE Basket					

Sem.	Courses (Number, Abbreviated Title, L-T-P, credits)							Lecture courses	Contact h/week				Credits
	L	T	P	Total									
I	TXL712 Polymer & Fibre Physics (3-0-0) 3	TXL747 Colour Science (3-0-0) 3	TXL749 Theory and Practice of Dyeing (3-0-0) 3	TXL753 Advanced Textile Printing Technology (2-0-0) 2	TXP749 Textile Coloration Lab (0-0-2) 1	TXP751 Characterization of Chemicals and Finished Textiles Lab (0-0-2) 1	TXR752 Professional Practices (0-0-2) 1	4	11	0	6	17	14
II	TXL748 Advances in Finishing of Textiles (3-0-0) 3	TXL754 Sustainable Chemical Processing of Textiles (2-0-0) 2	TXL783 Design of Experiments and Statistical Techniques (3-0-0) 3	TXP748 Textile Preparation and Finishing Lab (0-0-2) 1		PE-1 (3-0-0) 3	PE-2 (3-0-3) 3	5	14	0	2	16	15
Summer	TXR800 Industrial Summer Training												
III	TXD805 Major Project Part-I (TCP) (0-0-12) 6			TXS751 Research Seminar (0-0-2) 1	PE-3* (3-0-0) 3	PE-4 (3-0-0) 3		2	6	0	14	18	13
IV	TXD806 Major Project Part-II (TCP) (0-0-24) 12							0	0	0	24	24	12

* From TCP PE Basket

Total = 54

Master of Technology in Fibre Science and Technology

Department of Textile and Fibre Engineering

The overall credits structure

Category	PC	PE	OE	Total
Credits	42	12	0	54

Program Core

TXD802	Major Project Part-I	0	0	12	6	TXL741	Env. Manag. in Textile and Allied Industries	3	0	0	3
TXD804	Major Project Part-II	0	0	24	12	TXL747	Colour Science	3	0	0	3
TXL711	Polymer and Fibre Chemistry	3	0	0	3	TXL750	Science of Clothing Comfort	3	0	0	3
TXL712	Polymer and Fibre Physics	3	0	0	3	TXL752	Design of Functional Clothing	3	0	0	3
TXL713	Technology of Melt Spun Fibres	3	1	0	4	TXL754	Sustainable Chemical Processing of Textiles	2	0	0	2
TXL714	Advanced Materials Characterization Techniques	1	0	0	1	TXL772	Computational Methods for Textiles	2	0	2	3
TXL715	Technology of Solution Spun Fibres	3	0	0	3	TXL773	Medical Textiles	3	0	0	3
TXL748	Advances in Finishing of Textiles	3	0	0	3	TXL775	Technical Textiles	3	0	0	3
TXL749	Theory and Practice of Dyeing	3	0	0	3	TXL777	Product Design and Development	3	0	0	3
TXP711	Polymer and Fibre Chemistry Laboratory	0	0	2	1	TXL780	Principles of Characterization of Functional and Technical Textiles	3	0	0	3
TXP712	Polymer and Fibre Physics Laboratory	0	0	2	1	TXL781	Project Appraisal and Finance	3	0	0	3
TXP716	Fibre Production and Post Spinning Operation Laboratory	0	0	4	2	TXL782	Production and Operations Management in Textile Industry	3	0	0	3
Total Credits		42				TXL783	Design of Experiments and Statistical Techniques	3	0	0	3

Program Electives

TXL700	Modelling and Simulation in Fibrous Assemblies	2	0	2	3	TXS806	Independent Study (TTF)	0	3	0	3
TXL710	High Performance and Specialty Fiber	3	0	0	3	TXV701	Process Cont. and Econ. in Manmade Fibre Prod.	1	0	0	1
TXL719	Functional & Smart Textiles	3	0	0	3	TXV702	Management of Textile Business	1	0	0	1
TXL724	Textured Yarn Technology	3	0	0	3	TXV703	Special Module in Textile Product Mgmt.	1	0	0	1
TXL734	Nonwoven Processes and Products	3	0	0	3	TXV706	Special Module in Fibre Science	1	0	0	1
TXL740	Science & App. of Nanotechnology in Textiles	3	0	0	3	TXV707	Special Module in Textile Chemical Processing	1	0	0	1

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)							Lecture courses	Contact h/week				Credits
									L	T	P	Total	
I	TXL711 Polymer & Chemistry (3-0-0) 3	TXP711 Polymer & Fibre Chemistry Lab (0-0-2) 1	TXL712 Polymer & Fibre Physics (3-0-0) 3	TXP712 Polymer & Fibre Physics Lab (0-0-2) 1	TXL713 Technology of Melt Spun Fibres (3-1-0) 4	TXL749 Theory and Practice of Dyeing (3-0-0) 3		4	12	1	4	17	15
II	TXL715 Technology of Soln Spun Fibres (3-0-0) 3	TXP716 Fibre Production & Post Spinning Operation Lab (0-0-4) 2	TXL748 Advances in Finishing of Textiles (3-0-0) 3	TXL714 Characterization of advanced materials (1-0-0) 1		PE-2 (3-0-0) 3	PE-1 (3-0-0) 3	5	12	0	6	18	15
Summer													
III	TXD802 Maj. Proj. Part-I (TTF) (0-0-12) 6				PE-3 (3-0-0) 3	PE-4 (3-0-0) 3		2	6	0	12	18	12
IV	TXD804 Maj. Proj. Part-II (TTF) (0-0-24) 12							0	0	0	24	24	12

Total = 54

Master of Technology in Radio Frequency Design and Technology

Centre for Applied Research and Electronics

The overall credits structure

	Core		Elective		Total
Category	PC	BC	PE	OE	Total
Credits	24	3	24*/21**	0*/3**	51

* For students with M.Tech Dissertation

** For students without M.Tech Dissertation

Program Core

CRD802	Minor Project		0	0	6	3
CRD811	Major Project-I		0	0	12	6
CRL702	Architectures and Algorithms for DSP Systems		2	0	4	4
CRL711	CAD of RF and Microwave Circuits		3	0	2	4
CRL724	RF and Microwave Measurements		3	0	0	3
CRP718	RF and Microwave Measurement Lab		1	0	6	4
Total Credits			24			

Bridge Course (Core)

CRL601	Basics of Statistical Signal Analysis		2	0	2	3
CRL611	Basics of RF and Microwaves		2	1	0	3
CRL621	Fundamentals of Semiconductor Devices		3	0	0	3

Program Electives

ELL711	Signal Theory		3	0	0	3
ELL712	Digital Communications		3	0	0	3
ELL714	Basic Information Theory		3	0	0	3
ELL718	Statistical Signal Processing		3	0	0	3
ELL719	Detection and Estimation Theory		3	0	0	3
ELL720	Advanced Digital Signal Processing		3	0	0	3
ELL725	Wireless Communications		3	0	0	3
ELL731	Mixed Signal Circuit Design		3	0	0	3
ELL734	MOS VLSI design		3	0	0	3
ELL735	Analog Integrated Circuits		3	0	0	3
ELL784	Introduction to Machine Learning		3	0	0	3
ELL815	MIMO Wireless Communications		3	0	0	3
ELL833	CMOS RF IC Design		3	0	0	3

ELP725	Wireless Communication Laboratory		0	1	4	3
CRD802	Minor Project		0	0	6	3
CRD812	Major Project-II		0	0	24	12
CRD814	Major Project-III		0	0	12	6
CRL704	Sensor Array Signal Processing		3	0	0	3
CRL706	Selected Topics in Radars and Sonars		3	0	0	3
CRL707	Human & Machine Speech Communication		3	0	0	3
CRL708	Sonar Systems Engineering		3	0	0	3
CRL709	Underwater Electronic Systems		3	0	0	3
CRL712	RF and Microwave Active Circuits		3	0	0	3
CRL715	Radiating Systems for RF Communication		3	0	0	3
CRL722	RF and Microwave Solid State Devices		3	0	0	3
CRL725	Technology of RF and Microwave Solid State Devices		3	0	0	3
CRL726	RF MEMS Design and Technology		3	0	0	3
CRL727	Introduction to Quantum Electron Devices		3	0	0	3
CRL729	Sensors and Transducers		3	0	0	3
CRL731	Selected Topics in RFDT-I		3	0	0	3
CRL732	Selected Topics in RFDT-II		3	0	0	3
CRL733	Selected Topics in RFDT-III		3	0	0	3
CRL734	Selected Topics in RFDT-IV		3	0	0	3
CRP723	Fabrication Techniques for RF and Microwave Devices		1	0	4	3
CRS735	Independent Study		0	3	0	3
CRV741	Acoustic Classification using Passive Sonar		1	0	0	1
CRV742	Special Module in Radio Frequency Design and Technology-I		1	0	0	1
CRV743	Special Module in Radio Frequency Design and Technology-II		1	0	0	1

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)				Lecture courses	Contact h/week				Credits
						L	T	P	Total	
I	CRL711 CAD of RF and Microwave Circuits (3-0-2) 4	CRL718 RF and Microwave Measurement Lab (1-0-6) 4	PE-1 (3-0-0) 3	Bridge course : Any one of the following three : CRL601 Basics of Stat. Signal Analysis (2-0-2) 3/ CRL611 Basics of RF and Microwaves (2-1-0) 3/ CRL621 Fund. of Semiconductor Devices (3-0-0) 3	4	9-10	10	8-10	18-19	14
II	CRL724 RF and Microwave Measurements (3-0-0)	CRL702 Architectures and Algorithms for DSP Systems (2-0-4) 4	PE-2 (3-0-0) 3	CRD802 Minor Project (0-0-6) 3	3	8	0	10	18	13
Summer										
III	CRD811 Major Project-I (0-0-12)		PE-3 (3-0-0) 3	PE-4 (3-0-0) 3	2	6	0	12	18	12
IV Project Option OR			CRD812 (0-0-24)*		0	0	0	24	24	12
IV Course Option	PE-5 (3-0-0) 3		CRD814 Major Project-III (0-0-12) 6	OE-1 (3-0-0) 3	2	6	0	12	18	12

* **Note** : Minimum eligibility criterion for doing CRD812 (M.Tech. Project 2) in final semester leading to M.Tech. with Dissertation shall be B grade in CRD811. However, additional/higher criteria may be set CFB based on which CRC shall approve/disapprove this option for each student.

Total = 51

Master of Technology in Atmospheric-Oceanic Science and Technology

Centre for Atmospheric Sciences

The overall credits structure

Category	PC	PE	OE	Total
Credits	33	21	0	54

Program Core

ASD881	Project-I		0	0	12	6
ASL730	Introduction to Weather, Climate and Air Pollution		1	0	0	1
ASL732	Mathematical and Computational Methods for Atmospheric and Oceanic Sciences		2	0	2	3
ASL733	Physics of the Atmosphere		3	0	0	3
ASL734	Dynamics of the Atmosphere		3	0	0	3
ASL735	Atmospheric Chemistry and Air Pollution		3	0	0	3
ASL736	Science of Climate Change		3	0	0	3
ASL737	Physical and Dynamical Oceanography		3	0	0	3
ASL738	Numerical Modeling of the Atmosphere and Ocean		2	0	2	3
ASP731	Data Analysis Methods for Atmospheric and Oceanic Sciences		0	0	4	2
ASP820	Advanced Data Analysis for Weather and Climate		1	0	4	3
Total Credits			33			
ASL761	Earth System Modeling		3	0	0	3
ASL762	Air-Sea Interaction		3	0	0	3
ASL763	Coastal Ocean and Estuarine Processes		3	0	0	3
ASL765	Impacts of Climate Change and Air Pollution on Human Health		3	0	0	3
ASL821	Advanced Dynamic Meteorology		3	0	0	3
ASL822	Climate Variability		3	0	0	3
ASL823	Geophysical Fluid Dynamics		3	0	0	3
ASL824	Parameterization of Physical Processes		3	0	0	3
ASL826	Ocean Modeling		2	0	2	3
ASL827	Advanced Dynamic Oceanography		3	0	0	3
ASL851	Special Topics in Climate		3	0	0	3
ASL852	Special Topics in Oceans		3	0	0	3
ASL853	Special Topics in Atmosphere		3	0	0	3
ASL854	Special Topics in Air Pollution Studies		3	0	0	3
ASL856	Special Topics in Atmospheric and Oceanic Observations		2	0	2	3
ASP766	Atmospheric Measurements and Analysis Hands-on		1	0	4	3
ASP825	Mesoscale Modeling		0	0	6	3
ASP855	Special Topics in Atmosphere and Ocean		1	0	4	3
ASP867	Special Module in Weather Forecasting		0	0	2	1
ASP868	Special Module in Atmospheric and Oceanic Observations		0	0	2	1
ASS800	Independent Study		0	3	0	3
ASV862	Special Module in Climate		1	0	0	1
ASV863	Special Module in Oceans		1	0	0	1
ASV864	Special Module in Atmosphere		1	0	0	1
ASV865	Special Module in Air Pollution Studies		1	0	0	1
ASV866	Special Module in Atmosphere and Ocean		1	0	0	1
ASV892	An Introduction to Renewable Energy Meteorology		1	0	0	1

Program Electives

ASC869	Atmospheric and Oceanic Science Colloquium		0	1	0	1
ASD882	Project-II		0	0	24	12
ASL750	Boundary Layer Meteorology		3	0	0	3
ASL751	Dispersion of Air Pollutants		3	0	0	3
ASL752	Mesoscale Meteorology		3	0	0	3
ASL753	Atmospheric Aerosols		3	0	0	3
ASL754	Cloud Physics		3	0	0	3
ASL755	Remote Sensing of the Atmosphere and Ocean		3	0	0	3
ASL756	Synoptic Meteorology		3	0	0	3
ASL757	Tropical Weather and Climate		3	0	0	3
ASL758	General Circulation of the Atmosphere		3	0	0	3
ASL759	Land-Atmosphere Interactions		3	0	0	3
ASL760	Renewable Energy Meteorology		3	0	0	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)						Lecture courses	Contact h/week				Credits
								L	T	P	Total	
I	ASL730 Introduction to Weather, Climate and Air Pollution (1-0-0) 1	ASP731 Data Analysis Methods for Atmospheric and Oceanic Sciences (0-0-4) 2	ASL732 Mathematical and Computational Methods for Atmospheric and Oceanic Sciences (2-0-2) 3	ASL733 Physics of the Atmosphere (3-0-0) 3	ASL734 Dynamics of the Atmosphere (3-0-0) 3	ASL735 Atmospheric Chemistry and Air Pollution (3-0-0) 3	5	12	0	6	18	15
II	ASL736 Science of Climate Change (3-0-0) 3	ASL737 Physical and Dynamical Oceanography (3-0-0) 3	ASL738 Numerical Modeling of the Atmosphere and Ocean (2-0-2) 3	ASP820 Advanced Data Analysis for Weather and Climate (1-0-4) 3	PE-1 (3-0-0) 3		5	12	0	6	18	15
Summer												
III	ASD881 Project-I (Core) (0-0-12) 6	PE-2 (3-0-0) 3	PE-3 (3-0-0) 3				2	6	0	12	18	12
IV Project Option OR	ASD882 Project-II (0-0-24) 12						0	0	0	24	24	12
IV Course Option	PE-4 (3-0-0) 3	PE-5 (3-0-0) 3	PE-6 (3-0-0) 3	PE-7 (3-0-0) 3			4	12	0	0	12	12

Total = 54

Master of Technology in Electric Mobility

Centre for Automotive Research and Tribology

The overall credits structure

Category	PC	PE	OE	Total
Credits	24	18	6	48

Program Core

Program Core						CTL711	Embedded Systems for Automotive Sector	3	0	0	3
CTL703	Introduction to Electric Vehicles	3	0	0	3	CTL719	Energy Storage Systems for EVs	3	0	0	3
CTL705	Engineering of Electric Vehicles	3	0	0	3	CTL720	Vehicle System Dynamics and Control	3	0	0	3
CTL707	Batteries for Electric Vehicles: Multidisciplinary Perspectives	3	0	0	3	CTL725	Recycling, Reusability and Remanufacturing	3	0	0	3
CTL717	Power Electronics and Drives for Electric Vehicles	3	0	0	3	CTL727	Materials for Electric Vehicle Applications	3	0	0	3
CTP702	Electric Vehicle Laboratory-I	1	0	4	3	CTL729	Automotive Reliability and Life Testing	3	0	0	3
CTP704	Electric Vehicle Laboratory-II	1	0	4	3	CTL731	Automotive Noise and Condition Monitoring	3	0	0	3
CTD 801	M.Tech. Major Project Part-I	0	0	12	6	CTL732	Advanced Vehicle Propulsion	3	0	0	3
Total Credits		24				CTL717	Electric Energy Storage Systems	3	0	0	3
						CTL735	Computer Aided Analysis of Power Electronics	3	0	0	3
						CTL713	Connected and Autonomous Vehicles	3	0	0	3
						CTL736	Design of Motors for Automotive Application	3	0	0	3
						CTL741	Advanced Motors for Electric Mobility	3	0	2	4
						CTD802	M.Tech. Major Project Part-II	0	0	24	12
							Minor Project	0	0	6	3
Program Electives											
CTL704	Electrical Engineering and Tribology	3	0	0	3						
CTL709	Charging Infrastructure for EVs	3	0	0	3						

Program Electives

CTL704	Electrical Engineering and Tribology	3	0	0	3
CTL709	Charging Infrastructure for EVs	3	0	0	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)				Lecture courses	Contact h/week				Credits
						L	T	P	Total	
I	CTL703 Introduction to Electric Vehicles (3-0-0) 3	CTL705 Engineering of Electric Vehicles (3-0-0) 3	CTP702 Electric Vehicle Laboratory-I (1-0-4) 3	PE-1 (3-0-0) 3	3	10	0	4	14	12
II	CTL707 Batteries for Electric Vehicles: Multidisciplinary Perspectives 3-0-0) 3	CTL717 Power Electronics and Drives for Electric Vehicles (3-0-0) 3	CTP704 Electric Vehicle Laboratory-II (1-0-4) 3	PE-2 (3-0-0) 3	3	10	0	4	14	12
III	CTD801 Major Project Part-I (0-0-12) 6	OE-1 (3-0-0) 3	OE-2 (3-0-0) 3		2	6	0	12	18	12
M.Tech. Option I (With Major project Part – II with a Thesis Submission*)										
IV	CTD802 Major Project Part-II (0-0-24) 12				0	0	0	24	24	12
OR										
IV	PE-3 (3-0-0) 3	PE-4 (3-0-0) 3	PE-5 (3-0-0) 3	PE-6 (3-0-0) 3	4	12	0	0	12	12

* **Note** : Major Project Part - II option has a requirement of Min. CGPA 8 at the end 3rd Sem and B Grade in Major Project Part - I. In exceptional cases CRC may waive the CGPA requirement

Total = 48

Master of Technology in Biomedical Engineering

Centre for Biomedical Engineering

The overall credits structure

	Core			Elective		Total
Category	BC	CC	CP	PE	OE	Total
Credits	02	18	21	09	03	53

Bridge Courses (Core)

BMV701	Basic Electronics	1	0	0	1
BMV702	Basic Mathematics for Biologists	1	0	0	1
BMV703	Basic Biology & Physiology	1	0	0	1
BMV705	Mechanics of Biomaterials	1	0	0	1
Total Credits					2¹

¹Students shall take any two core bridge courses based on their background (Engg./ Biology) on suggestion of the program adviser.

Program Core

BML770	Fundamentals of Biomechanics	3	0	0	3
BML710	Industrial Biomaterial Technology	3	0	0	3
BML720	Medical Imaging	3	0	0	3
BML737	Application of Mathematics in Biomedical Engineering	2	0	0	2
BML760	Biomedical Ethics, Safety and Regulatory Affairs	2	0	0	2
BML740	Biomedical Instrumentation	3	0	0	3
BMP743	Basic Biomedical Laboratory	0	0	4	2
BMD801	Major Project-I	0	0	0	9
BMD802	Major Project-II	0	0	0	12
Total Credits					39

Program Electives

BVQ706	Graduate Students Sesearch Seminar	0	0	2	1
BML707	Molecular Bioengineering for Human Health	3	0	0	3
BML713	Pharmaceuticals, Efficacy and Delivery	3	0	0	3
BML714	Advanced Neuromechanics	3	0	2	4
BML734	Physiological Signal Processing	3	0	0	3
BML735	Biomedical Image and Signal Processing	2	0	2	3
BML741	Medical Device Design	2	0	4	4
BML774	Soft Tissue Characterization and Applications	3	0	2	4
BML750	Point of Care Medical Diagnostic Devices	3	0	0	3
BML771	Orthopaedic Device Design	2	0	0	2
BML772	Biofabrication	2	0	2	3
	Computational Physiology	2	0	2	3
BML781	Orthopedic Device Design and Prototyping	2	0	2	3
BML790	Modern Medicine: An Engg. Perspective	2	1	0	3
BML800	Research Techniques in Biomedical Engg.	3	0	0	3
BML810	Tissue Engineering	3	0	0	3
BML815	Selected Topics in Biomedical Engineering	2	0	0	2
BML820	Biomaterials	3	0	0	3
BML860	Nanomedicine	3	0	0	3
BML830	Biosensor Technology	3	0	2	4
BML850	Cancer: Diagnosis and Therapy	3	0	0	3
BML880	Healthcare Wearables: Design and Applications	2	0	2	3
BMV704	Fundamentals of Neuromechanics	1	0	0	1

Sem.	Courses (Number, Abbreviated Title, L-T-P, credits)						Lecture courses	Contact h/week				Credits
								L	T	P	Total	
I	BML770 Fundamentals of Biomechanics (3-0-0) 3	BML710 Industrial Biomaterial Technology (3-0-0) 3	BML720 Medical Imaging (3-0-0) 3	PE-1# (2-4)	BC-1 (1)	BC-2 (1)	6	13-15	0	0	13-15	13-15'
II	BML737 Application of Mathematics in Biomedical Engineering (2-0-0) 2	BML760 Ethics, Safety and Regulatory Affairs (2-0-0) 2	BML740 Biomedical Instrumentation (3-0-0) 3	BMP743 Basic Biomedical Laboratory (0-0-4) 2	PE-2# (2-4)	PE-3# (2-4)	5	11-13	0	4	15-17	13-15'
Summer												
III	BMD801 Major Project-I (0-0-18) 9			OE-1 (2-4)			2	2-4	0	18	20-22	11-13
IV	BMD802 Major Project-II (0-0-24) 12						0	0	0	24	24	12

PE-1, 2 & 3; OE-1: Minimum 2 to maximum 4 credits can be taken by students towards each program or open elective courses.

Total credits for three program electives and one open electives should be a minimum of 12.

! Total course credits for students in each semester should not exceed 15 for the first two semesters.

Total = 53

COL770	Advanced Artificial Intelligence	3	0	2	4	MTL785	Natural Language Processing	3	0	0	3
COL786	Advanced Functional Brain Imaging	3	0	2	4	AIL861	Special Topics in AI Applications	3	0	0	3
ELL718	Statistical Signal Processing	3	0	0	3	COL864	Special Topics in Artificial Intelligence	3	0	0	3
ELL793	Computer Vision	3	0	0	3	COL873	Special Topics in Natural Language Processing	3	0	0	3
COL780	Computer Vision	3	0	0	3	AIL862	Special Topics in Computer Vision	3	0	0	3
COL778	Principles of Autonomous Systems	3	0	2	4	COL878	Special Topics in AI-based Robotics	3	0	0	3
ELL881	Special Topics in Computers - II*	3	0	0	3	Program Electives (Other)					
ELL883	Embedded Intelligence	3	0	0	3	AID799	Minor Project	0	0	6	3
ELL885	Machine Learning for Computational Finance	3	0	0	3	AID710	Mini Project in Artificial Intelligence	0	0	6	3
ELL890	Computational Neuroscience	3	0	0	3	AIS710	Independent Study in Artificial Intelligence	0	3	0	3
ELL891	Advances in Deep Learning	3	0	0	3						
MTL733	Stochastic of Finance	3	0	0	3						

*Subject to approval of topic/specialization in a particular semester by the School Research Committee (ScRC).

Master of Technology in Cyber Security

Interdisciplinary Programme

The overall credits structure

Category	PC	PE	OE	Total
Credits	32	16-24	00	48-56

Program Core

JCD891	Minor Project	0	0	6	3
JCD892	M.Tech. Project Part-I	0	0	12	6
JCD893	M.Tech Project Part-II	0	0	24	12
COL702	Advanced Data Structures/ ELL781 Software Fundamentals for Computer Tech**	3	0	2	4*
COL759	Cryptography & Computer Security/ MTL730 Cryptography	3	0	0	3
SIL765	Network and System Security/ ELL810 Cyber Security and Information Assurance	3	0	2	4*

** Note the CSE background students will do COL702 and non-CS background students will do ELL781 or as instructed by PEC.

* Note that since ELL781 & ELL810 being 3-0-0 courses the lab component will be made up with registering for JCP781(0-0-2)1 & JCP810 (0-0-2)1 respectively.

Total Credits 32

Bridge Courses

Min. 6 credits, may be waived in exceptional cases on recommendation by PEC

SIL618	Computer Architecture	3	0	2	4
COL633	Resources Mgmt. in Computer Systems	3	0	2	4
COL671	Artificial Intelligence	3	0	2	4
COL672	Computer Networks	3	0	2	4

Streamed Program Electives (PE)

*Program Electives can be added on the recommendation by PEC

*Note - Students are required to do three electives in their respective streams and one each in the other two streams to ensure breadth.

Streamed Electives(JCS) in System Security and Cyber Forensics

JCS816	Independent Study	0	3	0	3
COL731	Advanced Compiler Techniques for Optimization, Safety and Security	3	0	2	4
COL718	Architecture of high performance Computers	3	0	2	4
COL724	Advanced Computer Networks	3	0	2	4
COL728	Compiler Design	3	0	3	4.5
COL729	Compiler Optimisation	3	0	3	4.5
COL732	Virtualization and Cloud Computing	3	0	2	4
COL733	Cloud Computing Technology Fundamental	3	0	2	4
COL768	Wireless Networks	3	0	2	4
COL851	Special Topics in Operating Systems	3	0	0	3
COL861	Special Topics in Hardware Systems	3	0	0	3
COL864	Special Topics in Artificial Intelligence	3	0	0	3
COL865	Special Topics in Computer Applications	3	0	0	3
COL867	Special Topics in High Speed Networks	3	0	0	3
COL870	Special Topics in Machine Learning	3	0	0	3
COL871	Special Topics in Programming Language	3	0	0	3
COL874	Special Topics in Compiler and Lang Impl	3	0	0	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)					Lecture courses	Contact h/week				Credits
							L	T	P	Total	
I	COL702/ ELL781+JCP781 Advanced Data Structures/ Software Fundamentals for Computer Tech (3-0-2) 4 / (3-0-0)3+ (0-0-2)1	Bridge-1 (3-0-2)4	COL759/MTL730 Cryptography & Computer Security / Cryptography (3-0-0) 3	PE-1 (3-4)		3-4	9-12	0	2-6	11-18	10-15
II	JCD891 Minor Project (0-0-6) 3	Bridge-1 (3-0-2)4	SIL765/ELL810 +JCP810 Cyber Security and Information Assurance/ Network and System Security (3-0-2) 4 / (3-0-0)3+ (0-0-2)1	PE-2 (3-4)		2-3	6-9	0	8-12	14-21	10-15
III	JCD892 MTP-I (0-0-12) 6	PE-3 (3-4)	PE-4 (3-4)			2	6	0	12-16	18-24	12-15
IV	JCD893 MTP-II (0-0-24) 12	PE-5 (3)				1	3	0	20-22	23-25	15

NOTE: 1. Registration for MTP-II has a requirement of Min. CGPA 7.5 at the end 3rd sem and B Grade in JCD892. In exceptional cases PEC may waive the CGPA requirement

2. MTP – II can also be done in a collaborative manner with Industry/University.

In future joint degree programmes with other institutes/universities can be considered.

We are proposing M. Tech in Cyber security with 20 seats with additional Full-time/Part-time sponsored candidates from Industry/Government. The admissions will be done through GATE in CS/EC/MA/EE as per Institute norms for other interdisciplinary programs.

Total = 48-56

COL876	Special Topics in Formal Methods	3	0	0	3	ELL881	Special Topics in Computers-II	3	0	0	3
COV881	Special Module in Hardware Systems	1	0	0	1	MTL729	Comp. Algebra & its Applications	3	0	0	3
COV882	Special Module in Software Systems	1	0	0	1	MTL735	Advanced Number Theory	3	0	0	3
COV887	Special Module in High Speed Networks	1	0	0	1	MTL744	Mathematical Theory of Coding	3	0	0	3
COL886	Special Topics in Operating Systems	3	0	0	3	MTL782	Data Mining	3	0	2	4
ELV710	Special Module in Cyber Security	1	0	0	1	MTL811	Mathematical Foundation of AI	3	0	0	3
ELL714	Basic Information Theory	3	0	0	3	SIL771	Special Topics in Cyber Security	3	0	0	3
ELL785	Computer Communication Networks	3	0	0	3	SIL773	Digital Watermarking and Steganography	3	0	0	3
ELL800	Numerical Linear Algebra and Optimization in Engineering	3	0	0	3	SIL775	Biometric Security	3	0	0	3
ELL880	Special Topics in Computers-I	3	0	0	3	SIL779	Data Privacy	3	0	0	3
ELL881	Special Topics in Computers-II	3	0	0	3	SIV810	Special Module in Cyber Security	1	0	1	1.5
ELL892	Internet Technologies	3	0	0	3	SIV812	Special Module in Computer Forensics	1	0	1	1.5
ELL893	Cyber-Physical Systems	3	0	0	3	SIV814	Special Module in Application Security	1	0	1	1.5
ELL895	Network Security	3	0	0	3	SIL763	Introduction to Blockchains, Cryptocurrencies and Smart contracts	3	0	2	4
ELL897	Network Management	3	0	0	3	Streamed Electives(JCS) in Embedded System and Hardware Security					
MDL804	Behavioral Finance	1.5	0	0	1.5	JCS816	Independent Study	0	3	0	3
MSL852	Network System: Applications and Mgmt.	3	0	0	3	COL718	Architecture of High-Performance Computer	3	0	2	4
MSL855	Electronic Commerce	3	0	0	3	COL719	Synthesis of Digital Systems	3	0	2	4
MSL878	Electronic Payments	1.5	0	0	1.5	COL720	Real Time Systems	3	0	2	4
MTL744	Mathematical Theory of Coding	3	0	0	3	COL750	Foundations of Automatic Verification	3	0	2	4
SIL771	Special Topics in Cyber Security	3	0	0	3	COL788	Advanced Topics in Embedded Computing	3	0	0	3
SIL773	Digital Watermarking and Steganography	3	0	0	3	COL861	Special Topics in Hardware Systems	3	0	0	3
SIL775	Biometric Security	3	0	0	3	COL862	Special Topics in Software Systems	3	0	0	3
SIL777	Secure Programming Methodologies	3	0	2	4	COV889	Special Module in Concurrency	1	0	0	1
SIL779	Data Privacy	3	0	0	3	COL812	System Level Design and Modelling	3	0	0	3
SIV810	Special Module in Cyber Security	1	0	1	1.5	ELV710	Special Module in Cyber Security	1	0	0	1
SIV812	Special Module in Computer Forensics	1	0	1	1.5	ELL720	Advanced Digital Signal Processing	3	0	0	3
SIV814	Special Module in Application Security	1	0	1	1.5	ELL733	Digital ASIC Design	3	0	2	4
SIV895	Special Module on Intelligent Information Processing	1	0	0	1	ELL748	System-on-Chip Design and Test	3	0	0	3
Streamed Electives(JCS) in Cryptography and Cryptanalysis						ELL765	Smart Grid Technology	3	0	0	3
JCS816	Independent Study	0	3	0	3	ELL772	Planning and Operation of Smart Grid	3	0	0	3
COL730	Parallel Programming	3	0	2	4	ELL787	Embedded Systems and Applications	3	0	0	3
COL774	Machine Learning	3	0	2	4	ELL880	Special Topics in Computers-I	3	0	0	3
COL864	Special Topics in Artificial Intelligence	3	0	0	3	ELL881	Special Topics in Computers-II	3	0	0	3
COL865	Special Topics in Computer Applications	3	0	0	3	ELL883	Embedded Intelligence	3	0	0	3
COL870	Special Topics in Machine Learning	3	0	0	3	MSL855	Electronic Commerce	3	0	0	3
COL872	Special Topics in Cryptography	3	0	0	3	MSL878	Electronic Payments	1.5	0	0	1.5
COV878	Special Module in Machine Learning	1	0	0	1	MDL804	Behavioral Finance	1.5	0	0	1.5
COV884	Special Module in Artificial Intelligence	1	0	0	1	SIV810	Special Module in Cyber Security	1	0	1	1.5
ELL710	Coding Theory	3	0	0	3	SIV814	Special Module in Application Security	1	0	1	1.5
ELL712	Digital Communications	3	0	0	3	SIL771	Special Topics in Cyber Security	3	0	0	3
ELL711	Signal Theory	3	0	0	3	SIL773	Digital Watermarking and Steganography	3	0	0	3
ELL718	Statistical Signal Processing	3	0	0	3	SIL775	Biometric Security	3	0	0	3
ELL720	Advanced Digital Signal Processing	3	0	0	3	SIL777	Secure Programming Methodologies	3	0	2	4
ELL800	Numerical Linear Algebra and Optimization in Engineering	3	0	0	3	SIL781	Secure Hardware-based Systems Design	3	0	2	4
ELL880	Special Topics in Computers-I	3	0	0	3	SIL763	Introduction to Blockchains, Cryptocurrencies and Smart Contracts	3	0	2	4

Master of Technology in Energy Studies

Interdisciplinary Programme

The overall credits structure

Category	PC	PE	OE	Total
Credits	30	18	06	54

Program Core

ESL710	Energy, Ecology and Environment	3	0	0	3
ESL711	Fuel Technology	3	0	0	3
ESL720	Energy Conservation	3	0	0	3
ESL730	Direct Energy Conversion	3	0	0	3
ESL740	Non-conventional Sources of Energy	3	0	0	3
ESL750	Economics and Planning of Energy Systems	3	0	0	3
ESL760	Heat Transfer	3	0	0	3
ESP713	Energy Laboratory	0	0	6	3
JSD801	Major Project Part-I (JES)	0	0	12	6
Total Credits		30			

ESL770	Solar Energy Utilization	3	0	0	3
ESL779	Li-ion Batteries: Technology and Thermal Management	3	0	0	3
ESL796	Operation and Control of Electrical Energy Systems	3	0	0	3
ESL810	MHD Power Generation	3	0	0	3
ESL840	Solar Architecture	3	0	0	3
ESL850	Solar Refrigeration and Air Conditioning	3	0	0	3
ESL860	Electrical Power Systems Analysis	3	0	0	3
ESL870	Fusion Energy	3	0	0	3
ESL871	Advanced Fusion Energy	3	0	0	3
ESL880	Solar Thermal Power Generation	3	0	0	3
JSD799	Minor Project (JES)	0	0	6	3
JSD802	Major Project Part-II (JES)	0	0	24	12
JSS801	Independent Study (JES)	0	3	0	3

Program Electives

ESL714	Power Plant Engineering	3	0	0	3
ESL718	Power Generation, Transmission and Distribution	3	0	0	3
ESL722	Integrated Energy Systems	3	0	0	3
ESL732	Bioconversion and Processing of Waste	3	0	0	3
ESL734	Nuclear Energy	3	0	0	3
ESL737	Plasma Based Materials Processing	3	0	0	3
ESL746	Hydrogen Energy	3	0	0	3
ESL755	Solar Photovoltaic Devices and Systems	3	0	0	3
ESL768	Wind Energy and Hydro Power Systems	3	0	0	3

Program Electives (other Departments)

MEL815	Applied Combustion	2	0	2	4
MEL816	Analysis of I.C. Engine Processes	3	0	2	4
EEL748	Power Quality	3	0	0	3
EEL899	Distribution Automation	3	0	0	3
CHL722	Fundamentals of Fuel Cell Technology	3	0	2	4

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)					Lecture courses	Contact h/week				Credits
							L	T	P	Total	
I	ESL740 Non-conventional Sources of Energy (3-0-0) 3	ESL711 Fuel Technology (3-0-0) 3	ESL760 Heat Transfer (3-0-0) 3	ESL720 Energy Conservation (3-0-0) 3	PE-1 (3-0-0) 3	5	15	0	0	15	15
II	ESL750 Economics and Planning of Energy Systems (3-0-0) 3	ESL710 Energy, Ecology and Environment (3-0-0) 3	ESP713 Energy Laboratory (0-0-6) 3	ESL730 Direct Energy Conversion (3-0-0) 3	PE-2 (3-0-0) 3	4	12	0	6	18	15
Summer	JSD801 Major Project Part-1 (JES)					0					
III	JSD801 Major Project Part-I (JES) (0-0-12) 6	OE-1 (3-0-0) 3		OE-2 (3-0-0) 3		2	6	0	12	18	12
IV	JSD802 Major Project Part-II (JES) (0-0-24) 12					0	0	0	24	24	12
	PE-3 (3-0-0) 3	PE-4 (3-0-0) 3	PE-5 (3-0-0) 3	PE-6 (3-0-0) 3		4	12	0	0	12	

NOTE: for part time students, the above credits would be completed over 6 semesters with the Major Project Part I & II being offered in the 5th and 6th semesters respectively

Total = 54

Master of Technology in Energy & Environment Technologies and Management

Interdisciplinary Programme

The overall credits structure

Category	PC	PE	OE	Total
Credits	30	18	06	54
Bridge (Audit) Courses	1	0	0	1

Program Core

ESL711	Fuel Technology	3	0	0	3
ESL715	Electrical Energy Management Systems	3	0	0	3
ESL727	Energy and Environment	3	0	0	3
ESL740	Non-conventional Sources of Energy	3	0	0	3
ESL774	Quantitative Methods for Energy Management and Planning	3	0	0	3
ESL776	Industrial Energy and Environment Analysis	3	0	0	3
ESL784	Cogeneration and Energy Efficiency	3	0	0	3
ESP728	Energy & Environment Laboratory	0	0	6	3
ESD801	Major Project Part-I (ESN)	0	0	12	6

Compulsory (Audit) Courses (CA)

ESN704	Basic Thermal Engineering	1	0	0	0
ESN712	Basic Electrical Engineering	1	0	0	0
Total Credits		30			

Program Electives

ESL721	Pulse Width Modulation Techniques and AC Motor Drives	3	0	0	3
ESL724	Energy Audit	3	0	0	3
ESL733	Organic Waste to Energy Conversion Technology	3	0	0	3
ESL735	Hazardous Waste Management	3	0	0	3
ESL738	Power System Planning and Operation	3	0	0	3
ESL742	Economics and Financing of Renewable Energy Systems	3	0	0	3
ESL743	Optimal Design of Energy Systems	3	0	0	3
ESL744	Plasmas for Energy and Environment	3	0	0	3
ESL745	Environmental Audit and Impact Assessment	3	0	0	3
ESL749	Developing Energy Efficiency and Renewable Energy Projects	3	0	0	3
ESL752	Carbon Audit and Management	3	0	0	3
ESL754	Nanomaterials for Energy Applications	3	0	0	3
ESL756	Energy Policy and Planning	3	0	0	3
ESL758	Carbon Capture and Storage	3	0	0	3
ESL764	Environmental Economics	3	0	0	3
ESL766	Environmental Regulations	3	0	0	3
ESL771	Instrumentation and Control in Energy Systems	3	0	0	3
ESL772	Energy Storage	3	0	0	3
ESL775	Liquid Sprays for Energy Sector and Industrial	3	0	0	3

Applications

ESL779	Li-ion Batteries: Technology and Thermal Management	3	0	0	3
ESL781	Alternative Fuels for Aircraft and Rocket Propulsion	3	0	0	3

Program Electives/Open Electives

ESL773	Battery Storage	3	0	0	3
ESL778	Industrial Waste Management and Recycling	3	0	0	3
ESL780	Zero Emission Vehicles	3	0	0	3
ESL782	Emission Control in Internal Combustion Engines	3	0	0	3
ESL796	Operation & Control of Electrical Energy Systems	3	0	0	3
ESL797	Operation of Electrical Energy Systems with Large Scale Integration of Renewable Energy Sources.	3	0	2	4
ESL798	Distributed and Decentralized Energy Systems	3	0	0	3
ESL804	Pollution Control in Power Plants	3	0	0	3
ESL875	Alternative Fuels for Transportation	3	0	0	3
ESV891	Special Topics on Emerging Trends in Energy and Environmental Technologies	1	0	0	1
ESD799	Minor Project (ESN)	0	0	6	3
ESS801	Independent Study (ESN)	0	3	0	3
ESD802	Major Project Part-II (ESN)	0	0	24	12

Program Electives (other Departments)

CLL723	Hydrogen energy and Fuel Cell Technology	3	0	0	3
CLL724	Environmental Engg. and Waste Mgmt.	3	0	0	3
CLL725	Air Pollution Control Engineering	3	0	0	3
CLL706	Petroleum Production Engineering	3	0	0	3
CLL794	Petroleum Refinery Engineering	3	0	0	3
CVL820	Environmental Impact Assessment	3	0	0	3
CVL822	Emerging Technologies for Environmental Mgmt.	3	0	0	3
CVL823	Thermal Techniques for Waste Mgmt.	3	0	0	3
CVL824	Life Cycle Analysis & Design for Environment	3	0	0	3
CVL721	Solid Waste Engineering	3	0	0	3
CVL720	Air Pollution and control	3	0	0	3
CVL847	Transportation Economics	3	0	0	3
ELL765	Smart Grid Technology	3	0	0	3
MCL812	Combustion	3	0	0	3
MCL825	Design of Wind Power Farms	3	0	0	3
PYL727	Energy Materials and Devices	3	0	0	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, credits)						Lecture courses	Contact h/week				Credits
								L	T	P	Total	
I	ESL740 Non-conventional Sources of Energy (3-0-0) 3	ESL711 Fuel Technology (3-0-0) 3	ESL715 Electrical Energy Mgmt. Systems (3-0-0) 3	ESL727 Energy and Environment (3-0-0) 3	PE-1 (3-0-0) 3	ESN704 Basic Thermal Engineering/ ESN712 Basic Electrical Engineering (1-0-0) 0	6	16	0	0	16	15
II	ESL776 Industrial Energy and Environment Analysis (3-0-0) 3	ESL784 Cogeneration and Energy Efficiency (3-0-0) 3	ESP728 Energy & Environment Laboratory (0-0-6) 3	ESL774 Quantitative Methods for Energy Mgmt. and Planning (3-0-0) 3	PE-2 (3-0-0) 3		4	12	0	6	18	15
Summer	ESD801 Major Project Part-I											
III	ESD801 Major Project Part-I (0-0-12) 6	OE-1 (3-0-0) 3	OE-2 (3-0-0) 3				2	6	0	12	18	12
IV	ESD802 Major Project Part-II (0-0-24) 12						0	0	0	24	24	12
	PE-3 (3-0-0) 3	PE-4 (3-0-0) 3	PE-5 (3-0-0) 3	PE-6 (3-0-0) 3			4	12	0	0	12	

NOTE: for part time students, the above credits would be completed over 6 semesters with the Major Project Part I & II being offered in the 5th and 6th semesters respectively

Total = 54

Master of Technology in Renewable Energy Technologies and Management

Interdisciplinary Programme

The overall credits structure

Category	PC	PE	OE	Total
Credits	27	24	0	51
Bridge (Audit) Courses	0-4	0	0	0-4

Program Core

ESL739	Bio-energy : Resources, Technologies and Applications	3	0	0	3	ESL744	Plasma for Energy and Environment	3	0	0	3
ESL742	Economics and Financing of Renewable Energy Systems	3	0	0	3	ESL746	Hydrogen Energy	3	0	0	3
ESL753	Solar Thermal Technologies and Systems	3	0	0	3	ESL749	Developing Energy Efficiency and Renewable Energy Projects	3	0	0	3
ESL755	Solar Photovoltaic Devices and Systems	3	0	0	3	ESL751	Renewable Energy Resource Assessment and Forecasting	3	0	0	3
ESL768	Wind Energy and Hydro Power Systems	3	0	0	3	ESL752	Carbon Audit and Management	3	0	0	3
ESP705	Renewable Energy Laboratory	0	0	6	3	ESL756	Energy Policy and Management	3	0	0	3
ESD798	Minor Project (ESR)	0	0	6	3	ESL757	Energy Policy and Planning	3	0	0	3
ESD851	Major Project-I (ESR)	0	0	12	6	ESL771	Instrumentation and Control in Energy Systems	3	0	0	3
						ESL772	Energy Storage	3	0	0	3
						ESL773	Battery Storage	3	0	0	3

Bridge (Audit) Courses (BA): (Based on Student's background and preparedness)

ESN702	Introduction to Project Management	1	0	0	0	ESL774	Quantitative Methods for Energy Mgmt. & Planning	3	0	0	3
ESN703	Technical Writing	1	0	0	0	ESL775	Liquid Sprays for Energy Sector and Industrial Applications	3	0	0	3
ESN704	Basic Thermal Engineering (for Non-mechanical Students)	1	0	0	0	ESL779	Li-ion Batteries: Technology and Thermal Management	3	0	0	3
ESN712	Basic Electrical Engineering (for Non-electrical Students)	1	0	0	0	ESL780	Zero Emission Vehicles	3	0	0	3
ESN791	Applied Mathematics and Computational Methods	1	0	0	0	ESL781	Alternative Fuels for Aircraft and Rocket Propulsion	3	0	0	3
	Total Credits				27	ESL790	Policy and Regulatory Aspects of Power System Operation with Increasing Renewable Energy Share	3	0	0	3
						ESL791	Renewable energy Integration and Power Systems	3	0	0	3
						ESL796	Operation and Control of Electrical Energy Systems	3	0	0	3
						ESL797	Operation of Electrical Energy Systems with Large Scale Integration of Renewable Energy Sources	3	0	2	4

Program Electives

ESL718	Power Generation, Transmission and Distribution	3	0	0	3	ESL798	Distributed and Decentralized Energy Systems	3	0	0	3
ESL721	Pulse Width Modulation Techniques and AC Motor Drives	3	0	0	3	ESL799	Essentials of Electrical Power Generation by Renewable Energy Sources	2	0	2	3
ESL729	Renewable Energy and Environment	3	0	0	3	ESL840	Solar Architecture	3	0	0	3
ESL730	Direct Energy Conversion	3	0	0	3	ESL842	Negative CO ₂ Emission Technologies	3	0	0	3
ESL732	Bioconversion and Processing of Waste	3	0	0	3						
ESL737	Plasma Based Materials Processing	3	0	0	3						

FULL TIME (4-Semester Schedule)

Sem.	Courses (Number, Abbreviated Title, L-T-P, credits)							Lecture courses	Contact h/week				Credits
	L	T	P	Total									
I	ESL739 Bio-energy : Resources, Technologies & Applications (3-0-0) 3	ESL742 Economics and Financing of Renewable Energy Systems (3-0-0) 3	ESL755 Solar Photovoltaic Devices and Systems (3-0-0) 3	ESL753 Solar Thermal Technologies and Systems (3-0-0) 3	PE-1 (3-0-0) 3	BA-1 (1-0-0) 0 [If required]	BA-2 (1-0-0) 0 [If required]	5-7	15-17	0	0	15-17	15
II	ESP705 Renewable Energy Laboratory (3-0-0) 3	ESL768 Wind Energy and Hydro Power Systems (3-0-0) 3	PE-2 (3-0-0) 3	PE-3 (3-0-0) 3	PE-4 (3-0-0) 3	BA-3 (1-0-0) 0 [If required]	BA-4 (1-0-0) 0 [If required]	4-6	12-14	0	6	18-20	15
Summer	Internship and ESD801 [Major Project Part-I (0-0-12) 6]												
III	ESD851 Major Project Part-I (ESR) (0-0-12) 6 [and Internship]					ESD798 Minor Project (ESR) (0-0-6) 3		0	0	0	18	18	09
IV	ESD852 Major Project Part-II (ESR) (0-0-24) 12 [with option to carry out in the home country for a foreign national student with provision for presentations through video conference]							0 or 4	0 or 12	0	0 or 24	12 or 24	12
	PE-5 (3-0-0) 3	PE-6 (3-0-0) 3		PE-7 (3-0-0) 3		PE-8 (3-0-0) 3							

* Possibility of inviting foreign students about a month in advance to take bridge courses is being explored

Total = 51

ESL845	Net Zero Energy Buildings	3	0	0	3	ESS851	Independent Study (ESR)	0	3	0	3
ESL850	Solar Refrigeration and Air-conditioning	3	0	0	3	ESD852	Major Project Part – 2 (ESR)	0	0	24	12
ESL852	Emerging Materials for Next Generation Photovoltaic Applications	3	0	0	3	Program Electives (From outside Centre)					
ESL855	Solar Photovoltaic Power Generation	3	0	0	3	ASL760	Renewable Energy Meteorology	3	0	0	3
ESL875	Alternative Fuels for Transportation	3	0	0	3	CLL723	Hydrogen Energy and Fuel Cell Technology	3	0	0	3
ESL880	Solar Thermal Power Generation	3	0	0	3	EEL758	Power Quality	3	0	0	3
ESL885	Solar Industrial Process Heating	3	0	0	3	EEL765	Smart Grid Technology	3	0	0	3
ESP706	Renewable Energy Simulation Laboratory	0	0	6	3	MCL825	Design of Wind Power Farms	3	0	2	4
ESV891	Special Topics on Emerging Trends in Energy and Environmental Technologies	1	0	0	1	PYL727	Energy Materials and Devices	3	0	0	3

PART TIME (6-Semester Schedule)

Sem.	Courses (Number, Abbreviated Title, L-T-P, credits)				Lecture courses	Contact h/week				Credits
						L	T	P	Total	
I	ESL739 Bio-energy : Resources, Technologies & Applications (3-0-0) 3	ESL742 Economics and Financing of Renewable Energy Systems (3-0-0) 3	ESL753 Solar Thermal Technologies and Systems (3-0-0) 3	BA-1 (1-0-0) 0 [If required]	3-4	9-10	0	0	9-10	9
II	ESL768 Wind Energy and Hydro Power Systems (3-0-0) 3	ESP705 Renewable Energy Laboratory (0-0-6) 3	ESL755 Solar Photovoltaic Devices and Systems (3-0-0) 3	BA-2 (1-0-0) 0 [If required]	2-3	6-7	0	6	9-10	9
Summer										
III	PE-1 (3-0-0) 3	PE-2 (3-0-0) 3	BA-3 (1-0-0) 0 [If required]		2-3	6-7	0	0	6-7	6
IV	PE-3 (3-0-0) 3	PE-4 (3-0-0) 3	BA-4 (1-0-0) 0 [If required]		2-3	6-7	0	6	6-7	6
Summer	Internship and ESD851 [Major Project Part-I (0-0-12) 6]									
V	ESD851 Major Project Part-I (0-0-12) 6 [and Internship]			ESD798 Minor Project (0-0-6) 3	0	0	0	18	18	9
VI	ESD852 Major Project Part (0-0-24) 12				0 or 4	0 or 12	0	0 or 24	12 or 24	12
	PE-5 (3-0-0) 3	PE-6 (3-0-0) 3	PE-7 (3-0-0) 3	PE-8 (3-0-0) 3						

Total = 51

Master of Technology in Industrial Tribology and Maintenance Engineering

Interdisciplinary Programme

The overall credits structure

Category	PC	PE	OE	Total
Credits	33	09	06	48

Program Core

ITL702	Diagnostic Maintenance and Condition Monitoring	3	0	2	4	ITL710	Design of Tribological Elements	3	0	0	3
ITL703	Fundamentals of Tribology	3	0	2	4	ITL711	Reliability, Availability and Maintainability (RAM) Engineering	3	0	0	3
ITL705	Materials for Tribological Applications	3	0	0	3	ITL717	Corrosion and its Control	3	0	0	3
ITL714	Failure Analysis and Repair	3	0	2	4	ITL730	Lubricants	2	0	2	3
JIT801	Major Project Part-I (JIT)	0	0	12	6	ITL740	Risk Analysis and Safety	2	1	0	3
JIT802	Major Project Part-II (JIT)	0	0	24	12	ITL752	Bulk Materials Handling	2	0	2	3
Total Credits		33				ITL760	Noise Monitoring and Control	2	0	2	3

Program Electives

ITL709	Maintenance Planning and Control	3	0	0	3	ITL810	Bearing Lubrication	3	0	0	3
						JIS800	Independent Study	0	3	0	3
						JID800	Minor Project	0	0	6	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)							Lecture courses	Contact h/week				Credits
									L	T	P	Total	
I	ITL703 Fundamentals of Tribology (3-0-2) 4	ITL705 Materials for Tribological Applications (3-0-0) 3				PE-1 (3-0-0) 3	OE-1 (3-0-0) 3	4	12	0	2	14	13
II	ITL702 Diagnostic Maintenance & Condition Monitoring (3-0-2) 4	ITL714 Failure Analysis & Repair (3-0-2) 4				PE-2 (3-0-0) 3	PE-3 (3-0-0) 3	4	12	0	4	16	14
III	JID801 Major Project Part-I (JIT) (0-0-12) 6					OE-2 (3-0-0) 3		1	3	0	12	15	9
IV	JID802 Major Project Part-II (JIT) (0-0-24) 12							0	0	0	24	24	12

Total = 48

Master of Technology in Instrument Technology

Interdisciplinary Programme

The overall credits structure

Category	PC	PE	OE	Total
Credits	39	15	0	54

Program Core

DSD801	Major Project Part-I	0	0	12	6		PYL780	Diffractive and micro optics	3	0	0	3
DSD802	Major Project Part-II	0	0	24	12		PYL790	Integrated Optics	3	0	0	3
DSL711	Sensors and Transducers	3	0	0	3		PYL792	Optical Electronics	3	0	0	3
DSL712	Electronic Techniques for Signal Conditioning and Interfacing	3	0	0	3		PYL793	Photonic Devices	3	0	0	3
DSL714	Instrument Design and Simulations	2	0	2	3		PYP761	Optical fabrication and Metrology Laboratory	0	0	6	3
DSL731	Optical Components and Basic Instruments	3	0	0	3		CRL725	Technology of RF and Microwave Solid State Devices	3	0	0	3
DSL734	Laser Based Instrumentation	3	0	0	3		DSC812	Term Paper and Seminar	0	3	0	3
DSP703	Instrument Technology Laboratory-I	0	0	6	3		DSL601	Electronic Components and Circuits (for Non-electrical Students only)	3	0	0	3
DSP704	Instrument Technology Laboratory-II	0	0	6	3		DSL603	Material and Mechanical Design (for electrical students only)	3	0	0	3
Total Credits		39					DSL722	Precision Measurement Systems	3	0	0	3

Program Electives

ELL746	Biomedical Electronics	3	0	0	3		DSL733	Optical Material and Optical Techniques in Instrumentation	2	0	2	3
ELL783	Operating System	3	0	2	4		DSL737	Display Devices and Technology	3	0	0	3
ELL787	Embedded Systems and Applications	3	0	0	3		DSL740	Instrument Organization and Ergonomics	2	0	2	3
ELL735	Analog Integrated Circuits	3	0	0	3		DSL742	Integrated Quantum Photonics	3	0	0	3
ELL734	MOS VLSI design	3	0	0	3		DSL750	Opto-electronic Detectors and Image Sensors	3	0	0	3
ELL784	Introduction to Machine Learning	3	0	0	3		DSL755	Sensing and Imaging Techniques	3	0	0	3
ELL883	Embedded Intelligence	3	0	0	3		DSL811	Selected Topics in Instrumentation-I	3	0	0	3
MCL705	Experimental Methods	3	0	2	4		DSL814	Selected Topics in Instrumentation-II	3	0	0	3
MCL749	Mechatronic Product Design	3	0	2	4		DSL815	Special Topics in Instrumentation	1	0	0	1
MCL781	Machining Processes and Analysis	3	0	2	4		DSP705	Advanced Instrument Technology Lab	0	0	6	3
MCL783	Automation in Manufacturing	3	0	2	4		DSS720	Independent Study	0	3	0	3
PYL755	Basic optics and optical instrumentation	3	0	0	3							

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)						Lecture courses	Contact h/week				Credits
								L	T	P	Total	
I	DSP703 Instrument Technology Laboratory-I (0-0-6) 3	DSL711 Sensors and Transducers (3-0-0) 3	DSL731 Optical Components and Basic Instruments (3-0-0) 3	DSL601*/ DSL603** Electronic Components and Circuits/Material and Mechanical Design (3-0-0) 3	PE-1 (3-0-0) 3		4	12	0	6	18	15
II	DSP704 Instrument Technology Laboratory-II (0-0-6) 3	DSL712 Electronic Techniques for Signal Conditioning and Interfacing (3-0-0) 3	DSL714 Instrument Design and Simulations (2-0-2) 3	DSL734 Laser Based Instrumentation (3-0-0) 3	PE-2 (3-0-0) 3		4	11	0	8	19	15
Summer												
III	DSD801 (0-0-12) 6	PE-3 (3-0-0) 3	PE-4 (3-0-0) 3				2	6	0	12	18	12
IV	DSD802 (0-0-24) 12						0	0	0	24	24	12

*For students with non-electrical Engineering, Electronics, Instrumentation, Electronics and Communication background.

**For students with Electrical Engineering, Electronics, Instrumentation, Electronics and Communication background.

Total = 54

Master of Technology in Optoelectronics and Optical Communication

Interdisciplinary Programme

The overall credits structure

Category	PC	PE	OE	Total
Credits	24	27	0	51

Program Core

ELL717	Optical Communication Systems	3	0	0	3	ELL819	Introduction to Plasmonics	3	0	0	3
ELL727	Digital Communication & Information Systems	3	0	0	3	ELL820	Photonic Switching and Networking	3	0	0	3
JOD801	Major Project Part-I	0	0	12	6	JOD802	Major Project Part-II	0	0	24	12
JOP791	Laboratory-I (Fiber Optics Lab/Opt. Comm. Lab)	0	0	6	3	JOL793	Selected Topics-I	3	0	0	3
JOP792	Laboratory-II (Fiber Optics Lab/Opt. Comm. Lab)	0	0	6	3	JOL794	Selected Topics-II	3	0	0	3
PYL791	Fiber Optics	3	0	0	3	JOS795	Independent Study	0	3	0	3
PYL792	Optical Electronics	3	0	0	3	JOV796	Selected Topics in Photonics	1	0	0	1
Total Credits		24									
						PYL748	Quantum Optics	3	0	0	3
						PYL749	Quantum Information and Computation	3	0	0	3
						PYL756	Fourier Optics and Holography	3	0	0	3
						PYL757	Statistical and Quantum Optics	3	0	0	3
						PYL758	Advanced Quantum optics and application	3	0	0	3
						PYL760	Biomedical optics and Bio-photonics	3	0	0	3
						PYL770	Ultra-fast Optics and Applications	3	0	0	3
						PYL771	Green Photonics	3	0	0	3
						PYL790	Integrated Optics	3	0	0	3
						PYL793	Photonic Devices	3	0	0	3
						PYL795	Optics and Lasers	3	0	0	3
						PYL891	Fiber Optic Components and Devices	3	0	0	3
						PYL892	Guided Wave Photonic Sensors	3	0	0	3

Program Electives

ELL716	Telecommunication Switching and Transmission	3	0	0	3						
ELL720	Advanced Digital Signal Processing	3	0	0	3						
ELL723	Broadband Communication Systems	3	0	0	3						
ELL726	Nano-Photonics and Plasmonics	3	0	0	3						
ELL728	Optoelectronic Instrumentation	3	0	0	3						
ELL738	Micro and Nano Photonics	3	0	0	3						
ELL785	Computer Communication Networks	3	0	0	3						
ELL814	Wireless Optical Communications	3	0	0	3						
ELL817	Access Networks	3	0	0	3						

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)						Lecture courses	Contact h/week				Credits
								L	T	P	Total	
I	PYL791 Fibre Optics (3-0-0) 3	ELL727 Digital Comm. & Information Systems (3-0-0) 3	JOP791 Laboratory-I (Fibre Optics Lab/Opt. Comm. Lab) (0-0-6) 3	PYL/ELL PE-1 (3-0-0) 3	PYL/ELL PE-2 (3-0-0) 3		4	12	0	6	18	15
II	PYL792 Optical Electronics (3-0-0) 3	ELL717 Optical Communication System (3-0-0) 3	JOP792 Laboratory-II (Fibre Optics Lab/Opt. Comm. Lab) (0-0-6) 3	PYL/ELL PE-3 (3-0-0) 3	PYL/ELL PE-4 (3-0-0) 3		4	12	0	6	18	15
Summer												
III	PYL/ELL PE-5 (3-0-0) 3	JOD801 Major Project Part-I (0-0-12) 6					1	3	0	12	15	9
IV	JOD802 Major Project Part-II Or 12 Credits PE Courses in lieu of Major Project Part-II) (0-0-24) 12						0	0	0	24	24	12

Total = 51

Master of Technology in Telecommunication Technology & Management

Interdisciplinary Programme

The overall credits structure

Category	PC	PE	OE	Total
Credits	36	12	0	48

Program Core

ELL711	Signal Theory	3	0	0	3
ELL712	Digital Communications	3	0	0	3
ELL785	Computer Communication Networks	3	0	0	3
BSP710	Communication & Signal Processing Tech. Lab.	0	1	4	3
ELP725	Wireless Communication Laboratory	0	1	4	3
MSL723	Telecommunications Systems Management	3	0	0	3
JTD801	Major Project-I	0	0	12	6
JTD802	Major Project-II	0	0	24	12
Total Credits					36

Program Electives

Streamed Electives (JTM) in (Signal and Information Processing)

CRL707	Human & Machine Speech Communication	3	0	0	3
ELL706	Optimization for Electrical Engineers	3	0	0	3
ELL715	Digital Image Processing	3	0	2	4
ELL718	Statistical Signal Processing	3	0	0	3
ELL719	Detection and Estimation Theory	3	0	0	3
ELL720	Advanced Digital Signal Processing	3	0	0	3
ELL724	Multichannel Signal Processing	3	0	0	3
ELL729	Stochastic Control and Reinforcement Learning	3	0	0	3
ELL784	Introduction to Machine Learning	3	0	0	3
ELL786	Multimedia Systems	3	0	0	3
ELL792	Computer Graphics	3	0	0	3
ELL793	Computer Vision	3	0	0	3
ELL823	Selected Topics in Information Processing-I	3	0	0	3
JTD792/ SMD792	Minor Project	0	0	6	3

Streamed Electives (JTM) in (Communication Systems and Networks)

COL724	Advanced Computer Networks	3	0	2	4
ELL706	Optimization for Electrical Engineers	3	0	0	3
ELL710	Coding Theory	3	0	0	3
ELL714	Basic Information Theory	3	0	0	3
ELL716	Telecommunication Switching & Transmission	3	0	0	3
ELL717	Optical Communication Systems	3	0	0	3
ELL718	Statistical Signal Processing	3	0	0	3
ELL719	Detection and Estimation Theory	3	0	0	3
ELL720	Advanced Digital Signal Processing	3	0	0	3
ELL723	Broadband Communication Systems	3	0	0	3
ELL729	Stochastic Control and Reinforcement Learning	3	0	0	3
ELL813	Advanced Information Theory	3	0	0	3
ELL814	Wireless Optical Communications	3	0	0	3
ELL815	MIMO Wireless Communications	3	0	0	3
ELL816	Satellite Communication	3	0	0	3
ELL817	Access Networks	3	0	0	3
ELL818	Telecommunication Technologies	3	0	0	3
ELL820	Photonic Switching and Networking	3	0	0	3
ELL824	Selected Topics in Information Processing-II	3	0	0	3
ELL894	Network Performance Modeling and Analysis	3	0	0	3
ELL895	Network Security	3	0	0	3

ELL896	Mobile Computing	3	0	0	3
JTD792/ SMD792	Minor Project	0	0	6	3

Streamed Electives (JTM) in (Artificial Intelligence and Machine Learning)

COL761	Data Mining	3	0	0	4
COL762	Database Implementation	3	0	0	4
COL770	Advanced Artificial Intelligence	3	0	0	4
COL776	Learning Probabilistic Graphical	3	0	0	4
ELL706	Optimization for Electrical Engineers	3	0	0	3
ELL729	Stochastic Control and Reinforcement Learning	3	0	0	3
ELL784/ COL774	Introduction to Machine Learning	3	0	0	3
ELL791	Neural Systems and Learning Machines	3	0	0	4
ELL795	Swarm Intelligence	3	0	0	3
ELL882	Large-Scale Machine Learning	3	0	0	3
ELL886	Big Data Systems	3	0	0	3
ELL888	Advanced Machine Learning	3	0	0	3
JTD792/ SMD792	Minor Project	0	0	6	3

Streamed Electives (JTM) in (Embedded Systems)

COL718	Architecture of High Performance Computers	3	0	2	4
COL719	Synthesis of Digital Systems	3	0	2	4
ELL766	Appliance Systems	3	0	0	3
ELL787	Embedded Systems and Applications	3	0	0	3
ELL790	Digital Hardware Design	3	0	0	3
ELL887	Cloud Computing	3	0	0	3
ELL898	Pervasive Computing	3	0	0	3
ELL899	Testing and Fault Tolerance	3	0	0	3
ELP721	Embedded Telecommunication Systems Laboratory	0	1	4	3
ELP781	Digital Systems Lab	0	1	4	3
JTD792/ SMD792	Minor Project	0	0	6	3

Streamed Electives (JTM) in (Telecom Management)

MSL700	Fundamentals of Management of Technology	3	0	0	3
MSL701	Strategic Technology Management	2	0	2	3
MSL707	Management Accounting	3	0	0	3
MSL713	Information Systems Management	2	0	2	3
MSL726	Telecom Systems Analysis, Planning and Design	3	0	0	3
MSL728	International Telecommunication Management	3	0	0	3
MSL760	Marketing Management	2	0	2	3
MSL814	Data Visualization	1.50	0	1.5	
MSL815	Decision Support and Expert Systems	2	0	2	3
MSL850	Management of Information Technology	3	0	0	3
MSL878	Electronic Payments	1.50	0	1.5	
MSL883	ICTs Development and Business	1.50	0	1.5	
MSL886	IT Consulting and Practice	3	0	0	3
MSL888	Data Warehousing for Business Decisions	1.50	0	1.5	
MSL893	Public Policy Issues in the Information Age	1.50	0	1.5	
MSL894	Social Media and Business Practices	3	0	0	3
JTD792/ SMD792	Minor Project	0	0	6	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)				Lecture courses	Contact h/week				Credits
						L	T	P	Total	
I	ELL711 Signal Theory (3-0-0) 3	ELL712 Digital Communication (3-0-0) 3	ELL785 Computer Communication Networks (3-0-0) 3	BSP710 Communication & Signal Processing Technologies (0-1-4) 3	3	9	1	4	14	12
II	ELP725 Wireless Comm. Lab (0-1-4) 3	JTD792 Minor Project (0-0-6) 3 / PE-1 (3-0-0) 3	MSL723 Telecom Syst. Mgmt. (3-0-0) 3	PE-2 (3-0-0) 3	3/2	6/9	1	10/4	17/13	12
Summer										
III	JTD801 Major Project Part-I (0-0-12) 6	PE-3 (3-0-0) 3	PE-4 (3-0-0) 3		2	6	0	12	18	12
IV	JTD802 Major Project Part-II (0-0-24) 12				0	0	0	24	24	12

Total = 48

Master of Technology in VLSI Design Tools and Technology

Interdisciplinary Programme

The overall credits structure

Category	PC	PE	OC	Total
Credits	18	30	0	48

Program Core

ELL734	MOS VLSI design	3	0	0	3
ELP736	Physical Design Laboratory	0	0	6	3
JVD811	Major Project-I	0	0	24	12
Total Credits					18

JVS801	Independent Study	0	3	0	3
MTL704	Numerical Optimization	3	0	0	3
CRL702	Architectures and Algorithms for DSP Systems	2	0	4	4
CRL711	CAD of RF and Microwave Circuits	3	0	2	4
CRL712	RF and Microwave Active Circuits	3	0	0	3

Program Electives

BSP710	Communication & Signal Processing Tech. Lab.	0	1	4	3
COL702	Advanced Data Structures	3	0	2	4
COL718	Architecture of High Performance Computers	3	0	2	4
COP820	Processor Design Lab	0	0	8	4
ELL712	Digital Communications	3	0	0	3
ELL737	Flexible Electronics	3	0	0	3
ELL742	Introduction to MEMS Design	3	0	0	3
ELL743	Photovoltaics	3	0	0	3
ELL745	Quantum Electronics	3	0	0	3
ELL746	Biomedical Electronics	3	0	0	3
ELL747	Active and Passive Filter Design	3	0	0	3
ELL830	Issues in Deep Submicron VLSI Design	3	0	0	3
ELL831	CAD for VLSI, MEMS, and Nanoassembly	3	0	0	3
ELL832	Selected Topics in IEC-I	3	0	0	3
ELL833	CMOS RF IC Design	3	0	0	3
ELL883	Embedded Intelligence	3	0	0	3
ELP831	IEC Laboratory-I	0	0	6	3
ELV830	Special Module in Low Power IC Design	1	0	0	1
ELV831	Special Module in VLSI Testing	1	0	0	1
JVD799	Minor Project	0	0	12	6
JVD812	Major Project-II	0	0	24	12

Streamed Electives (JVL) in (ASIC and SoC Design)

COL719	Synthesis of Digital Systems	3	0	2	4
COL812	System Level Design and Modelling	3	0	0	3
COP745	Digital System Design Laboratory	0	0	6	3
ELL731	Mixed Signal Circuit Design	3	0	0	3
ELL735	Analog Integrated Circuits	3	0	0	3
ELL749	Semiconductor Memory Design	3	0	0	3

Streamed Electives (JVL) in (Micro and Nano Devices)

ELL730	I.C. Technology	3	0	0	3
ELL732	Micro and Nanoelectronics	3	0	0	3
ELL738	Micro and Nano Photonics	3	0	0	3
ELL739	Advanced Semiconductor Devices	3	0	0	3
ELL740	Compact Modeling of Semiconductor Devices	3	0	0	3
ELL744	Electronic and Photonic Nanomaterials	3	0	0	3

Streamed Electives (JVL) in (Embedded Intelligent Systems)

COL788	Advanced Topics in Embedded Computing	3	0	0	3
COL821	Reconfigurable Computing	3	0	0	3
ELL720	Advanced Digital Signal Processing	3	0	0	3
ELL741	Neuromorphic Engineering	3	0	0	3
ELL784	Introduction to Machine Learning	3	0	0	3
ELL797	Energy-Efficient Computing	3	0	0	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)				Lecture courses	Contact h/week				Credits
						L	T	P	Total	
I	ELP736 Physical Design Laboratory (0-0-6) 3	ELL734 MOS VLSI Design (3-0-0) 3	PE-1 (3-0-0) 3	PE-2 (3-0-0) 3	3	9	0	6	14	12
II	PE-3 (3-0-0) 3	PE-4 (3-0-0) 3	PE-5 (3-0-0) 3	PE-6 (3-0-0) 3	4	12	0	0	12	12
Summer										
III (OR)	JVD811 Major Project-I (0-0-24) 12				0	0	0	24	24	12
IV	JVD812 Major Project-II (0-0-24) 12				0	0	0	24	24	
III (OR)	PE-7 (3-0-0) 3	PE-8 (3-0-0) 3	JVD799 Minor Project (0-0-12) 6		2	6	0	12	18	
IV	JVD811 Major Project-I (0-0-24) 12				0	0	0	24	24	
III (OR)	JVD811 Major Project-I (0-0-24) 12				0	0	0	24	24	
IV	PE-7 (3-0-0) 3	PE-8 (3-0-0) 3	PE-9 (3-0-0) 3	PE-10 (3-0-0) 3	4	12	0	0	12	12

Total = 48

Master of Technology in Robotics

Interdisciplinary Programme

The overall credits structure

Category	PC	PE	OC	Total
Credits	42	6	3	51+ Bridge

Program Core

Foundational Concepts

JRL704	Robotics Lab	1	0	4	3
ELL701	Mathematical Methods Incontrol	3	0	0	3

Dynamics and Control

MCL799	Mechanics and Control of Robots	3	0	0	3
JRL747	Degrees of Freedom and Constraints in Devices	2	0	0	2

Reasoning/Cognition (any one)

COL778	Principles of Autonomous Systems	3	0	2	4
COL770	Advanced Artificial Intelligence	3	0	2	4

Sensing and Perception

JRL780	Computer Vision	3	0	2	4
ELL705	Stochastic Filtering and Identification	3	0	0	3

Project

JRD891	Minor Project	0	0	4	2
JRD892	Major Project Part-I	0	0	12	6
JRD893	Major Project Part-II	0	0	24	12

Bridge (BR: At least one of the following two, but waived for people with preparation in both)

COL671	Artificial Intelligence	3	0	2	4
ELL225/	Control Engineering-I	3	1	0	4
MCL212	Control Theory and Applications	3	0	2	4

Total Credits

42

Program Electives

JRL880	Special Topics in Robotics-I	3	0	0	3
JRL882	Special Topics in Robotics-II	2	0	0	2
JRV880	Special Module in Robotics-I	1	0	0	1
JRS799	Independent Study	0	3	0	3
ELL893	Cyber-Physical Systems	3	0	0	3

ELL794	Human-Computer Interface	3	0	0	3
ELL800	Numerical Linear Algebra and Optimization in Engineering	3	0	0	3
ELL706	Optimization for Electrical Engineers	3	0	0	3
MTL704	Numerical Optimization	3	0	0	3
MTL729	Computational Algebra and its Applications	3	0	0	3
ELL788	Computational Perception and Cognition	3	0	0	3
COL783	Digital Image Analysis	3	0	3	4.5
ELL715	Digital Image Processing	3	0	2	4
COL772	Natural Language Processing	3	0	2	4
MTL785	Natural Language Processing	3	0	0	3
ELL720	Advanced Digital Signal Processing	3	0	0	3
DSL711	Sensors & Transducers	3	0	0	3
MCL731	Analytical Dynamics	3	0	0	3
MCL738	Dynamics of Multibody Systems	2	0	2	3
MCL749	Mechatronics Product Design	3	0	2	3
MCL837	Advanced Mechanisms	2	0	2	3
MCL845	Advanced Robotics	2	0	2	3
ELL700	Linear Systems Theory	3	0	0	3
MCL741	Control Engineering	3	0	2	3
ELL702	Nonlinear Systems	3	0	0	3
ELL703	Optimal Control Theory	3	0	0	3
ELL704	Advanced Robotics	3	0	0	3
ELL729	Stochastic Control and Reinforcement Learning	3	0	0	3
ELL767	Mechatronics	3	0	0	3
ELL801	Nonlinear Control	3	0	0	3
ELL802	Adaptive and Learning Control	3	0	0	3
ELL803	Model Reduction in Control	3	0	0	3
ELL804	Robust Control	3	0	0	3
ELL805	Networked and Multi-Agent Control Systems	3	0	0	3
ELL806	Modeling and Control of Distributed Parameter Systems	3	0	0	3
ELL807	Stochastic Control	3	0	0	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, credits)					Lecture courses	Contact h/week				Credits
							L	T	P	Total	
I	JRL704 Robotics Lab (1-0-4) 3	ELL701 Mathematical Methods Incontrol (3-0-0) 3	MCL799 Mechanics and Control of Robots (3-0-0) 3	COL671 or ELL225/ MCL212 (3-0-2) 4		4	10	0	6	16	13
II	COL778 or COL770 (3-0-2) 4	JRL780 Computer Vision (3-0-2) 4	ELL705 Stochastic Filtering and Identification (3-0-0) 3	JR891 Minor Project (0-0-4) 2		3	9	0	8	17	13
III	PE-1 (3-0-0) 3	PE-2 (3-0-0) 3	JRD892 Major Project-I (0-0-12) 6	JRL747 Degrees of Freedom and Constraints in Devices (2-0-0) 2	OC (3-0-0) 3	5	11	0	12	23	17
IV	RD893 Major Project-II (0-0-24) 12					0	0	0	24	24	12

Total = 51 + 4 (Bridge Course)

Master of Public Policy

Schools of Public Policy

The overall credits structure

Category	PC	PE	OE	Total
Credits	39	9	6	54

Program Core

SPL703	Research Methods for Public Policy	3	0	0	3
SPL704	Introduction to Public Policy and STI	3	0	0	3
SPL705	Public Institutions in India: Theory and Practice	3	0	0	3
SPL706	STI and Sustainable Development	3	0	0	3
SPL707	Tools for Policy Analysis	3	0	0	3
SPL708	Statistics for Public Policy	3	0	0	3
SPL709	Principles of Economics for Public Policy	3	0	0	3
SPL779	Thesis-I	0	0	8	4
SPL780	Thesis-II	0	0	28	14
Total Credits					39

Program Electives

SPL711	Modelling Complex Adaptive Systems for Policy Analysis	3	0	0	3
SPL712	Comparative Industrial Policy	3	0	0	3
SPL719	Public Interest Technologies	3	0	0	3
SPL720	Energy and Infrastructure Finance: A Public Policy Lens	3	0	0	3
SPL721	Perspective on Climate Change: Implications for Policy	3	0	0	3
SPL 722	Governance Challenges in Energy Systems in Transition	3	0	0	3
SPL723	Understanding Public Policy Making through Case Studies	3	0	0	3
SPL724	Electricity Sector Reforms in the Renewables Era	3	0	0	3

SPL730	Science, Technology & Innovation Policy and Agriculture	3	0	0	3
SPL740	Casual Interference and Impact Assessment	3	0	0	3
SPL741	Socio-Economic Data Analysis	3	0	0	3
SPL750	Qualitative Research Methods for Public Policy	3	0	0	3
SPL752	Techs and the City - An urban policy perspective	3	0	0	3
SPL753	Urban Planning Practicum	2	0	4	3
SPL754	Geographical Information Systems (GIS) for Public Policy	2	0	2	3
SPL781	Special Topics in Climate Change Policy	3	0	0	3
SPL783	Special Topics in Natural Resource Management Policy	3	0	0	3
HSL781	Potential and Perils of the Digital Welfare	3	0	0	3
SPV791	Special Module on Ethics in Policy Making	1	0	0	1
SPV792	Inclusive City and Accessibility Audit	1	0	0	1
SPV793	Special Module on Public Policy in Management of Natural Resources	1	0	0	1
SPV794	Special Module on Public Policy in Data, Communication and Computation	1	0	0	1
SPV795	Special Module on Public Policy for Technical Higher Education	1	0	0	1
SPV796	Special Module on Public Policy for Sustainable Habits and Livelihoods	1	0	0	1
SPV797	Scenario Design for Managing Uncertainty	1	0	0	1
SPV798	Special Module on Public Policy in Transport and Infrastructure	1	0	0	1
SPV799	Special Module is Policy Studies	1	0	0	1
SPL810	Selected Topics in Policy Studies	3	0	0	3

Sem.	Courses (Number, Abbreviated Title, L-T-P, Credits)				Lecture courses	Contact h/week				Credits
						L	T	P	Total	
I	SPL704 Introduction to Public Policy and STI (3-0-0) 3	SPL706 STI and Sustainable Development (3-0-0) 3	SPL708 Statistics for Public Policy (3-0-0) 3	SPL709 Principles of Economics for Public Policy (3-0-0) 3	4	12	0	0	12	12
II	SPL703 Research Methods for Public Policy (3-0-0) 3	SPL705 Public Institutions in India: Theory and practice (3-0-0) 3	SPL707 Tools for Policy Analysis (3-0-0-0) 3	PE-1 (3-0-0) 3	4	12	0	0	12	12
Summer										
III	PE-2 (3-0-0) 3	PE-3 (3-0-0) 3	OE-1 (3-0-0) 3	Thesis-I (0-0-8) 4	3	9	0	8	17	13
IV	OE-1 (3-0-0) 3	Thesis-II (0-0-28) 14			1	3	0	28	31	17

Total = 54